



****Service Manual****

Last Update on 3/17/2020

DEALER MEMO

TO: All Oasis Service Centers
SUBJECT: Model Year Spa Serial Numbers

DATE: January 1, 2002
FROM: Technical Service

The guidelines for assigning serial numbers to Oasis spas have been modified to prevent duplicate serial numbers from being produced.

To understand the serial number Watkins is implementing, it is important to know how to read a serial number, so the following section describes how to read spa serial numbers, in addition to specifying the change we are implementing. The serial number may be found on the spa label inside of most equipment compartment in the upper right hand side on the filter can. See Figure 1 for a generic example.

2000 – 2001 Models:

Oasis spa serial #'s have 9 characters comprised of a combination of the selections below.

M	M	S	#	Y	#	#	#	#
Model Designator		Series	Qtr	Year	Sequence			

Model Designators/Model	Series Codes	Year
OA - Oasis	I – Watkins	2000 B
		2001 C
		2002 D

Example:

OA	I	1	B	1	0	0	1
Oasis	Watkins	1st Qtr	2000	First spa built in the first quarter			

2002 – Current Models:

Oasis spa serial #'s have 10 characters comprised of a combination of the selections below.

M	M	S	#	#	Y	#	#	#	#
Model Designator		Series	Number	Qtr	Year	Sequence			

Model Designators/Model	Series Codes	Number	Year
OA - Oasis	I – Watkins	1 – Oasis	2000 B
		2 – Oasis II	2001 C
			2002 D

Example:

OA	I	1	1	B	1	0	0	1
Oasis	Watkins	Number	1st Qtr	2000	First spa built in the first quarter			

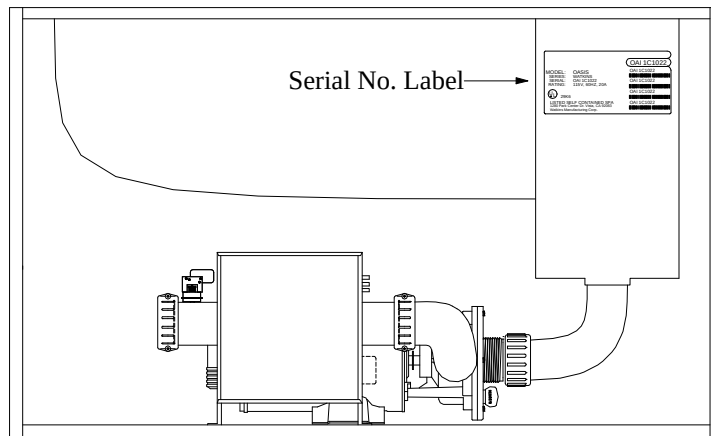


Figure 1

If you have any questions, do not hesitate to contact the Technical Support Department at (800) 669-1881 @ extension 115.

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**WATKINS MANUFACTURING
2002 OASIS DOMESTIC SERVICE PARTS WHOLESALE/RETAIL PRICE LIST
EFFECTIVE APRIL 9, 2002**

PART NUMBER	ITEM DESCRIPTION	WHOLESALE PRICE	SUGGESTED RETAIL PRICE
005005	JET, BLASTER BODY	\$1.00	\$2.00
005006	JET, BLASTER, SPINNER	\$2.50	\$5.00
005007	JET, BLASTER, FACEPLATE INSERT	\$1.00	\$2.00
005023	JET, BLASTER, FACEPLATE	\$1.00	\$2.00
005024	JET, BLASTER, WALLFITTING	\$1.00	\$2.00
005039	JET, MAGNA, DUAL ROTO	\$8.00	\$16.00
005080	JET, MAGNA GRAY 10-4875UNLG	\$20.00	\$40.00
005097	JET, SUCTION COVER GREY 30133	\$1.15	\$2.30
005099	JET, WALL FITTING 1 1/2" 30147	\$1.00	\$2.00
005101	JET, STANDARD SUCTION NUT	\$1.00	\$2.00
007002	MANUAL, OASIS	\$6.95	\$13.90
008019	SENSOR SET, 9710	\$26.90	\$53.80
008104	TRANSFORMER, 9130	\$16.50	\$33.00
008122	FUSE, 20A	\$5.00	\$10.00
009047	PUMP, IMPELLER 1HP	\$10.75	\$21.50
009060	PUMP, DRAIN PLUG	\$1.00	\$2.00
010001	MANIFOLD, 1.5" w/ 10.5" PORTS	\$4.40	\$8.80
010002	MANIFOLD, 1.5" w/ 10.5" PORTS	\$3.95	\$7.90
010005	MANIFOLD, 2Sx2x(6) 3/4" S PORT	\$1.95	\$3.90
010006	MANIFOLD, 2Sx2Sx(6) 1/2" PORTS	\$1.85	\$3.70
010007	MANIFOLD, 2Sx2x2(4) 1/2" PORT	\$1.50	\$3.00
010009	MANIFOLD, 1" BARB ADAP (2)3/8"	\$0.75	\$1.50
010010	MANIFOLD, 4 PORT .75x1.5	\$1.55	\$3.10
010011	MANIFOLD, 2Sx2SPGx(4) 3/4 S	\$1.50	\$3.00
010012	MANIFOLD, 2 PORT 1 S1/2" Sx3/4"	\$1.45	\$2.90
010013	MANIFOLD, 5 PORT 2Sx3/4"	\$2.95	\$5.90
010017	MANIFOLD, 10 PORT 1-3/4S X 1-3/4 SPG X 3/4 P	\$2.35	\$4.70
010018	MANIFOLD, 6 PORT	\$2.05	\$4.10
010019	MANIFOLD, 8 BARB MANIFOLD	\$1.70	\$3.40
010020	MANIFOLD, 4 BARB MANIFOLD	\$1.20	\$2.40
011052	FITTINGS, PVC3/4 X 3/4 BARB10-4578	\$0.50	\$1.00
011075	SENSOR FITTING	\$3.60	\$7.20
017008	SCREW, 1/4"-20 INSERT ZINC	\$0.15	\$0.30
017066	MOUNT, RUBBER-PUMP	\$0.75	\$1.50
017081	BALBOA TERM STRIP 50AMP	\$6.50	\$13.00
017082	COVERLOCKS IDEAL	\$18.60	\$37.20
017086	SCREW. #8 X 1 5/8 P-BGL SS D/W	\$0.10	\$0.20
017089	SCREW, 1/4"-20X1-1/2 P-TRUSS	\$0.10	\$0.20
020120	CORD, GFCI	\$15.75	\$31.50
022005	AIR CONTROL WATERDROP A/C GRAY	\$3.75	\$7.50
022030	VALVE OZONE, CHECK VALVE	\$2.10	\$4.20
023001	LIGHT, WALL FIT NUT, GASKET	\$2.95	\$5.90
023010	LIGHT, WIRE HARNESS	\$2.00	\$4.00

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023013	LIGHT BULB	\$2.50	\$5.00
023016	REFLECTOR, SMALL	\$0.40	\$0.80
024014	HEATER, SPLIT NUT	\$2.00	\$4.00
025001	CORD, 15 AMP GFCI 125v	\$40.00	\$80.00
031003	OZONE, GOLD CLAMPS	\$0.25	\$0.50
031007	OZONE, BRASS BARB FITTING	\$1.25	\$2.50
033014	FILTER, 50 S.F., OASIS	\$20.00	\$39.00
30627	TEE, 1" SxSxS	\$0.60	\$1.20
30628	TEE, 1-1/2" SxSxS	\$1.05	\$2.10
30634	TEE, 1"x1"x1-1/2" SxSxS	\$1.65	\$3.30
30636	TEE, 1-1/2"x1-1/2"x3/4" SxSxS	\$1.60	\$3.20
30637	TEE, 2"x2"x1/2" SxSxS	\$1.70	\$3.40
30638	TEE, 2"x2"x1" SxSxS	\$1.70	\$3.40
30639	TEE, 2"x2"x1-1/2" SxSxS	\$2.75	\$5.50
30644	ELBOW, 90D 1" SLIPxSLIP	\$0.50	\$1.00
30645	ELBOW, 90D 1-1/2" SLIPxSLIP	\$0.80	\$1.60
30649	ELBOW, ST 90 1/2" SPIGxSLIP	\$0.65	\$1.30
30650	ELBOW, ST 90 3/4" SPIGxSLIP	\$0.75	\$1.50
30651	ELBOW, ST 90 1" SPIGxSLIP	\$1.10	\$2.20
30652	ELBOW, ST 90 1-1/2" SPIGxSLIP	\$1.60	\$3.20
30657	ELBOW, 45D 1" SLIPxSLIP	\$0.70	\$1.40
30659	ELBOW, 45D 1-1/2" SLIPxSLIP	\$1.10	\$2.20
30660	COUPLING, 1/2" SLIPxSLIP	\$0.15	\$0.30
30661	COUPLING, 3/4" SLIPxSLIP	\$0.25	\$0.50
30662	COUPLING, 1" SLIPxSLIP	\$0.40	\$0.80
30664	COUPLING, 1-1/2" SLIPxSLIP	\$0.55	\$1.10
30665	COUPLING, 2" SLIPxSLIP	\$0.85	\$1.70
30675	BUSHING, 3/4"x1/2" SPIGxSLIP	\$0.25	\$0.50
30676	BUSHING, 1"x1/2" SPIGxSLIP	\$0.50	\$1.00
30677	BUSHING, 1"x3/4" SPIGxSLIP	\$0.50	\$1.00
30680	BUSHING, 1-1/2"x3/4" SPxSL	\$0.60	\$1.20
30681	BUSHING, 1-1/2"x1" SPxSL	\$0.60	\$1.20
30691	447-010 CAP SL 1"	\$0.30	\$0.60
30693	CAP 1-1/2 SL	\$0.65	\$1.30
30696	PLUG, 1/2" SPIG	\$0.30	\$0.60
30697	PLUG, 3/4" SPIG	\$0.60	\$1.20
30698	PLUG, 1" SPIG	\$0.70	\$1.40
30699	PLUG, 1-1/2" SPIG	\$1.90	\$3.80
30702	ADAPTER, 3/4"x1/2" MHTxSLIP	\$0.70	\$1.40
31288	REDUCER, 2"x1-1/2" SPIGxSL	\$0.75	\$1.50
31487	TEE, 1/2"x1/2" BARB	\$0.55	\$1.10
31574	TEE, 1-182"x1-1/2"x1"	\$1.00	\$2.00
31932	ELBOW, 90D 2" SLIPxSLIP	\$1.05	\$2.10
32465	TEE, 2"	\$0.90	\$1.80
32467	ELBOW, 45D 2"	\$0.90	\$1.80

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PART NUMBER	ITEM DESCRIPTION	WHOLESALE PRICE	SUGGESTED RETAIL PRICE
33048	TEE, 2 X 2 X 3/4	\$0.95	\$1.80
33761	HEATER NUT, 2" SPLIT	\$1.60	\$3.20
34277	AP 100 SPEARS HOSECAP 3/4 WHITE	\$2.10	\$4.20
34391	ELBOW 2"	\$1.80	\$3.60
34918	RED AND BLUE LENS OS LIGHT ASSEMBLY	\$2.10	\$4.20
35514	ELBOW 2" STREET 45 DEG SPXSL	\$1.15	\$2.30
38612R	COVER, CD-13 OASIS RUST REPLACEMENT	\$190.00	\$380.00
38613	ENC, CD-13 OASIS RUST	\$140.00	\$280.00
38661	BREAKER GFCI, SUBPANEL BOX	\$100.00	\$200.00
38668	OWNER MAN, OASIS (DOM) - 2000	\$6.95	\$13.90
61625	SERVICE MANUAL, OASIS	\$15.00	\$30.00
70220	RING, WEAR 1SP	\$0.45	\$0.90
70308	KIT, TUBING FLEX 2 CREAM	\$4.40	\$8.80
70309	KIT, TUBING FLEX 1-1/2 CREAM	\$2.95	\$5.90
70311	KIT, TUBING FLEX 1 CREAM	\$2.15	\$4.30
70312	KIT, TUBING FLEX 3/4 CREAM	\$2.20	\$4.40
70313	KIT, TUBING FLEX 1/2 CREAM	\$1.90	\$3.80
70317	KIT, TUBING CLEAR 3/8 X 9/16	\$1.40	\$2.80
70322	KEY, COVERLOCK - PUSH TYPE	\$0.55	\$1.10
70922	ASSY, GFCI	\$60.40	\$120.80
70996	GFCI W/O CORD	\$32.90	\$65.80
71036	FITTING, COMPRESS 2" W/O-RING	\$4.00	\$8.00
71038	FITTING,COMPRESS.1.5"W/O-RING	\$2.58	\$5.16
71409	WEAR RING, XP	\$0.55	\$1.10
71415	MANIFOLD, XP MAGNETEK	\$5.70	\$11.40
71447	KIT, SHAFT SEAL ASSEMBLY	\$6.60	\$13.20
71515	KIT COVERLOCK, BLACK	\$4.40	\$8.80
71554	O-RING, VOLUTE 50HZ	\$0.55	\$1.10
71578	ASSY, SENSOR	\$13.40	\$26.80
71580	KIT, HEATER FLOW THRU '99	\$61.50	\$123.00
71586	PRESS SWITCH	\$16.50	\$33.00
71612	KIT, LIGHT RECEPTACLE	\$3.85	\$7.70
71693	VOLUTE 48FR XP2	\$15.95	\$31.90
71696	COVER, SUCTION XP2	\$15.25	\$30.50
71928	CORE ASSY, CD-12, W/BKT	\$40.00	\$80.00
72102	KIT, LITE LDR CPA, HIGLAND	\$50.85	\$101.70
72122	VOLUTE FMCP & TMCP SERIES	\$9.40	\$18.80
72135	KIT XMFR REPLACE (008102)	\$30.20	\$60.40
72137	O-RING, VOLUTE	\$0.40	\$0.80
72138	COVER, SUCTION	\$5.40	\$10.80
72140	KIT, PVC 1/4" TUBING	\$2.45	\$4.90
72154	KIT, OASIS CORNER PIECE	\$2.60	\$5.20
72155	KIT, LILY BSKT ASSY W/FLOATS	\$12.25	\$24.50
72156	KIT, HOUSING W/FLANGES	\$59.05	\$118.10
72161	KIT PRESSURE SWITCH	\$10.60	\$21.20

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PART NUMBER	ITEM DESCRIPTION	WHOLESALE PRICE	SUGGESTED RETAIL PRICE
72162	KIT LITE LEADERS SYST PRE-'01	\$222.45	\$444.90
72163	KIT HEATER, REPLACEMENT	\$53.40	\$106.80
72168	BREAKER, 50-AMP 2 POLE	\$72.00	\$144.00
72172	VOLUTE FMHP SERIES	\$6.90	\$13.80
72186	KIT, BEZEL/LEVER DIV VALVE '01	\$6.65	\$13.30
72187	KIT, OZONE JET ASSY PRE-'01	\$3.25	\$6.50
72200	KIT JET PUMP (009064)	\$140.00	\$280.00
72201	KIT, JET PUMP	\$174.25	\$348.50
72225	KIT, PRESSURE SWITCH	\$12.05	\$24.10
72227	KIT, OZONE WATER QUALITY	\$80.00	\$160.00
72281	KIT, FRNT/BACK PANEL-OAS '01	\$110.95	\$221.90
72282	KIT, SIDE PANEL-OAS '01	\$105.15	\$210.30
72285	KIT, LABEL,OASIS OVERLAY	\$5.10	\$10.20
72286	KIT, CEDAR SLAT, OASIS-24"	\$4.20	\$8.40
72287	KIT, OASIS OZONE JET ASSY	\$3.30	\$6.60
72290	KIT, OASIS JET ASSY 98 - CURR	\$3.50	\$7.00
72341	KIT, OASIS STANDPIPE	\$2.50	\$5.00
72344	KIT, CAL PUMP HOSE DUCT	\$3.50	\$7.00
72353	BACK PANEL 24 1/2" X 77 3/4"	\$109.62	\$219.24
72354	SIDE PANEL 24 1/2" X 67 3/4"	\$105.10	\$210.20
72355	SMALL DOOR PANEL, 24 1/2" x 24 7/8"	\$61.99	\$123.98
72356	LARGE DOOR PANEL 24 1/2" X 53 1/8"	\$95.05	\$190.10
72369	KIT, TUBING, CLEAR 1/4 X 3/8	\$2.90	\$5.80
72370	KIT, TUBING, CLEAR 1/2 X 5/8	\$3.15	\$6.30
72373	KIT, TUBING BRAID 3/8	\$3.20	\$6.40
72374	KIT, TUBING BRAID 3/4 X 1	\$4.85	\$9.70
72469	KIT, CBA W/O HEATER	\$359.00	\$718.00
72471	KIT, CNTRL HD 4B OASIS '02	\$42.50	\$85.00
72516	KIT, LG "PUMP" DR OASIS II	\$116.00	\$232.00
72517	KIT, SM "EQUIP" DR OASIS II	\$76.00	\$152.00
72518	KIT, SD DR OASIS II '02	\$125.00	\$250.00
72519	KIT, BK DR OASIS II '02	\$130.00	\$260.00

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PART NUMBER	ITEM DESCRIPTION	WHOLESALE PRICE	SUGGESTED RETAIL PRICE
10066	THERMOMETER	\$7.65	
10069	ELECTRIC METER	\$135.00	
50275	TOOL, WALLFITTING	\$5.00	
71129	TEST LEAD, NEEDLE BLACK	\$7.50	
71131	TEST LEAD, NEEDLE RED	\$7.50	



Delivery And Installation

DELIVERY AND INSTALLATION

Before Delivery

1. Inspect the Site – Visiting the site is highly recommended before the spa is delivered to insure that installation is done quickly and professionally. Perform the following when visiting a site:

- A concrete slab with a 1 inch per 10 feet slope is preferred so the rain water and water spillover will run off and not puddle underneath the spa (water under the spa for long periods of time may cause the wood to deteriorate).
- Determine where the customer wants to place the spa. Make accurate and detailed notes that can be used to find both the customers' home, and the exact location where the spa is to be installed.
- Identify any obstructions that might create delivery problems (i.e. embankments, animals that could hinder delivery, unmovable vehicles, heating/cooling systems, trees, and/or gates).
- Use a vertical delivery cart.

2. Determine the number of installers needed. At least two installers are needed for every installation.

3. Check the electrical set-up at the site. Make sure the circuit that is to be used is wired in accordance with all applicable local electrical codes or standards. More than 90% of the sites do not have adequate electrical circuits. If a dedicated circuit or sub-panel needs to be installed, set up an appointment for a qualified electrician to install the circuit or sub-panel and wire the spa while you are installing it. If that is not

possible, have an electrician come after the delivery, accompanied by a sales person or a service technician. (A dedicated circuit is a circuit that has only one receptacle, which is solely for the use of the spa).

NOTE: All electrical work must be done by a licensed electrician, and all wiring should be copper (sub-panels must be rated as outdoor and rain-tight enclosures).

4. Determine the tools needed for delivery and installation.

- For different year model spas, the spa may prime another way. Refer to the specific owner's manual for filling instruction. Use a garden hose for filling the spa. Two hoses will fill the spa faster.
- A 3-ft (1 m) level for leveling the spa.

Delivery and Installation

There are 12-steps that outline the proper delivery and installation procedure. Completing all 12 steps should take approximately 1—2 hours. The steps are listed on the delivery checklist, and a copy of this checklist should be taken on each delivery and each step should be completed. The checklist is included at the back of this section. Go over the checklist with the customers and have them sign it after they understand that all 12 steps have been completed.

Detailed instructions for completing each step are on the pages that follow.

Step 1: Place the Spa

Verify that the location where the spa is to be placed is where the spa owner wants it. Check the notes taken on the pre-delivery visit to the site and verify the location with the spa owner before installation.

Unload and unwrap the spa

When the spa is moved from the truck to the installation site, be careful not to drag it against gates or eaves. Place no heavy objects on or in the spa.

Every spa is packed in special wrapping materials. Refer to the following instructions to remove the wrapping from the spa:

CAUTION – Be careful when unwrapping the spa not to damage the spa skirting.

1. Tip the spa onto one side so you can reach under the spa.
2. Start from the edge and cut the packing material towards the center of the spa.
3. Slowly pull the packaging material out from the indented lip and away from the bottom of the spa.
4. Tip the spa back upright.
5. When completed, the wrapping will end up in a large bundle inside the spa.

Step 2: Visually inspect the spa

Carefully check the surface of the spa and the cover for any defects. For example, check for damage to the skirting.

Step 3: Level the spa

The spa must be level before it is filled, otherwise it will not function properly. It is recommended that a level be used across the

top of the spa that spans the width both ways and diagonally across the spa. Consult the Leveling Instructions included with the spa if any questions arise.

Step 4: Remove the equipment compartment door, and give the start-up package to the spa owner

Advise spa owners to carefully read the Owner's Manual and warranties. If they have any questions about the material, let them know that they can call the dealership or the manufacturer. Place the spa's serial number sticker from the equipment compartment, peel-off sticker without the barcode, and write the date of installation on page 1 of the Owner's Manual. Advise the spa owner to fill out the warranty card and send it in.

Step 5: Fill the spa with water

Make sure the drain caps are tightly secured. Use an ordinary garden hose to fill the spa two inches above the highest jet.

Step 6: Connect power to the spa

Plug the spa into the outlet or close the sub-panel breaker to connect power to the spa.

Step 7: Test the spa's functions

Verify that the following items are functioning properly:

Test the GFCI Device

Trip the GFCI device by pushing in the test button. Wait 30 seconds then reset the GFCI.

Adjust the Thermostat

Raise and lower the set temperature of the thermostat to ensure the temperature controls are functioning.

Activate the Spa Light

Switch the spa light on to see if it is functioning.

Check the Jet Menus

Verify the jet menus using the Jet Menu diagrams located in the Owner's Manual.

Once the Jet Menus have been verified, check the operation of each jet in the spa.

- Direct Jets – Verify that the direct jet is activated.
- Rotary Jets (optional) – Increase and decrease the jet pressure of each jet using the jet's air control.
- Mini Rotary Jets (optional) – Increase and decrease the jet pressure of each jet using the jet's air controls.

Step 8: Sanitize the spa water

Prepare the water for sanitation by checking the following:

Check the total alkalinity. The total alkalinity should be in the 125 - 150 parts per million range. Use sodium bicarbonate (baking soda) to raise total alkalinity.

- Check the calcium hardness. The calcium hardness level should be 200 parts per million (ppm). Add soft water, stain and scale control, or calcium hardness increaser to adjust the calcium hardness.
- Check the pH. The pH should be between 7.4 - 7.6. Use a pH increaser or decreaser to adjust the pH.

Sanitize the water

If the spa owner is using chlorine as their primary sanitizer, then use a super-chlorinating treatment (10 parts per million or above) to kill any bacteria that may be present in the spa and/or plumbing. Activate the jet pump, sprinkle the chemicals slowly into the filter compartment and leave the jet pump on for approximately 10 minutes.

Advise the spa owners to wait until the sanitizer is below 5 parts per million before entering the spa.

WARNING – Be sure to notify the customer of improper chemical administration (misuse of the bromine floater or super chlorination with the cover in place).

Step 9: Orient the spa owner to the spa

Show the spa owner how to activate each of the spa functions, and go over the maintenance schedule. The goal is not only to demonstrate the functions and benefits of the spa, but also to teach the customer the importance of maintenance. The spa owner who fully understands the spa's operation and maintenance procedures is more likely to be a satisfied customer.

NOTE: More information about spa functions can be found in the Owner's Manual. Point out the appropriate sections to the spa owner for future reference.

Thermostat

Show the customer how to adjust the spa's water temperature.

Spa components

In easy to understand language, explain the general purposes of the jet pump, heater, etc.

Spa Light

Show the spa owner where the light bulb is located and how to replace it.

Jet adjustment

Show the customer how to adjust the jets. It is usually effective to encourage the customer to place a hand into the water so they can feel for themselves the difference in jet pressure and direction.

Water chemistry maintenance

Stress to the customer the importance of daily and weekly chemical maintenance program. The Owner's Manual will explain all aspects of chemical usage and safety. Explain the damage that can be done when the water is not maintained properly. Show the customer how to use chemical test kits.

Filter maintenance

Tell the spa owner that the filter needs to be cleaned at least once a month and that failure to maintain the filter will result in reduced water flow. Show the spa owner how to remove and install the filter. Then, explain that the filter should always be cleaned with a filter degreaser to remove mineral and oil buildup. Tell the spa owner to soak the filter in a degreaser (according to the manufacture's instructions), then thoroughly spray the filters with a garden hose until clean.

Draining the spa

Explain that the spa water should be changed once every four months. Show the

spa owner where the drain is located in the equipment compartment, and how to hook a garden hose to it.

Vinyl cover care

Explain how and when (every month) the vinyl cover should be cleaned. Explain that snow must not be allowed to build up on the spa cover, and that no one should sit or stand on the cover. Also, explain that chemicals or cleaners not recommended by Watkins Manufacturing should not be used to clean the cover, and that the cover should not be lifted by the flaps or the coverlock straps.

Skirt maintenance

To keep the skirt looking new, a natural wood-color oil-based stain should be applied twice a year or as needed. Have the spa owner consult their Owner's Manual for further information.

Shell maintenance

A non-abrasive, non-foaming cleaner should be used to clean the spa surface. Baking soda can be used for minor surface cleaning. Have the spa owner consult their Owner's Manual for further information on shell maintenance.

Spa safety features

Explain the spa safety features, which include the high-limit and the GFCI device. Tell the spa owner that the entire spa will shut down if the high-limit or the GFCI trips or if a fuse blows. Let the spa owner know that the heater will be inoperative if a flow-monitoring device (pressure switch or heater thermal cut-off) interrupts the heater circuit.

- **High Limit** – Explain that the high limit(s) will trip when the water exceeds the high limit value, which typically indicates a low flow condition (usually caused by clogged or dirty filters). Show the spa owner how to reset the high limit(s).
- **GFCI** - Explain that the GFCI trips when there is a ground fault in the spa and protects the customer against electrical shocks. Show the customer how to reset it and how to test it. Stress to the customer to test the GFCI at least once a month. If repeated tripping occurs, the source should be identified and corrected before continued spa use. Never circumvent the problem.
- **Flow monitoring devices** (pressure switch or heater thermal cut-off) – Explain to the customer that the flow monitoring devices interrupt the heater circuit when flow is restricted, but that the reset of the spa will remain operational.

Step 10: Winterizing

In cold weather areas, explain the importance of winterizing the spa by emptying the spa; removing all of the water from the interior plumbing; preparing the filter compartment; and closing the suction opening.

Step 11: Place the Cover on the spa

Place the cover on the spa and install the cover locks if they are not already installed. Show the customer how to use the cover locks.

Step 12: Restore area to original condition

Put all trash in a suitable place. If there is no place for the trash at the spa owner's home, collect the trash and take it with you for later disposal. Because of the amount of space that the spa wrapping takes, you may want to take it to take it with you for later disposal rather than leave it with the customer.

Put everything back into place that was moved when the spa was delivered. For example, if a table was moved, put it back in place.

Good clean up is very important as it leaves a lasting impression on the customer.

DELIVERY AND INSTALLATION CHECKLIST

Spa delivery and installation involves a 12-step procedure, which should take approximately 1 - 2 hours to complete. The steps are on this Delivery and Installation Checklist. A copy of this checklist should be taken on each spa delivery and each step should be completed. Go over the checklist with spa owners and have them sign it after they understand that all 12 steps have been completed.

Spa Owner's Name _____

Spa Model _____ Serial Number _____ Date _____

- ☐ Spa placed in position desired by the spa owner.
- ☐ Spa visually inspected: ☐ Spa surface ☐ Skirting ☐ Cover
- ☐ Spa leveled.
- ☐ Equipment door is opened and start-up package given to the spa owner.
- ☐ Install the drain plug.
- ☐ Spa filled with water.
- ☐ Spa plugged into its power source.
- ☐ Spa's functions tested and/or adjusted:
 - ☐ GFCI ☐ Jets
 - ☐ Thermostat ☐ Spa Light
 - ☐ Air Valve(s)
- ☐ Spa water sanitized.
- ☐ Spa owner oriented to the spa:
 - ☐ Thermostat ☐ Draining of the spa
 - ☐ Spa components ☐ Cover care
 - ☐ Jet adjustment ☐ Redwood skirting maintenance
 - ☐ Water chemistry ☐ Shell maintenance
 - ☐ Filter maintenance ☐ Spa safety features
- ☐ Winterizing (in cold weather areas)
- ☐ Install the coverlock bases and place the on the spa.
- ☐ Area returned to the original condition

Spa Owner's Signature _____



***Component Explanation
& Diagnosis***



Component Explanation and Diagnosis

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Section 1 : Ground Fault Circuit Interrupter

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GFCIs/GFCI BREAKERS

PART I: CURRENT LEAKAGE

Ground Fault Circuit Interrupters (GFCI) serve an important purpose on OASIS Spas or any electrical circuit. They provide protection against current leakage to ground, which can cause injury or, in extreme cases, death. GFCI breakers, along with providing current leakage to ground protection, protect the spa's equipment and components from overloads and short circuits (excessive current draw). This is the reason GFCIs must always be installed properly and never be removed or altered. 115-volt model spas are protected by a Class A GFCI, which is included with the spas; 230-volt (optional model spas must have GFCI breakers, which are included in a subpanel. Both the Class A GFCI and GFCI breakers must be installed by a qualified electrician. It is important that ALL installation instructions be followed.

Monthly testing must be stressed to spa owners, both for your and their protection. When a GFCI trips, it normally indicates current leakage to ground caused by faulty insulation, wet wiring or wet equipment connected to the circuit. Tripping of GFCI breakers may also indicate an excessive current draw on the circuit.

There is no known device that can offer complete protection against the hazards of an electrical accident under all conceivable conditions. For example, the GFCI does not protect a person who comes in contact with both hot wires on 230-volt models, or in contact with either one of the hot wires and the neutral wire on 115-volt models. There is also no protection from voltage introduced from an exterior source (radio, hair dryer, etc.).

Even with a GFCI installed, an electrical shock may be felt, but such a shock will be less than what is normally considered dangerous. However, individuals with acute heart problems or other conditions that may make someone susceptible to electrical shock may still be seriously injured. Remember, electricity, when handled carelessly or misused, is dangerous.

DIAGNOSIS AND TESTING

TESTING GFCIS

To test a GFCI, push the test button. The GFCI should trip (current will not flow). Then push the reset button. This should activate the GFCI, which will activate that particular circuit. If either of these tests fail, replace the GFCI.

CAUTION: A GFCI breaker provides protection only to the circuit to which it is connected. It does NOT provide protection to other circuits.

MEASURING FOR CURRENT LEAKAGE

IMPORTANT: Always perform a current leakage test when experiencing GFCI tripping.

IMPORTANT: Be sure that any additional ground or bond wires are disconnected. If there are two paths to ground the current leakage. Will be split, which will give inaccurate readings.

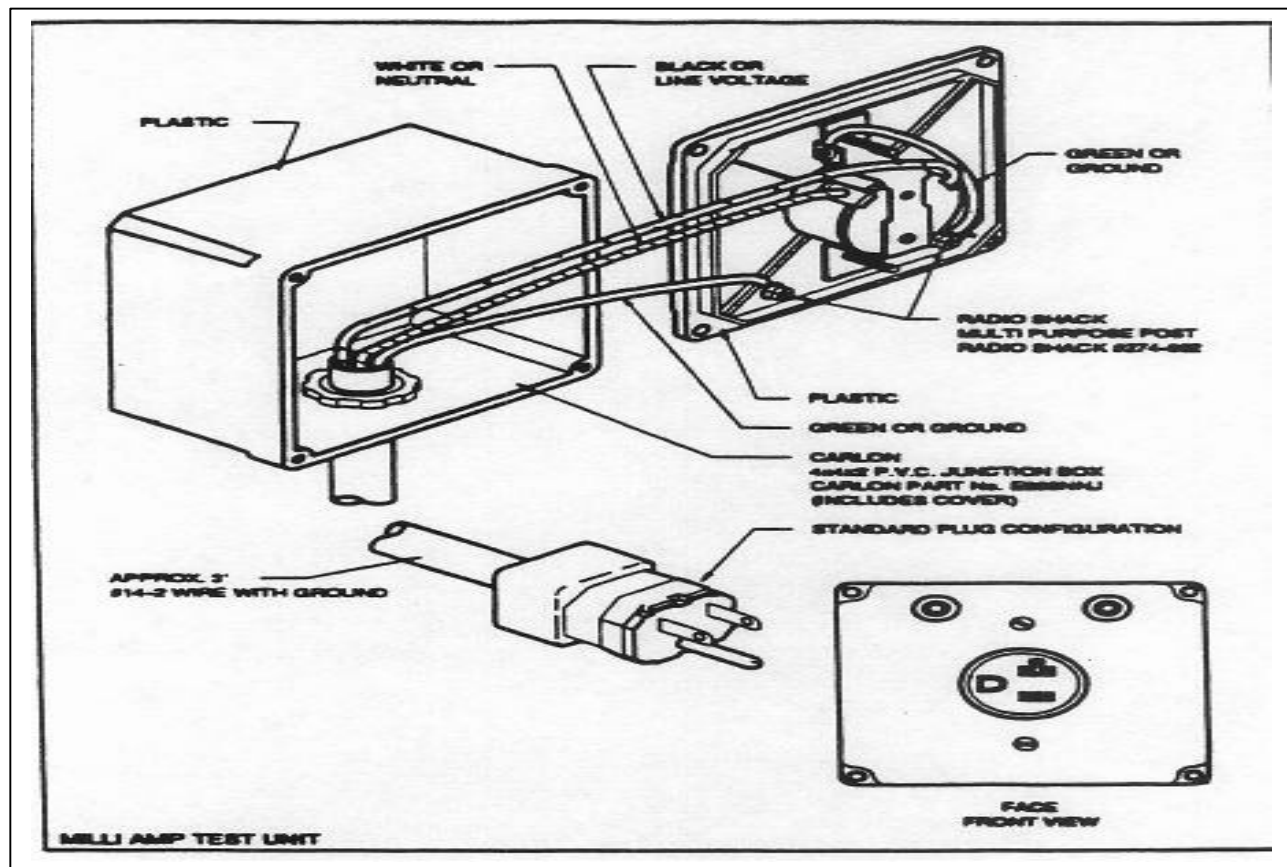
Note: GFCIs only test the load side of the circuit they are attached to.

These instructions are for use a meter that is capable of reading AC milliamps.

Refer to the following instructions to test for current leakage:

1. Set the volt/ohm meter to the AC milliamp function (20m on the meter).
2. Plug the black test lead into the "COM" receptacle on the meter. Plug the red test lead into the appropriate mA receptacle on the meter.
3. To test for AC milliamp leakage, break the ground connection of the circuit being tested and use the test leads with alligator clips to complete the circuit. On a 230-volt spa, testing is done at the subpanel grounding bar. On a 115-volt model spa, testing is done in the control box on the main grounding lug.

Note: To simplify testing current leakage on 115-volt spa models, build the test box illustrated below.



Determine current leakage from each of the spa's components (heater, jet pump, lamp and circulating pump) by isolating the individual components. Do this by operating each individual component alone (nothing else in the spa in operation). Check the meter for milliamp reading.

Any component registering more than .5 milliamps (mA) of current leakage is considered defective and should be replaced.

GFCI SPECIFICATIONS

Breakers

- Trips at 5 mA (+/- 1 mA) and within 1/4 of the sine wave or 8 milliseconds.
- Determines maximum current draw of a circuit, for example 15, 20 or 30 amps.

All Others

- All Class A GFCIs trip at 5 mA (+/- 1 mA) and within 1/4 of the sine wave or 8 milliseconds.

There is an industrial-grade GFCI available that is intended to protect equipment only. The trip point is 30 mA, which is beyond the safe level to protect people.

GFCI INSTALLATION

Receptacles

A receptacle is wired in series with the circuit it is protecting. The GFCI is generally marked "line or source" for the incoming wire and "load" for the wires going to the circuit. Both black and white wires, along with the ground wire need to be connected.

Note: Proper GFCI installation is imperative for the spa owner's safety and spa's protection.

Breakers

All wires of the circuit (hot and neutral wires for a 115-volt circuit; both hot wires for a 230-volt circuit) are to be attached to the breaker. The breaker is generally marked "load power" on the black wire(s) (power) and "load neutral" on the white wire (neutral).

Note: Failure to connect the neutral wire to the GFCI breaker will cause the breaker to trip immediately when power is applied.

230 Breakers - Additionally, it is important to connect the "pigtail" wire (white) on the breaker to the neutral buss bar in the electrical subpanel. This wire acts as a path to ground for the neutral wire in a 230-volt circuit and as a path for the trip circuit on the breaker.

Note: On 230-volt breakers if the pigtail is not installed to the neutral buss bar, the breaker will not manually test, but it will protect the circuit.

GFCI OPERATION

A GFCI measures the differential of current in an electrical circuit. When the differential of current across the load is greater than the trip value of the GFCI (5 mA, +/- 1 mA), the GFCI will trip. When tripped, the GFCI internally opens a set of switches, which will not allow current to flow through the incoming wires of the circuit. This is particularly important in a 115-volt circuit where reversed polarity may exist due to the circuit being improperly wired.

It is also important to understand that a GFCI will only test the load side of the attached circuit. It will not test the source side connected to it.

Note: A spa being protected by a GFCI must have a path to ground for the GFCI to operate. Complaints of spa owners receiving a low current shock is usually the sign of improperly grounded piece of equipment.

COMMON REASONS FOR GFCI TRIPPING

- Excessive current leakage from an electrical part.
- Moisture on electrical component(s).
- Incorrect wire installation to the GFCI.

OTHER MINOR REASONS FOR GFCI TRIPPING

- Low voltage. Some designs are more immune to this than others since the internal tripping circuit is not mechanically held.
- Radio frequency (RF).
 - A. Radiated RF (traveling through air)
 - B. Conducted RF (traveling through wires), such as that generated by chattering relays and contactors.
 - Excessive voltage spikes.
 - Scraping off wire insulation when installing wire into conduit.
 - Lightning strikes and high-energy utility switch noises.

G.F.C.I. TROUBLESHOOTING

IMPORTANT! Keep in mind that a majority of G.F.C.I. tripping problems can be attributed to incorrect wiring. G.F.C.I. troubleshooting usually finds the problem.

If Correct Wiring Is Verified

- Check to see if the proper G.F.C.I. is installed.
- Check the label in the system box near TB1 to determine the maximum amperage draw for the system.
- Be sure the G.F.C.I. is rated for more amperage than the system will draw.
- For a 230 V dedicated system, a 2-pole G.F.C.I. with a neutral.
- For a 120/230 V system, the G.F.C.I. must include a load neutral out.
- If the load neutral wire is routed from the sub-panel neutral bar directly to TB1 in the system box, then the G.F.C.I. will trip when a 120 V device is activated.
- If the wiring is correct and the G.F.C.I. will not reset, then unplug the pump and try to reset the G.F.C.I.
- If the G.F.C.I. trips again, then unplug the blower and push the reset button. If the G.F.C.I. continues to trip, then do the same procedure for the ozone generator.
- If the G.F.C.I. stops tripping after you unplugged one of the spa's components, turn off the power to the spa then plug in each component except the one that tripped the G.F.C.I.
- Power up the system. If the G.F.C.I. no longer trips, then you have correctly identified the problem.
- Repair or replace the component as instructed by the spa manufacturer.
- If you have unplugged all of the spa's components and the G.F.C.I. still doesn't reset then the problem is most likely a ground fault in the heater.

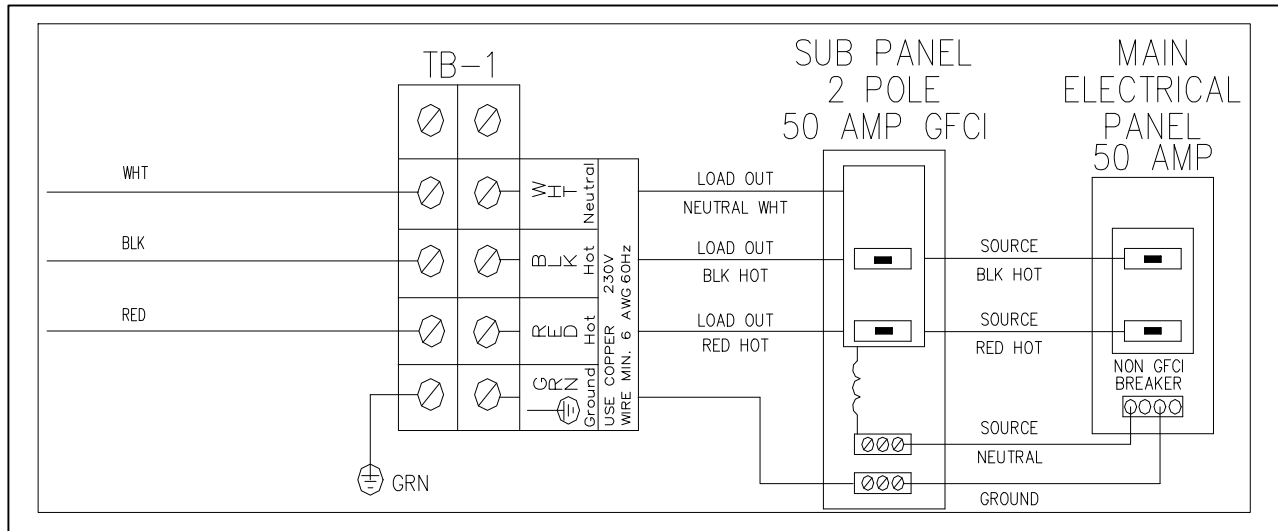
To Disconnect the Heater

- First, turn off the main circuit breaker, then remove both heater straps or wires from the terminal block, not the heater itself.
- After restoring the power, try to reset the G.F.C.I., again. If it no longer trips after the system calls for heat, then replace the heater.
- If the G.F.C.I. still trips, look for pinched or shorted wires at the transformer. Make sure that the screws that attach the transformer to the system box have not pinched or damaged the insulation of the transformer wires.
- If the transformer wires are undamaged, check for any other pinched wires. Refer to the wiring diagram to verify the correct wiring of the control system.
- If everything looks to be in perfect working order, then the G.F.C.I. may be defective.

IMPORTANT: More than one GFCI in series will not cause the GFCI to trip. A GFCI measures the differential of current across each side of an electrical circuit and in no way could contribute to the tripping of another GFCI.

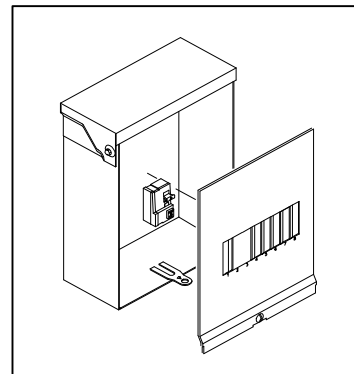
For example, if there is a GFCI breaker in the service panel and a GFCI cord connection on the spa it is possible that the GFCI breaker will trip first should a component develop current leakage. Keep in mind GFCI specifications - five (5) milliamps (+/- one (1) milliamp). Should this happen it is probably because the GFCI breaker is calibrated lower than the cord GFCI and not due to the proximity of the two GFCIs.

SUB PANEL WIRING SCHEMATIC



WIRING CHECK FOR G.F.C.I./SERVICE DISCONNECT

The National Electrical Code states that a service disconnect breaker box (a G.F.C.I. can be used for this purpose) must be located at least five feet away from the spa and should be conveniently located near the equipment bay. If it is not in plain sight, keep the disconnect padlocked when in the off position.



Precautions - Effective January 1994, G.F.C.I.'s are required for spa installations. Prior to that, G.F.C.I.'s had been recommended for spa installations, but were not mandatory. If the spa you are servicing was not installed with a G.F.C.I., strongly urge your customer to improve safety and comply with current standards by installing one.

IMPORTANT! Remember, high voltage is still accessible in the house breaker box even though you have turned off the spa breaker.

PART II: VOLTAGE TESTING

G.F.C.I. LINE-IN WIRING CHECK

Locate the proper circuit breaker and turn it off.

Remove the cover from the house breaker box.

Check the main service amperage rating to the breaker box.

Note: Typically, a house circuit will require at least a 100 Amp service when a spa is installed.

From the circuit breaker, locate the red load wire and the black load wire.

From the neutral bar the green ground wire.

Be sure there are no other appliances on the spa circuit.

If there are, service must be re-wired to supply the spa only.

Make sure all four wires exit the house breaker box via conduit, routed to the G.F.C.I. breaker box.

The black should be connected to the G.F.C.I. line-in 1. The red should be connected to the line-in 2.

The white load neutral connects to the neutral bar. The green ground wire should be attached securely to the ground lug inside the G.F.C.I. box.

G.F.C.I. LINE-OUT WIRING CHECK

If the spa is equipped with a 115/230 V control system, an additional white load neutral wire must connect to the load neutral out. This wire runs the others to the system box. Proper placement of this neutral wire is essential. If miswired, the G.F.C.I. will trip when a 115 V device is activated.

Once you have found all wiring correctly installed, begin to check for proper voltage.

VOLTAGE CHECKS - BREAKER BOX/G.F.C.I.

When checking for proper voltage, please keep in mind that the acceptable voltage range is $\pm 10\%$ of the recommended voltage. Acceptable voltage when 115 V is specified as the desired voltage, is between 108 and 132 V. Acceptable voltage when 230 V is specified as the desired voltage, is between 216 and 264 V.

IMPORTANT: This service must be single phase. Any abnormal voltage reading requires an electrician. Do not attempt to fix problems yourself. High voltage can seriously injure or kill.

BREAKER BOX VOLTAGE CHECK

Set your multi-meter or voltmeter for AC Volts.

Make sure the G.F.C.I. is off.

Carefully turn on the spa circuit breaker at the sub-panel.

At the house breaker box, probe the spa circuit breaker between the black and red wires. Your meter should read 230 V.

Probe the black and green ground wires. You should see 115 V.

Probe between the red and green ground wires. You should also see 115 V.

The voltage between the white load neutral and the green ground wire. You should see 0 volts.

G.F.C.I. LINE-IN VOLTAGE CHECK

230 V Dedicated System:

- Be sure the spa circuit breaker (located in the house breaker box) is on.
- Make sure the G.F.C.I. is off.
- Probe the black and red wires. The meter should read 230 V.
- Probe the black and green ground wires. The meter should read 115 V.
- Probe the red and green ground wires. This should also read 115 V.
- The voltage between the white load neutral and the G.F.C.I. box ground lug should read 0 V.
- Turn on the G.F.C.I. breaker before continuing to the system box.

115/230 V Convertible System:

- Be sure the spa circuit breaker (located in the house breaker box) is on.
- Make sure the G.F.C.I. is off.
- Probe the black and red wires. The voltage should be 230 V.
- Probe the black and white load neutral wires. The voltage should read 115 V.
- Probe the red and the white load neutral wires. The voltage should read 115 V.
- Turn on the G.F.C.I. breaker before continuing to the system box.

VOLTAGE CHECKS - SYSTEM BOX

G.F.C.I. LOAD OUT VOLTAGE CHECK

230 V Dedicated System:

- Be sure that the house breaker is on.
- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires at G.F.C.I. load out 1 and 2. The voltage should be 230 V.
- Probe the black wire and the G.F.C.I. neutral bar. The meter should read 115 V.
- Probe the red wire and G.F.C.I. neutral bar for 115 V.
- Probe the black load out wire and the box chassis ground lug. Your meter should read 115 V.
- Probe the red load out wire and the ground lug. The voltage should read 115 V.
- Recheck voltage under peak load conditions.*

115/230 V Convertible System Check:

- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires at G.F.C.I. load out 1 and 2. Look for 230 V.
- Probe the black and white load neutral wires. The meter should read 230 V.
- Probe the red load out and box chassis ground lug for 115 V.
- Probe the black load out wire and ground lug for 115 V.
- Recheck voltage under peak load conditions.*

*Peak Load Check - It is important to check the voltage again under peak load conditions. To reach peak load, turn on the blower, heater, lights and all pumps.

SYSTEM BOX CHECK (AT TB 1)

230 V Dedicated System Check:

- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires. The voltage should be 230 V.
- Probe the black wire and green ground wire. The meter should read 115 V.
- Probe the red wire and green ground wire. The voltage should read 115 V.
- Probe the white neutral wire and the green ground wire. This should be 0V.
- Recheck voltage under peak load conditions.*

115/230 V Convertible System Check:

- Be sure the G.F.C.I. breaker is on.
- Begin the voltage check at TB1.
- Probe the black and red wires. Your meter should read 230 V.
- Probe the black and white wires. Look for 115 V.
- Probe the black and green ground wires also 115 V.
- Probe the red and green ground wires again 115 V.
- Probe the red and white load neutral wires. This should read 115 V.
- Recheck voltage under peak load conditions.*

*Peak Load Check - It is important to check the voltage again under peak load conditions. To reach peak load, turn on the blower, heater, lights and all pumps.

Peak Load Check for 230 V System:

- Check the voltage between the black and red wires. The acceptable voltage range is between 216 and 264 V. Probe the black and white neutral wires. This voltage must be between 108 and 132 V.

Peak Load Check for 115/230 V Convertible System:

- Check the voltage between the black and red wires. Acceptable range is between 216 and 264 V.
- Probe the black and the white load neutral wires. The voltage must be between 108 and 132 V.
- Probe the red and the white load neutral wires. The voltage must be between 108 and 132 V.

IMPORTANT - If the voltage is not within the acceptable range, call an electrician or the local electric company to diagnose the problem.



Section 2: Spa Control System

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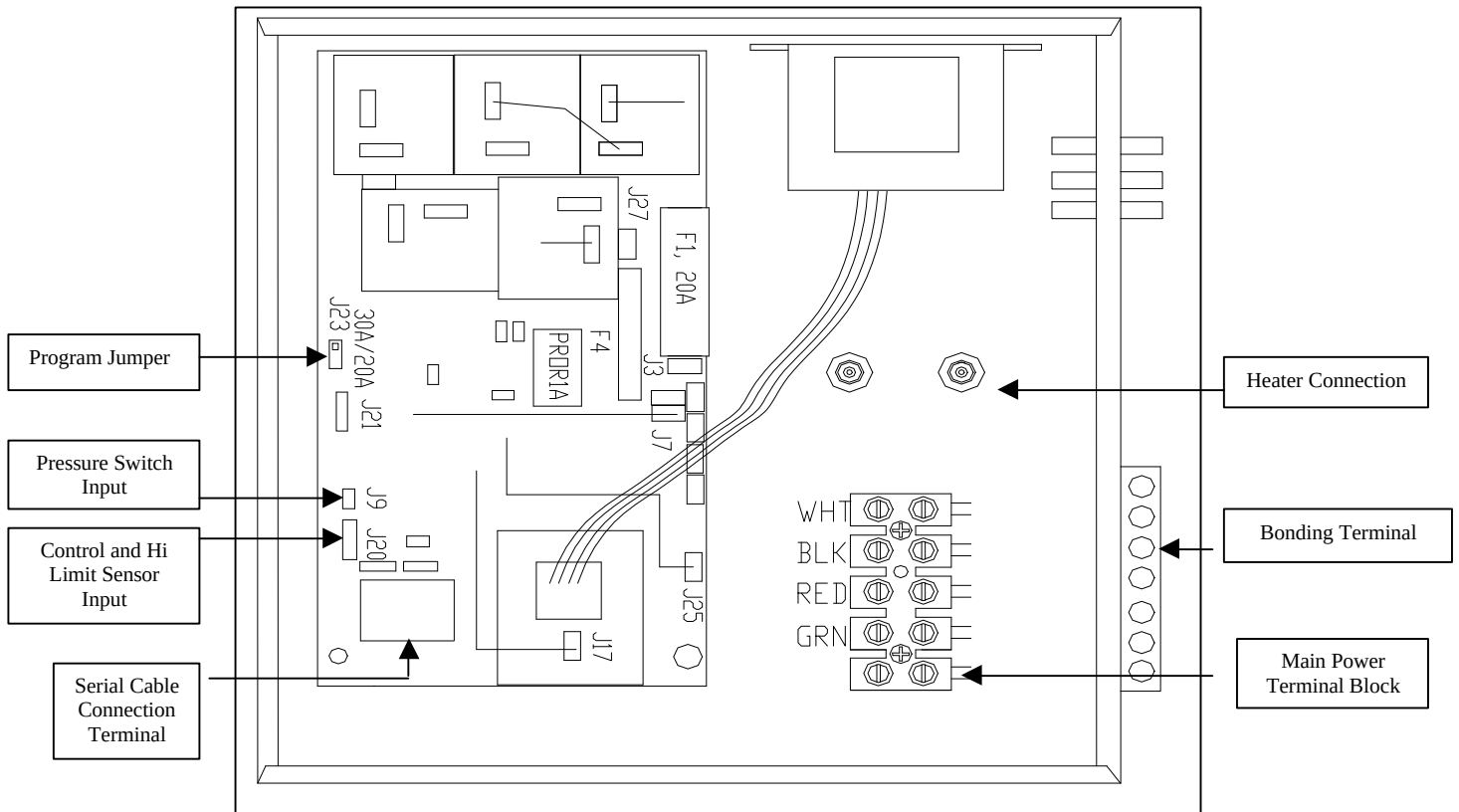
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Section 2: Spa Control System

CONTROL SYSTEM

The PROP1A control panel, used in the OASIS spa, allows the spa user to regulate and view the status of the spa functions. These functions are regulated by the topside control panel and include the temperature control, light, hydro and air jets. The topside control panel also displays the set temperature. The PROP1A topside control panel consists of two items: the digital display and control touch pads. It is connected to the control system, via a serial cable.

There is one control box, PROP1A, used for both series:



The control box is designed to control the spa functions (jets, light, set temperature) and monitor the status of the control system. Since the same box can be used in different models with different functions, the Program Jumpers must be set to “program” the control box for the spa model it is installed in. Additionally, the control box is installed for a 115 VAC model.

- **Serial Cable Connection Terminal (Topside Control Panel)** – seats the control panel’s serial cable.
- **Program Jumpers** – control the spa functions and settings (see *Jumper Configuration*).
- **Input (Control Sensor)** – receives the control sensor connection wire.
- **Input (High Limit Sensor)** – receives the high limit sensor connection wire.
- **Main Power Terminal Block** – receives the main power line and passes source voltage on to the circuit board and heater relays.
- **Power Connection** – supply power to the spa light, jet pump, and ozonator.
- **Heater Connector** – attaches to the via the hold straps.
- **Bonding Terminal** – connects all of the solid copper bonding wires to the main ground line.

All metal permanently placed near the spa must be bonded to the grounding lug as described in the Pre-Delivery instructions and the Homeowners Manual. Please refer to these documents for more complete information.

Power Connections For The Equipment Compartment Components

The jet pump and blower are wired via flag terminals to the appropriate relays.

Terminal Block (TB-1)

The terminal block (TB-1) receives the source voltage from the 115 VAC dedicated source and delivers voltage to the various circuits and electronic components within the control box.

Grounding (bonding) Lug

The minimum size bonding wire is #8 AWG (solid copper). The ozonator (if installed) must be bonded to the control box grounding lug.

Heater Ground

The heater is grounded to the box via the holding straps.

Program Jumpers

The setting of the program jumpers determines the operation of the spa functions. When replacing a control box, the program jumpers must be set for the specific model, according to the label on the control box cover or the instructions in this manual.

Serial Cable Input

The serial cable input is the interface between the topside control panel and the control box. The spa will be non-operational if the serial cable is broken or disconnected from the terminal.

Control Sensor Input

The control sensor is electronically connected to the control box circuitry at the J20 input terminal.

Control Sensor

The control sensor is an electronic component that senses the temperature of the spa water by changing its electrical resistance value (ohms, Ω). The control box compares the temperature reading (in ohms) of the control sensor to the setting of the thermostat, and heat is turned on or off.

Hi-Limit Sensor Input

The hi-limit sensor is electronically connected to the control box circuitry at the J20 input terminal.

Hi-Limit Sensor

The hi-limit sensor is an electronic component that senses the temperature of the spa water by changing its electrical resistance value (ohms, Ω). Within the safety features of the spa, the temperature reading (in ohms) of the hi-limit sensor is compared to the permanent value of the hi-limit (the hottest temperature which the spa can reach without shutting off).

If the spa water temperature exceeds the hi-limit value of 118° F, or the high limit sensor is faulty or unplugged, the high limit will trip and the spa functions will not work. “OH” is displayed as an indication that a high limit problem exists.

Jet Relay 1 / Jet Relay 2

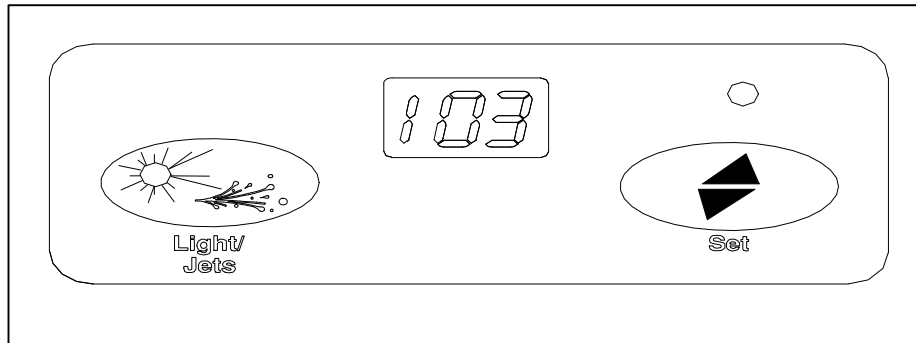
There are two (2) relays available on the board for low and high jet pump operation.

Lamp Assembly

The lamp assembly consists of the wire harness, reflector, bulb light, and lens which is plugged into the control box via a four-pin connector that is wired to the main circuit board via the J5 and J20 terminal.

Topside Control Panel

The topside control panel is a complete assembly that contains the control head (electronics portion) and control panel face (touch pads and display window) as one unit. The panel is used in conjunction with an overlay.



Diagnostic LEDs

There are three (3) diagnostic LEDs, *OH*, *FL*, and *Sn* are used to determine the current status of the spa's control system and are displayed in the display window.

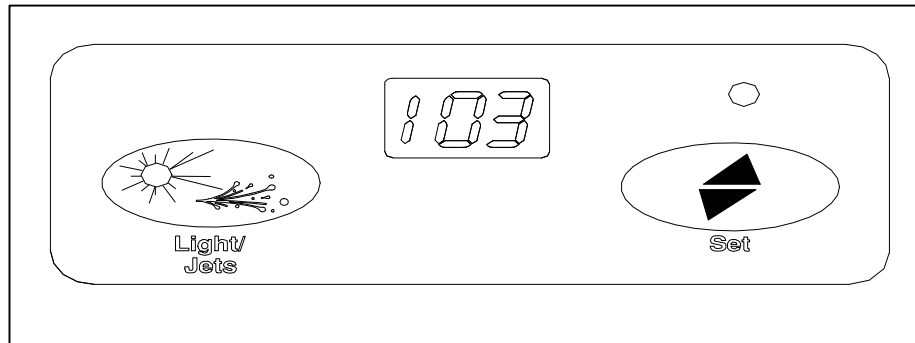
Section 2: Spa Control System

TOPSIDE CONTROL PANEL

The topside control panel is a complete assembly that contains the control head (electronics portion) and control panel face (touch pads, display, and indicator) as one unit. The panels are used in conjunction with overlays. The control panel is attached to the spa with four screws.

Overlay

Topside Control



Light and Pump Touch Pad – activates and deactivates the spa light and the pumps.

Temperature Control Touch Pad – raises and lower the thermostat setting (the maximum temperature setting is 104° F / 40° C; the lowest temperature setting is 80° F / 27° C).

Main Display - shows the set temperature value and LED messages.

Serial cable

The serial cable is the connection between the topside control panel and the control box. The cable is removable and replaceable.



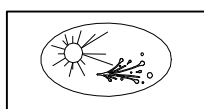
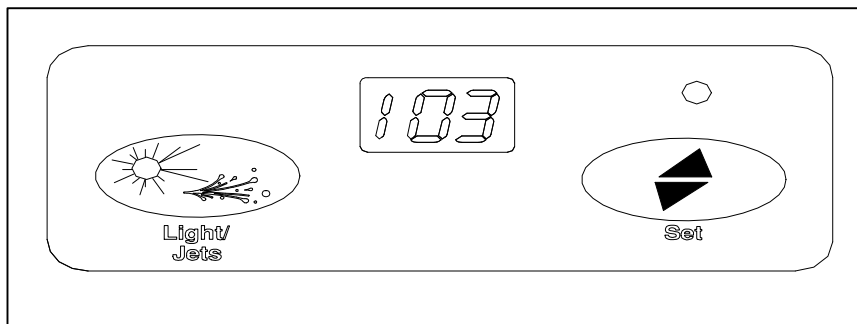
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Section 3: Operation

Topside Control Panel

Topside Control Panel Display and Spa Functions

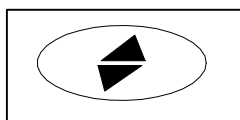
The topside control panel uses a display to show the temperature and other LED messages. When there is a demand for heat, the indicator will light up to show the call for heat.



LIGHT/JETS

Pressing the LIGHT/JET touch pad sequences the light and pump through the controls as follows:

- | | | |
|---|-----------------|-----|
| 1 | Light | On |
| 2 | Low Speed Pump | On |
| 3 | High Speed Pump | On |
| 4 | Light | Off |
| 5 | Pump | Off |



TEMPERATURE CONTROL

To adjust the temperature, press the SET touch pad again while the temperature is displayed. This will increase or decrease the temperature, depending on which direction was chosen last. Each successive press will change the set temperature by one degree in the same direction.

If the temperature is to be adjusted in the opposite direction, wait four (4) seconds to allow the display to revert to the actual water temperature. Press the SET touch pad again until to adjust in the desired directions.

MODE

Standard Mode – the heater and pump's low speed turn on as needed to reach and maintain the set temperature.

Economy Mode – the spa heats only during the filtration cycles.

To switch from the Standard mode to the Economy mode, press the SET touch pad then, while the temperature is displayed, press the LIGHT/JET touch pad. EC will flash alternately with the temperature signifying the Economy mode.



Section 4: Troubleshooting

IMPORTANT TROUBLESHOOTING TIP

Before beginning a troubleshooting procedure, if the control system has never operated properly (either from the factory or after a repair), check the setting of the JUMPERS. If the JUMPERS are set improperly, the spa will malfunction.

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Section 4: Trouble Shooting

MAIN PANEL MESSAGES

Panel messages are a quick clue toward solving a variety of problems. Here are the most common messages and what they mean.

OH

The Panel Display is Flashing OH or the Status/Heat LED is Flashing (1 second on, 1 second off) - This means that the control system is at a high-limit condition and the spa is deactivated.

Note: Overheating may occur if the low-speed pump is set to operate for extended periods of time, or if the incorrect pump is installed.

The following is a list of most probable causes of this message. Inspect these first:

- Check ball valves. Make sure that they are open.
- Make sure the correct pump and pressure switch are installed.
- Clean the filter/skimmer if there is any blockage.
- In extremely hot weather, check for proper cabinet ventilation.
- Make sure the temperature sensor is fully inserted into the sensor mount and that the sensor mount is properly insulated with foam.

Note: Improper insulation is one of the most common causes of spa overheating. Aerosol foam insulation is available at most hardware stores.

- Check for excessive filter duration.
- Check the water level.
- Check the water temperature with an accurate temperature thermometer:
- Remove the spa cover and allow the water to cool to below 110 F.
- Adding cool water may be necessary.
- Touch any button to reset the system.
- If the Problem Reoccurs, Test the Sensor Set:

Check sensor wires for cracks or damage that may indicate the presence of a rodent.

Inspect the connections of both sensors on the circuit board. Verify that the plugs are clean by unplugging the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test. If the problem is solved, replace the sensor set. If the problem is not solved, do not replace the sensor set. Plug in the original sensor set to verify that there is not a connection problem. If the problem continues after following the above steps, then replace the circuit board.



The Panel Display Shows Sn1

When this is shown it means that the high-limit or water temperature sensor has malfunctioned.

For the High-limit sensor:

- Check sensor wires for cracks or damage that may indicate the presence of a rodent.
- Inspect the connections of both sensors on the circuit board. Verify that the plugs are clean by unplugging the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test.
- If the problem is solved, replace the sensor set.
- If the problem is not solved, do not replace the sensor set.
- Plug in the original sensor set to verify that there is not a connection problem.
- If the problem continues after following the above steps, then replace the circuit board.

For the water temperature sensor:

- Unplug the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test.
- If the problem is solved, replace the sensor set. If the problem is not solved, do not replace the sensor set.
- Plug in the original sensor set to verify that there is not a connection problem.
- If the problem continues after following the above steps, then replace the circuit board



The Display Continually Shows FL

When this is happening, the software is detecting no pressure at the pressure switch with the pump. This may indicate a flow restriction, such as a dirty filter. Another cause of the flashing FL message could be a faulty or malfunctioned pressure switch

- Check the water level.
- Remove the filter cartridge, then run the spa. If the message disappears, then clean or replace the filter cartridge.
- Make sure the ball valves are open. If any valves are closed, opening them may remove the error message from the display and the spa should operate normally.
- Make sure the pump motor is functioning properly. If not, use a multi-meter to verify system box output to the pump.
- Check the floor/suction grills for obstructions.
- Check the pump to be sure it is primed and pumping properly.
- Check the pressure switch wire for cuts.
- Inspect the connection on the circuit board. Verify that the plug is clean by unplugging it (at jumper, J-9).

- If the spa does not operate normally, disconnect the wires from the pressure switch.
Disconnect the pressure switch from the circuit board at J-9
Set your voltmeter to the continuity setting.
Place the meter leads onto the two wires of the pressure switch.
Turn on the low-speed pump.
The meter should register continuity.
Turn the pump on from low-speed to high-speed if available and run for ten seconds.
After ten seconds has passed, turn off all pumps.
After four seconds, check the voltmeter and verify that there is no continuity registering.

Note: The most common pressure switch failure is poor water chemistry. Instruct the homeowner in proper water maintenance.

- If the spa operates normally, replace the switch.
- Clean the area where the pressure switch is located with a metal brush. Use silicone tape or sealant when installing a new pressure switch.
- If the problem continues, then replace the control.

MISCELLANEOUS MESSAGES

The Panel Display shows Unexplained Error Messages

Ec on - In Economy mode => heater will only operate during filtration cycles. Heater will operate if water temp is 20°F or more below set temp

St An - In Standard mode => heater will operate anytime the thermostat demands heat => water temp is always within 1°F of set temp

Section 4: Trouble Shooting

DIAGNOSIS

The center column (Diagnosis) is arranged so that the most likely cause of the malfunction is listed first, so the corrective actions should be performed in order from top to bottom until the problem is solved.

PROBLEM	DIAGNOSIS	CORRECTIVE ACTION
ENTIRE SPA INOPERATIVE The control panel does not activate any of the spa functions or components.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control box may not be receiving the proper source voltage. Many problems are associated with electronic control systems are caused by low voltage.	Check the GFCI. If the GFCI has tripped, perform the Leakage Current Test. If the GFCI has not tripped, Perform the Source Voltage Test.
	The control panel may be defective.	Perform the Control Panel Test.
	The spa light, heater, and/or jet pump may not be receiving the proper output voltage.	Perform the Output Voltage Test for the spa light, heater, and jet pump.
SPA LIGHT INOPERATIVE The spa light cannot be operated with the control panel.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The light bulb may be burned out.	Check the light bulb, and replace it if necessary.
	The control panel may be defective.	Perform the Control Panel Test.
	If the spa light will not turn ON, the spa light may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Spa light.
	If the spa light cannot be turned OFF from the control panel, and the control panel is not defective then the control box is defective.	

JET PUMP INOPERATIVE The jet pump cannot be operated with the control panel.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control panel may be defective.	Perform the Control Panel Test.
	If the jet pump will not turn ON (high-speed or low-speed), the jet pump may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Jet Pump.
	If the jet pump cannot be turned OFF from the control panel, and the control panel is not defective then the control box is defective.	
	If there is output voltage from the control box and the jet pump cannot be turned on, it is defective.	
HEATER INOPERATIVE The heater cannot be operated with the control panel.	If there is a message displayed in the control panel.	Refer to the Panel Message section.
	The set point may not be more than 2° F above the current water temperature displayed on the control panel.	Use the Temp button on the control panel to raise the set point more than 2° above the current water temperature displayed in the control panel.
	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control panel may be defective.	Perform the Control Panel Test.
	The jet pump may not be operative.	Perform the Output Voltage Test, Jet Pump.
	The jet pump may not be producing adequate flow.	Perform the Flow Test (restricted flow is usually the result of dirty or clogged filters).

HEATER INOPERATIVE – cont.	The pressure switch may be defective.	Perform the Pressure Switch Test.
	The sensors may be defective.	Perform the Sensor Test, Control, and the Sensor Test, High Limit.
	The heater may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Heater.
	If there is output voltage from the control box and the heater cannot be turned on, it is defective.	
	If the heater cannot be turned OFF, even though the set point is more than 2° F below the current water temperature displayed on the control panel and the control sensor is not defective then the control box is defective.	
	If there is output voltage from the control box and the heater cannot be turned on, it is defective.	
GFCI TRIPPING	The electrical connections in the control box may be bad.	Check the electrical connections.
	There may be a leak in the equipment compartment and/or excessive moisture on any of the electrical components	Check for a leak in the equipment compartment and/or excessive moisture on any of the electrical components, and repair the leak and/or dry the equipment compartment and electrical components.
	There may be a ground fault.	Perform the Leakage Current Test.

HIGH LIMIT TRIPPING

Immediately after the high-limit trips, it must be determined whether it was just the heater that overheated, the entire spa, or neither.

If the spa water temperature is equal to or below the temperature set point only the heater overheated.	Flow is restricted in conjunction with a defective control box and possibly a defective pressure switch.	Perform the Flow Test (restricted flow is usually the result of dirty or clogged filters). Then perform the Pressure Switch Test.
If the water temperature is above the maximum temperature set point (104° F) the entire spa overheated.	The control sensor may be defective.	Perform the Sensor Test, Control.
	The jet pump and/or heater may not be able to be switched OFF, due to a defective control box.	Verify that the heater and jet pump function normally (refer to the Heater Inoperative and Jet Pump Inoperative sections).
If neither the heater, nor the entire spa overheated.	The high limit sensor may be defective.	Perform the Sensor Test, High Limit.

JET PUMP LEAKING

There are two places where the source of the leak may be located, the wet end and the plumbing union.

Wet End	Damaged volute and/or loose or damaged volute cover.	Tighten volute cover and/or replace volute and/or volute cover.*
	Damaged shaft seal.	Replace the shaft seal.*
	Loose or damaged freeze drain plug and/or o'ring.	Tighten or replace the freeze drain plug and/or o'ring.*
	Damaged volute o'ring.	Replace the volute o'ring.*
Plumbing union, Inlet or Outlet	Damaged volute and/or volute cover.	Replace volute and/or volute cover.*
	Loose or damaged compression fitting.	Tighten or replace the compression fitting.
	Damaged compression fitting o'ring.	Replace the o'ring.

*Replace the entire pump if servicing its components is prohibited by the warranty, or if replacement components are not available.

HEATER LEAKING

There are two places where the source of the leak may be located, the heater body and the plumbing union.

Heater Body	Damaged heater body.	Replace the heater.
	Loose or damage pressure switch.	Remove the pressure switch and reinstall or replace.
	Loose or damaged compression fitting.	Tighten or replace the compression fitting.
	Damaged compression fitting o'ring.	Replace the o'ring.
	Bulk head fitting is leaking.	Replace the bulk head fitting.

FLOW RESTRICTED

The jet pump is weak or sporadic, and/or no air is mixing at the jets, and/or the pressure switch will not close when the jet pump is switched on.

	The filter may be dirty or clogged.	Perform the Flow Test.
	The air control valve may be closed.	Open the air control valve.
	There may be a leak in one of plumbing lines, or the jet pump or heater may be leaking.	Repair any leaks (refer to the Jet Pump Leaking or Heater Leaking sections of this manual).
	One of the air or plumbing lines may be obstructed.	Remove the obstruction from the air or plumbing line.

Section 4: Trouble Shooting

TEST

SOURCE VOLTAGE TEST

20 AMP 115 VAC SYSTEMS W/TWO POLE INLINE GFCI

1. Verify that there is $115 \pm 10\%$ VAC between the neutral wire and the hot wire in the in-line GFCI. If there is not 103.5 to 126.5 VAC between the neutral wire and the hot wire on the input side of the breaker, call an electrician. If the breaker is closed and there is not $115 \pm 10\%$ VAC between the neutral wire and the hot wire on the output side of the breaker, then the breaker is defective.
2. Check the GFCI test button, and reset the breakers. Perform the **Leakage Current Test** if the GFCI breaker test button is tripped.
Note: Every time power is disconnected from the spa (even when resetting the breakers), power should not be reconnected for at least 30 seconds.
3. Measure the voltage on the main power terminal block between the hot wire (black) and the neutral wire (white). There should be $115 \pm 10\%$ VAC present at the terminal block. If the terminal block is not receiving from 103.5 to 126.5 VAC between the neutral wire and the hot wire, then there is a problem with the wiring somewhere between the in-line GFCI and the terminal block. Inspect for damaged wiring or components and check the terminal block, main power cord, and in-line GFCI for loose wiring.

LEAKAGE CURRENT TEST

Note: Every time power is disconnected from the spa, power should not be reconnected for, at least, 30 seconds. If power is prematurely reconnected to the spa, then disconnect power, wait 30 seconds, and then reconnect power.

DANGER! RISK OF ELECTRICAL SHOCK – THE SPA’S GROUND WILL BE INTERRUPTED AND THE TEST METER WILL BE PLACED IN SERIES WITH THE SPA’S GROUND WHEN PERFORMING THIS TEST.

1. Verify that the program jumpers and electrical connections are properly configured, and that none of the fuses are blown.
2. Unplug the spa’s power cord (15 or 20 amp spas) or turn off the breakers at the sub-panel (50 amp spas).
3. Disconnect all of the No. 8 AWG solid copper wires attached to the bonding terminal, and disconnect any additional ground or bonding wires to eliminate all paths to ground but the main power source ground wire.

DANGER! RISK OF ELECTRICAL SHOCK – DISCONNECTING THE GROUND WIRES ELIMINATES ALTERNATE PATHS TO GROUND, USE EXTREME CAUTION WHEN TESTING.

4. Disconnect the heater’s power wires on the appropriate models as per follows:
Remove the heater wires from relays RL5 and RL4.
5. Unplug the jet pump from the right side of the control box.
6. If the spa is equipped with an ozonator, unplug it from the right side of the box.
7. Disconnect the bonding wires from the jet pump(s). Additionally, if the spa is equipped with an ozonator, disconnect its bonding wire. All connections should be removed at the bonding terminal block on the right side of the control box.
8. Disconnect all main power lines or cord wires from the main power terminal block (TB1).

9. Test for continuity between the main source power coming from the G.F.C.I. to the spa by testing between the black (hot) wire and the ground (green) wire in a 115V setup. In any test situation, if there is any continuity, there is a ground fault condition and either the source power line or cord are defective.

WARNING -The power line or cord must NOT be receiving any voltage when performing this test.

10. Test for continuity between the main source power coming from the G.F.C.I. to the spa by testing between the white (neutral) wire and the ground (green) wire in a 115V setup.

WARNING - If there is ANY continuity at all between the neutral and ground wire, there is a ground fault condition. Either the power cord or the G.F.C.I. is defective

WARNING - This SHOULD show continuity. If there is NO continuity, trace both the neutral (white) and ground (green) wires for a broken or loose wire.

WARNING - The power line or cord must NOT be receiving any voltage when performing this test.

Main Power Cord

11. Disconnect the main power cord hot, ground, and neutral wires from the main power terminal block.

12. Test for continuity between the main power cord hot and ground wires. If there is any continuity at all, there is a ground fault, and the power cord is defective.

WARNING - The power cord must NOT be receiving any voltage when performing this test.

13. Test for continuity between the main power cord neutral and ground wires. If there is any continuity at all, there is a ground fault, and the power cord is defective.

WARNING - The power cord must NOT be receiving any voltage when performing this test.

Control Box

14. Reconnect the main hot and neutral wires to the terminal block, and connect the test meter's leads to the ground wire and the terminal block, TB 1 at terminal 5.

15. Reconnect power to the spa. If the GFCI trips and there is no ground fault in the power cord, the control box is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage #1 (this value is the current leaking to ground from the control box). The control box is defective if current leakage #1 is equal to or greater than 0.5 mA.

16. Disconnect power to the spa.

Jet Pump (High and Low Speed)

17. Plug in the jet pump. If this is a two-pump spa, repeat the same procedure on second pump, treating it as a separate pump.

18. Set the meter to read AC milliamps (0.5 milliamps MAX). Place the meter in-line with the incoming ground wire for the spa and TB1, terminal 5.

19. Reconnect power to the spa and switch the jet pump to low speed. If the GFCI trips, the jet pump is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage #2 (this value is the current leaking from the jet pump when on low speed). The jet pump is defective if the value of current leakage #2 is equal to or greater than 0.5 mA.

20. Switch the jet pump to high speed. If the GFCI trips, the jet pump is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage #3 (this value is the current leaking from the jet pump when on high speed). The jet pump is defective if the value of current leakage #3 is equal to or greater than 0.5 mA.

21. Disconnect the power to the spa, disconnect the high and low speed jet pump hot and neutral wires, then disconnect the test meter from the ground wire and the flag terminal. Reconnect the pump upon the conclusion of the test.

Heater

22. Reconnect the heater load power connection to the proper heater terminals in the control box.
23. Reconnect power to the spa, raise the set point to the highest setting and verify that the set point is at least 2° F above the actual water temperature. If the GFCI trips, the heater is defective.
24. Disconnect the power to the spa, reconnect the spa light and jet pump wires. Then reconnect all of the No. 8 AWG solid copper wires that were previously attached to the bonding terminal, and any other ground wires that you disconnected.

Verify that the GFCI device is functioning properly

25. Add the current leakage values #1 (control box) and #2(jet pump), label this value current leakage B (this value is the current leaking to ground from the entire spa when the heater is on). If the value of current leakage B is 4 to 6 mA and the GFCI does not trip, then the GFCI device is defective.
26. Add the current leakage values #2 (jet pump) and #3 (this value is the current leaking from the jet pump when on high speed), label this value current leakage C (this value is the current leaking to ground from the entire spa when the jet pump is on high speed). If the value of current leakage C is 4 - 6 mA and the GFCI does not trip, then the GFCI device is defective.

OUTPUT VOLTAGE TEST

Spa Light

1. Disconnect power to the spa for 30 seconds to restore the default setting – spa light OFF.
2. Reconnect power to the spa and check the main power terminal block for $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC) to see if the spa is receiving the proper source voltage. If the terminal block is not receiving the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Press the **Light** button and measure the voltage in the control box between terminals: J25 and J30 (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). The spa light flag terminals should supply $12 \pm 10\%$ VAC (10.8 VAC to 13.2 VAC) output voltage.
 - If the flag terminals did not supply $12 \pm 10\%$ VAC when the voltage was measured, perform the Control Panel Test to determine whether or not the control panel is defective. If the control panel is not defective, then the control box is defective.
4. Measure the voltage in the light socket of the light harness, if the flag terminals did supply $12 \pm 10\%$ VAC (10.8 VAC to 13.2 VAC). The light socket should also supply $12 \pm 10\%$ VAC output voltage.
 - If the light socket did not supply $12 \pm 10\%$ VAC when the voltage was measured, the light harness is defective.

Jet Pump

1. Disconnect power to the spa for 30 seconds to restore the default setting – jet pump on STANDBY.
2. Reconnect power to the spa and check the main power terminal block for $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC) or $230 \pm 10\%$ VAC (207 VAC to 253 VAC), depending on the model, to see if the spa is receiving the proper source voltage. If the terminal block is not receive the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Use the **Temp**  button to lower the set point to the minimum setting 80° F (to keep the thermostat from switching the jet pump on unexpectedly).
4. Measure the voltage in the control box between the appropriate terminals, as illustrated in the *Electrical Schematics* section of this manual. The jet pump terminals should not supply any output voltage. If any voltage is supplied the control box is defective (provided that the thermostat did not switch the jet pump to low speed).
5. Press the **Jets** button (to switch the jet pump to low speed) and measure the voltage in the control box between the appropriate terminals as illustrated in the *Electrical Schematics* section of this manual or on the control box cover door. The jet pump terminals should supply $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC) or $230 \pm 10\%$ VAC (207 VAC to 253 VAC), depending on the model, output voltage (continue this test even if the proper output voltage is not supplied).
6. Press the **Jets** button (to switch the jet pump to high speed) and measure the voltage in the control box between the appropriate terminals as illustrated in the *Electrical Schematics* section of this manual. The jet pump terminals should supply $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC) or $230 \pm 10\%$ VAC (207 VAC to 253 VAC), depending on the model voltage.
7. For models with dual pumps, test the second pump in the similar manner as above.

Note: *If the terminals did not supply $115 \pm 10\%$ VAC or $230 \pm 10\%$ VAC when the jet pump was switched to high and low speed, perform the Control Panel Test to determine whether or not the control panel is defective. If the control panel is not defective, then the control box is defective.*

Heater

1. Use the **Temp**  button to lower the set point to the minimum setting 80° F, and verify that the set point is at least 2° F below the actual water temperature displayed on the control panel.
2. Check the main power terminal block for 115±10% VAC (103.5 VAC to 126.5 VAC) 230±10% VAC (207 VAC to 253 VAC), depending on the model voltage, to see if the spa is receiving the proper source voltage. If the terminal block is not receive the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Measure the voltage in the control box between the two heater connection (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). The heater connection should not supply any output voltage. If any voltage is supplied the control board is defective (provided that the thermostat did not switch the heater on).
4. Use the **Temp**  button to raise the set point to the maximum setting (104° F), and verify that the set point is at least 2° F above the actual water temperature displayed in the control panel.
5. Measure the voltage in the control box between the two heater connection bars P20 and P21 (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). The heater connection should supply 115±10% VAC (103.5 VAC to 126.5 VAC) or 230±10% VAC (207 VAC to 253 VAC) output voltage depending on the model of the spa. If the heater connection bars do not supply 115±10% VAC, refer to **Heater Inoperative** in the Diagnosis section of this manual.

CONTROL PANEL TEST

*Refer to **Entire Spa Inoperative** in the Diagnosis section of this manual, if the control panel will not power on, the display will not light up, and the control buttons do not function. If only some of the control panel buttons or control panel LEDs can be switched on and off, the control panel may be defective and should be considered suspect.*

1. Verify that the control panel's serial cable is properly connected to the circuit board.
2. Try to access the fiber optic/spa light, jet pump(s), and temperature set point from the control panel and note which components, buttons, and/or control panel LEDs do not function.
 - a) The **Light** button should switch the spa light and its indicator LED on and off.
 - b) The **Jets** button should access the following jet pump settings:
 - c) The temperature set point has a range from 80° to 104° F. The **Temp**  button should lower the temperature set point one degree at a time. The heater and low-speed jet pump should turn off when the set point is ± 2° F below the actual water temperature displayed on the control panel.
 - d) The **Temp**  button should raise the temperature set point one degree at a time. The heater and low-speed jet pump should turn on when the set point is ± 2° F above the actual water temperature displayed on the control panel.
3. Disconnect the control panel's serial cable and connect a test control panel.
4. Repeat step 2 with the test control panel installed, then review the following:
 - If the jet pump, heater, or spa light switch on when the spa is powered on and cannot be turned off with the test control panel, the control box is defective.
 - If the test control panel is fully functional and the suspect control panel is not, then the suspect control panel is defective.
 - If the test control panel responds exactly the same as the suspect control panel, then the control box and/or spa component(s) are defective. Refer to the Diagnosis section of this manual for each component that is inoperative.
 - If the test control panel is not fully functional, but more operational than the suspect control panel, then the suspect control panel is defective in conjunction with the control box and/or spa component(s). Refer to the Diagnosis section of this manual for each component that is inoperative.

CONTROL SENSOR TEST

Verify that the program jumpers are properly configured.

1. Check the temperature of the spa water with a thermometer.
2. Check the actual water temperature of the spa displayed in the control panel.
 - If the temperature displayed in the control panel coincides with the temperature obtained with the thermometer ($\pm 2^{\circ}\text{F}$) then the sensor is not defective (test complete).
 - If the temperature displayed in the control panel is not within 2°F of the temperature obtained from the thermometer then continue with this test.
3. Disconnect the sensor from the control circuit board.
4. Use the **Sensor Resistance vs. Temperature Chart or Graph** to verify that the ohmic value of the sensor coincides with the temperature of the water.
 - If ohmic value and water temperature do not coincide with the values on the graft, the sensor is defective.
 - If the ohmic value corresponds with thermometer but not with the control panel display, the control board is defective.

HIGH LIMIT SENSOR TEST

1. Lower the temperature set point to 40°F , and switch the jet pump to high speed.
2. Check the temperature of the spa water with a thermometer.
3. Check the actual water temperature of the spa displayed in the control panel.
 - If the temperature displayed in the control panel is not within 2°F of the temperature obtained from the thermometer then perform the **Control Sensor Test**.
4. Use the **Sensor Resistance vs. Temperature Chart or Graph** to verify that the ohmic value of the sensor coincides with the temperature of the water.
 - If ohmic value and water temperature do not coincide with the values on the graft, the sensor is defective.
 - If the ohmic value and water temperature do not coincide with the values on the graft, the control board is defective.

PRESSURE SWITCH TEST

*The control panel and jet pump must be operational and there must be good flow to perform this test. (Refer to the **Control Panel Test**, **Flow Test**, and **Jet Pump Inoperative** sections of this manual if necessary).*

1. Disconnect the wires from the pressure switch. If the FL diagnostic message is not displayed on the control panel, the control box is defective.
2. Make sure the jet pump is turned on and jump the disconnected pressure switch wires. If the FL diagnostic message is displayed on the control panel, the control board is defective.
3. Test for continuity three times – once with the jet pump off, once with the jet pump on low-speed, and once with the jet pump on high-speed.

Note: The wires from the circuit board to the pressure switch must be disconnected, otherwise continuity through the circuit board will be read.

- The pressure switch is working properly ONLY if there is continuity when there is adequate flow (jet pump on high and/or low speed) and no continuity when there is not pressure in the heater (jet pump on standby). The control box is defective if there are no flow problems and an FL0 diagnostic messages appears in the control panel after the pressure switch has been properly installed and the continuity check has determined that the pressure switch is working properly.
- If there is no continuity when there is adequate flow (jet pump on high and/or low speed), or if there is continuity when the jet is pump off, the pressure switch is defective.

FLOW TEST

1. There must be enough water in the spa to cover the filter completely.
 2. Turn the jet pump on (perform this test twice, once with the jet pump on low speed and once with the jet pump on high speed). Refer to the **Jet Pump Inoperative** section of this manual if the jet pump cannot be turned on.
 3. Feel the water pressure in front of each of the jets. The water pressure should feel the same from each of the jets.
 - If only one of the jets feels weak, it may be in the closed position, or there may be an obstruction behind the jet nozzle. Switch the jet to the open position, or turn the jet pump off, remove the jet nozzle and inspect it.
 - If all of the jets feel weak, probably the filter is dirty or clogged. Turn the jet pump off, remove the filter from the filter compartment, and clean or replace it, depending on how dirty it is. To verify that the filter is responsible for the restricted flow remove the filter from the filter compartment, turn the jets back on, and feel the water pressure from one of the jets (the water pressure should increase significantly when the dirty or clogged filter is removed).
- WARNING: DEBRIS IS FREE TO ENTER THE PLUMBING SYSTEM WHEN THE FILTERS ARE REMOVED, BE SURE THAT THE SPA IS CLEAN OR THIS COULD MAKE THE PROBLEM WORSE.**
- If all of jets feel sporadic the air control valve may be closed, or the jet pump may need to be primed if the water level was too low or if the spa was drained recently.
 - If all of the jets feel weak or sporadic and the pump is making an unusual noise the jet pump may be defective.

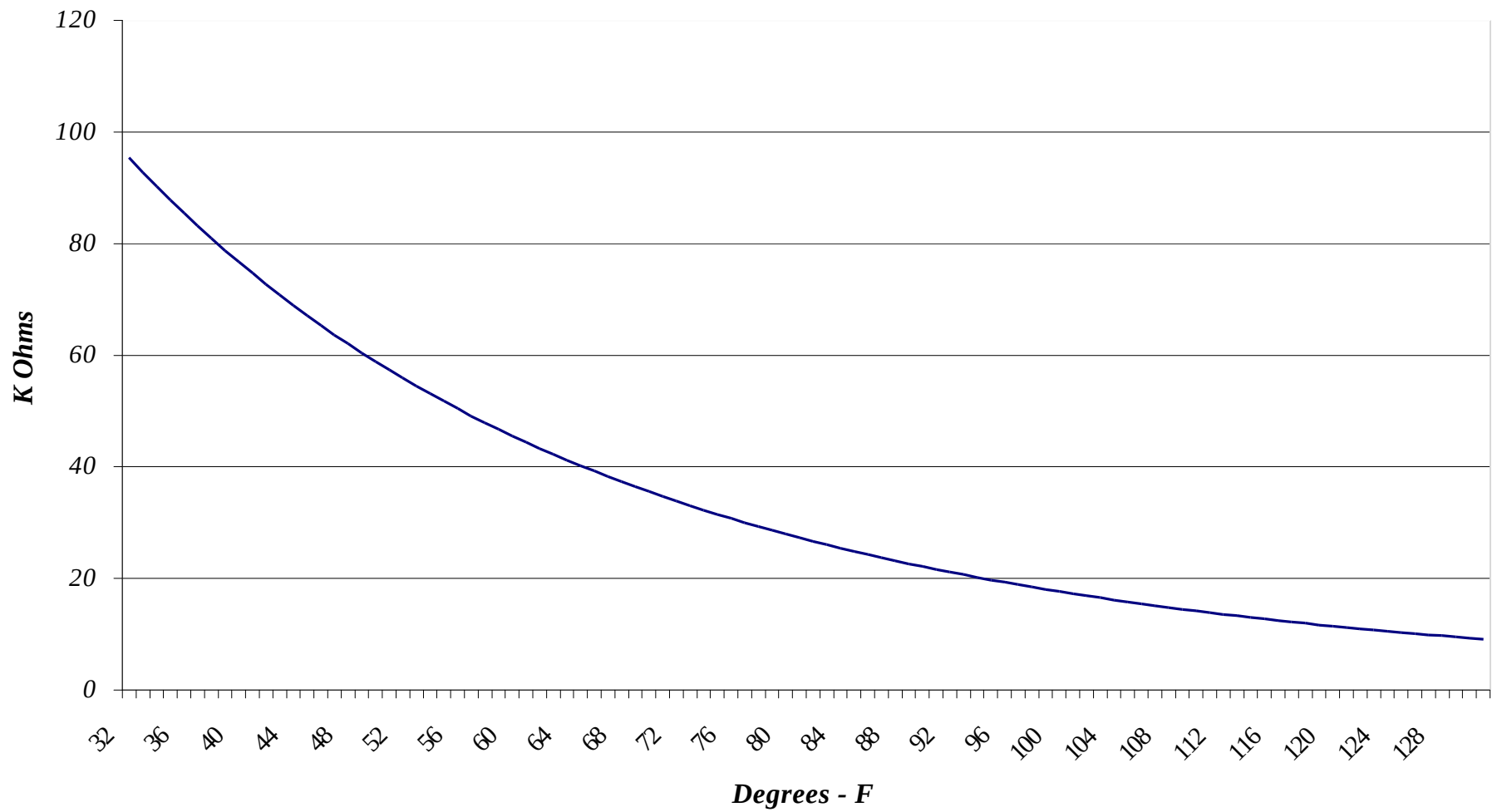
TRANSFORMER TEST:

1. Power up the spa.
2. At the transformer connector on the board and measure as follows:
For the 120V transformer use the voltmeter set to read volts.
Take a reading between the black (3) and white (4) terminals. The reading will be 120 VAC $\pm$ 10%.
Take a reading between the yellow (2) and yellow (1) terminals. The reading will be 12 VAC $\pm$ 10%.

Sensor Resistance vs. Temperature Chart

<i>Resistance in Degrees – °F</i>		<i>Resistance in Degrees – °F</i>		<i>Resistance in Degrees – °F</i>	
<i>K Ohms</i>		<i>K Ohms</i>		<i>K Ohms</i>	
95.35618	32	39.20837	66	18.03452	99
92.7654	33	38.25114	67	17.63626	100
90.25305	34	37.32016	68	17.24789	101
87.81653	35	36.41464	69	16.86924	102
85.45334	36	35.5338	70	16.5	103
83.16106	37	34.67692	71	16.13991	104
80.93736	38	33.84328	72	15.78872	105
78.77999	39	33.03217	73	15.4462	106
76.68679	40	32.24298	74	15.1121	107
74.65567	41	31.4749	75	14.78618	108
72.6846	42	30.72746	76	14.46824	109
70.77165	43	30	77	14.15806	110
68.91493	44	29.29193	78	13.85541	111
67.11263	45	28.60269	79	13.56011	112
65.363	46	27.93171	80	13.27195	113
63.66435	47	27.27848	81	12.99074	114
62.01506	48	26.64244	82	12.7163	115
60.41355	49	26.02613	83	12.44844	116
58.8583	50	25.42005	84	12.18699	117
57.34784	51	24.83273	85	11.93179	118
55.88076	52	24.26072	86	11.68265	119
54.44569	53	23.70358	87	11.43943	120
53.07131	54	23.16087	88	11.20196	121
51.72634	55	22.63219	89	10.9701	122
50.41955	56	22.11714	90	10.7437	123
49.14975	57	21.61532	91	10.55262	124
47.91579	58	21.12636	92	10.30671	125
46.71655	59	20.6499	93	10.09585	126
45.55097	60	20.18558	94	9.88989	127
44.418	61	19.73306	95	9.68872	128
43.31664	62	19.29201	96	9.49222	129
42.24592	63	18.86211	97	9.30025	130
41.2049	64	18.44305	98	9.1127	131
40.19268	65				

Sensor Resistance vs Temperature





Section 5: Removal & Installation

***IMPORTANT: BEFORE BEGINNING ANY EQUIPMENT REPAIR PROCESS ON THE SPA, ALWAYS
POWER DOWN THE SPA***

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Section 5: Removal and Installation

CONTROL PANEL

Removing the Topside Control Panel

Turn the power off to the spa.

Remove the equipment compartment door.

Access the control box.

Disconnect the modular connector from the control box panel.

On the bartop:

Make sure the water line is well below the bottom of the topside control panel.

Using a thin metal blade, i.e. a putty knife, gently pry around the edges of the panel.

Remove the panel and serial cable from the opening in the shell.

Installing the New Topside Control Panel

Clean the shell surface.

Put the serial cable through the opening in the shell.

Apply double-sided tape to the under edge of the topside control panel that will come in contact the spa shell.

Align the panel up with the opening.

Firmly press the panel in place.

In the control box, insert the modular connector into the receptacle refer to the Electrical Schematic, if necessary.

Apply power to the spa.

Make sure that the topside control panel is operating by running all functions.

Section 5: Removal and Installation

CONTROL BOX/HEATER

Removing the Control Box and Heater

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. Disconnect the connections on the right side of the box, i.e. ozone, pump, and spa light.
Optional: Marking and tagging wires for removal - Mark each wire prior to removal by using masking tape and a permanent marker.
4. On the right side of the control box, disconnect the copper bonding wire(s).
5. On the front of the control box, remove the screws and take off the control box cover.
6. On the bottom of the control box on the left side, disconnect the control panel cable.
7. On the left side of the control board, disconnect the sensors with needle nose pliers from J20.
Note: Mark the direction of the connector.
8. On the left side of the control board, disconnect the pressure switch with needle nose pliers from J9.
Note: Mark the direction of the connector.
9. Split nuts are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the split nut, a pipe wrench can be used.
Note: It is possible that there is approximately 2-5 gallons of water in the plumbing system that will need to be drained when the heater is disconnected.

Installing the Control Box and Heater

10. Refer to the Electrical Schematic for the correct configuration of the jumpers.
11. Reverse the removal procedure to install the control box / heater in the equipment compartment.

Section 5: Removal and Installation

HEATER

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. Disconnect the connections on the right side of the box, i.e. ozone, pump, and spa light.
Optional: Marking and tagging wires for removal - Mark each wire prior to removal by using masking tape and a permanent marker.
4. On the right side of the control box, disconnect the copper bonding wire(s).
5. On the front of the control box, remove the screws and take off the control box cover.
6. On the bottom of the control box on the left side, disconnect the control panel cable.
7. On the left side of the control board, disconnect the sensors with needle nose pliers from J20.
Note: Mark the direction of the connector.
8. On the left side of the control board, disconnect the pressure switch with needle nose pliers from J9.
Note: Mark the direction of the connector.
9. Split nuts are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the split nut, a pipe wrench can be used.
Note: It is possible that there is approximately 2-5 gallons of water in the plumbing system that will need to be drained when the heater is disconnected.
10. Inside the control box, disconnect the heater by removing the black heater wires and bonding screws using a nut driver.
11. On the bottom of the equipment compartment, remove the mounting screws.
12. Lift the control box/heater away from the base of the equipment compartment and remove.
13. On the backside of the control box, use the nut driver to remove the screws from the posts that hold the heater in place.
14. Remove the metal straps from around the heater.
15. Grasp the pressure switch and turn to the left, counter clockwise, to loosen. If necessary, channel lock pliers may be used.
16. Install the new heater onto the control box in the reverse order of this instruction.
Note: When installing the pressure switch, it may be necessary to apply a non-hardening pipe joint compound or teflon tape to the threads.

Section 5: Removal and Installation

PRESSURE SWITCH

Removal:

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. The pressure switch is located on the backside of the heater.
4. Using needle nose pliers, disconnect the wires from the flag terminals.
5. Grasp the pressure switch and turn to the left, counter clockwise, to loosen. If necessary, ½" wrench may be used.

Note: Some water, approximately 2-5 gallons, will come out when the pressure switch is removed.

Installation:

1. Apply a non-hardening pipe joint compound or teflon tape to the threads of the new pressure switch.
2. Align the threaded end of the switch perpendicular to the opening in the heater.
Important: The threads must be properly aligned so that installation will not break off the switch when tightening it in the heater.
3. Turn the pressure switch right, clock wise, and hand-tighten in place.
4. Replace the black wire into the upper flag terminal and the red wire into the lower flag terminal.
5. Fill the spa.
6. Test the setup as follows:
 - Disconnect the pressure switch from the circuit board.
 - Set your voltmeter to the continuity setting.
 - Place the meter leads onto the two wires of the pressure switch.
 - Open the gate valves before and after the heater.
 - Turn on the low-speed pump.
 - The meter should register continuity.
 - Turn the pump on from low-speed to high-speed if available and run for ten seconds.
 - After ten seconds has passed, turn off all pumps.
 - After four seconds, check the voltmeter and verify that there is no continuity registering.

Section 5: Removal and Installation

JET PUMP

Removal:

1. Power down the spa.
2. Access the equipment compartment.
3. Drain the water from the spa
4. Drain the water from the pump by removing the freeze/drain plug.
5. Access the control box.
6. Disconnect the connector from the right side of the control box.
7. There are approximately three (3) zip ties along the path of the power cord. These will need to be cut in order to free the cord.
8. Remove the bond wire from the jet pump.
9. Remove the four (4) butterfly bolts that hold the pump to the bottom of the equipment compartment.
10. The pump unions are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the pump union, a pipe wrench can be used.

Note: It is possible that there is a small amount of water in the pump that will need to be drained when the heater is disconnected.

Installation:

1. Reverse the removal procedure to install the jet pump in the equipment compartment.
2. Fill the spa.

Note: After installation, switch the jet pump to high speed and check the compression fittings for leaks.

Section 5: Removal and Installation

SPA LIGHT

Removal:

1. Disconnect power to the spa.
2. Remove the front panel to access the equipment compartment.
3. Locate the connector on the right side of the control box and remove the spa light wires from the control box.
4. Unscrew the spa light one half turn to the left to free it from the back wall of the equipment compartment.
5. Remove the spa light from the equipment compartment.

Installation:

Reverse the removal procedure to install the spa light in the equipment compartment.

Section 5: Removal and Installation

CONTROL/HI LIMIT SENSOR

Removal:

1. Disconnect power to the spa.
2. Drain the spa until the water level is below the thermowell in the filter compartment.
3. Remove the front access panel.
4. Open the control box lid.
5. Unplug the control sensor 4-pin connection wire (refer to the Electrical Schematic for the proper jumper location J-20) and remove it from the control box.
6. Trim any zip ties that hold the cords in place.
7. Unscrew the thermowell compression fitting, and remove the sensor. Use a wrench to loosen the compression fitting if necessary.
8. Remove the hi-limit sensor from the clip on the heater.

Installation:

1. Reverse the removal procedure to install the control sensor in the equipment compartment taking special care to apply insulation to the backside of the control sensor.
Note: After installation, switch the jet pump to high speed and check
2. Refill the spa.

Section 5: Removal and Installation

TRANSFORMERS

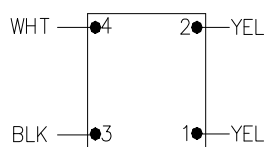
Removing the old transformer:

IMPORTANT: IT IS IMPERATIVE THAT THE POWER TO THE SPA BE DISCONNECTED BEFORE THIS PROCEDURE IS CARRIED OUT.

1. Access the equipment compartment.
2. On the front of the control box, remove the screws and take off the control box cover.
3. Locate the connector, J17, on the board and disconnect it.
4. The transformer is located on the top right part of the box.
5. On the outside of the box, take the screws out of the transformer using a nut driver.

Installing the new transformer:

1. Install the new transformer and using the nut driver, put the screws in place.
2. Install the connector into board using the electrical schematic illustrated below.



Testing the new transformer:

1. Power up the spa.
2. At the transformer connector on the board and measure as follows:
 - For the 120V transformer in the Highland series spa, use the voltmeter set to read volts. Take a reading between the black (3) and white (4) terminals. The reading will be 120 VAC $\pm$ 10%.
 - Take a reading between the yellow (2) and yellow (1) terminals. The reading will be 12 VAC $\pm$ 10%.

Section 5: Removal and Installation

AIR CONTROL VALVE

Removal

1. Disconnect power to the spa and drain the spa.
2. Remove the equipment compartment door.
3. Reach up underneath the bartop (from the equipment compartment) and unscrew the plastic lock nut that holds the air control valve in place against the spa shell.
4. Pull the air valve up through the hole in the spa shell, and cut it off using PVC cutters.
5. Pull the air control valve to free it from the spa.

Installation:

Reverse the removal procedure to install the air control valve.

Section 5: Removal and Installation

PLUMBING PIPE, FITTINGS AND CEMENT

Watkins Manufacturing uses standard sizes schedule 40 pipe and fittings ranging from ½ to 2 inch outside diameters.

Recommended Cements:

- Weld-on 1759 – This cement is used to bond PVC pipe or fitting to other PVC components. This cement has fillers to help seal loose PVC joints.
- Weld-on 794 – This cement is used to bond ABS fittings to rigid PVC pipe and fittings or other ABS components. Do not bond ABS fittings to flexible PVC hose.
- Weld-on 70 – This is a primer that should be used on flexible PVC hose before bonding to PVC fittings with Weld-on 1759 cement.

These cements are available at most hardware or plumbing stores.

Preparing pipes and fittings for bonding

1. Ensure that the pipe ends are free from burrs and cut perpendicular to the length of the pipe.
2. Check the fit of the joint before bonding. The fit of the joint will determine the drying time of the joint. To determine whether the fit of a joint is tight or loose, slip a piece of rigid PVC pipe into a fitting.

The pipe will feel snug half way into the fitting pocket if the fit is tight. The pipe will easily fit to the end of the fitting if the fit is loose.

Note: Flexible PVC tubing will fit into a PVC fitting ¼-inch when dry. The primer and cement will enable the hose to slide to the end of the fitting socket.

Bonding of Pipe and Fittings

1. When cementing PVC pipe, spread primer around the end of the pipe three times. Do not allow the primer to dry.
2. Using the glue applicator, spread the proper cement around the end of the pipe five times for full coverage. Avoid putting cement inside the socket.
Note: For ABS fittings this is imperative because of their thin walls.
3. Insert the pipe in the socket of the fitting. The pipe must seat on the bottom of the socket.
4. Hold the joint for 20 seconds to ensure the pipe does not back out of the socket before the cement is set.

Drying time depends on the fit of the joint. A tight fitting joint should dry within 15 minutes where a loose fitting joint could take 25 minutes or longer. The drying time also depends on weather conditions. Cold weather or high humidity may require longer drying times.

Visually inspect the outside of the glue joint. Be sure the cement is visible all around the joint. If possible, inspect the inside of the joint. Be sure the pipe is seated correctly and the joint is properly cemented.

Section 5: Removal and Installation

VINYL COVERS

The vinyl cover consists of a vinyl encasement and a foam core that insulate the spa when it is not in use. The cover also acts as a safety feature to keep young children out of the spa when the cover is fastened and locked into place. There are four (4) coverlocks hold and secure the cover on the spa.

Puddles on the Vinyl Cover

Puddles on the vinyl cover indicate that the cores are either broken or bowed. A small puddle could gradually grow and cause the cover to bow, which could cause the cover to sag. A sagging cover can create a gap between the cover and the spa, which can lead to heat loss in the water.

- If a core is suspected of being warped on initial delivery, test the cover by applying water on the cover while it is on the spa. If a large puddle forms, take the core out of the encasement, turn it over and reinstall it in the encasement.
- If the core is broken, replace the core.
- If the core is bowed or if there is an isolated depression, take the core out of the encasement, flip it over and reinstall it.

Heavy Foam Cores

A heavy core indicates that the core has absorbed water, usually through a tear in the vinyl. If there is a tear, the vinyl will have to be repaired or replaced.

Depending on the amount of water the core has absorbed it may be able to be dried by the customer. The core should be removed from the encasement and placed in direct sunlight.

Repairing Vinyl Tears

Many times when the encasement has a tear, it can be repaired with one of several vinyl repair kits available. Vinyl repair kits are typically found at major department stores. Refer to the instructions with the kit to make this repair.

Section 5: Removal and Installation

Replacing Foam Core

Refer to the following instructions to replace vinyl cover foam core:

1. Locate the zipper-pull in the pocket found on the side of the cover and open. Do not force the zipper. If the zipper gets stuck, lubricate the zipper with light oil (WD-40) or silicone spray. Gently work a stubborn zipper free.
Note: If the zipper breaks, it will not be covered under warranty.
2. Once the vinyl encasement is unzipped, equally pull on each corner of the foam core and remove it from the encasement.
3. Insert the new foam core into vinyl encasement. Be careful handling the new core so that the plastic covering does not tear. If the plastic rips, use two-inch pieces of quality tape to seal the rip.
 - Insert the core straight into the encasement. This is easier done with two people than with only one. It is easier to accomplish when the vinyl is warm so it will stretch.
 - If the core is slightly too large, use a large rubber mallet to pound the sides of the foam core. This will make the core smaller by compressing the foam along the edge.
 - If the core is too small put a foam strip along the bottom (along the zipper) edge of the foam.**Note: Replacing the foam core is easier on warm days.**
4. Close the zipper. Pull the zipper flap down and pull the zipper halves as close together as possible to make it as easy as possible to pull the zipper around the corners of the foam. Once the zipper is closed, pull up the zipper flap.

COVERLOCK REPLACEMENT

1. A notch must be made in the portion of the insert where the vinyl cover strap is placed. With a pair of dikes, make a cut at an angle from the edge of the insert to the center.
2. Make a straight cut in the center of the insert to meet the angled cut. Carefully remove the notch at the smallest part of the cut (should be about 3/16 inch).
NOTE: Be careful when making these cuts not to crack or break the insert.
3. Insert the strap from the vinyl cover through the notch.
4. Close the notch with the special tape supplied in the coverlock kit.



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Section 6: Appendix

JUMPER SETTING

The jumpers are located on the main circuit board inside the control box (refer to the Electrical Schematics for the exact locations). The setting of the jumpers determine how the spa's electronic control system will operate. Improper jumper setting will cause erratic spa functions, and may cause the house circuit breaker to trip.

As an example, imagine a control box is installed in a convertible spa wired for 115 VAC. If the jumpers are set for the proper spa model, but wrong voltage of **230 VAC** (converted), the control box will allow the heater to turn on while the jets are on. This will draw too much current, which can trip the house breaker.



Setting the Jumpers

1. Disconnect power from the control box.
2. Refer to the illustrations on the following page to find the jumper setting for your specific spa model.
3. Using a pair of needle-nose pliers, carefully remove or install the jumper in the appropriate location.

IMPORTANT: DO NOT ALTER THE SETTINGS OF JUMPERS NOT CALLED OUT ON THE ELECTRICAL SCHEMATICS.

JUMPER SETTINGS

JUMPER	AS LABELED ON BOARD	CALDERA FUNCTION	DEFAULT SETTING
J-23	50A/20A	SYSTEM VOLTAGE	115 POSITION
J-32	-----	-----	OPEN
J-33	-----	-----	CLOSED

Section 6: Appendix

CONVERTIBLE 115/230 VAC SPA MODELS

Required Tools:

Standard Screwdriver

Phillips Screwdriver

Needle-nose Pliers

Strain Relief Compression Tool or Vice-Grips

Required Parts:

38661 (1) Breaker GFCI, Subpanel Box

36021 (1) Program Jumper, Shunt 0.100

1. Disconnect the power.
2. Remove the screws and open the equipment compartment door.
3. Remove the screws and open the control box.
4. Locate TB-1, located in the lower right-hand corner of the control box.
5. If connected, remove the 115 VAC power cord wires from terminals 2, 3 and 5 then remove the strain relief and 115 VAC power cord from the access hole in the control box.
6. Refer to the picture below.
7. On the main circuit board (inside the control box), locate the Jumper J-23 (positioned on the upper left of the circuit board, next to the top).

WARNING! Do not allow the pliers to contact any electronic components inside the control box.

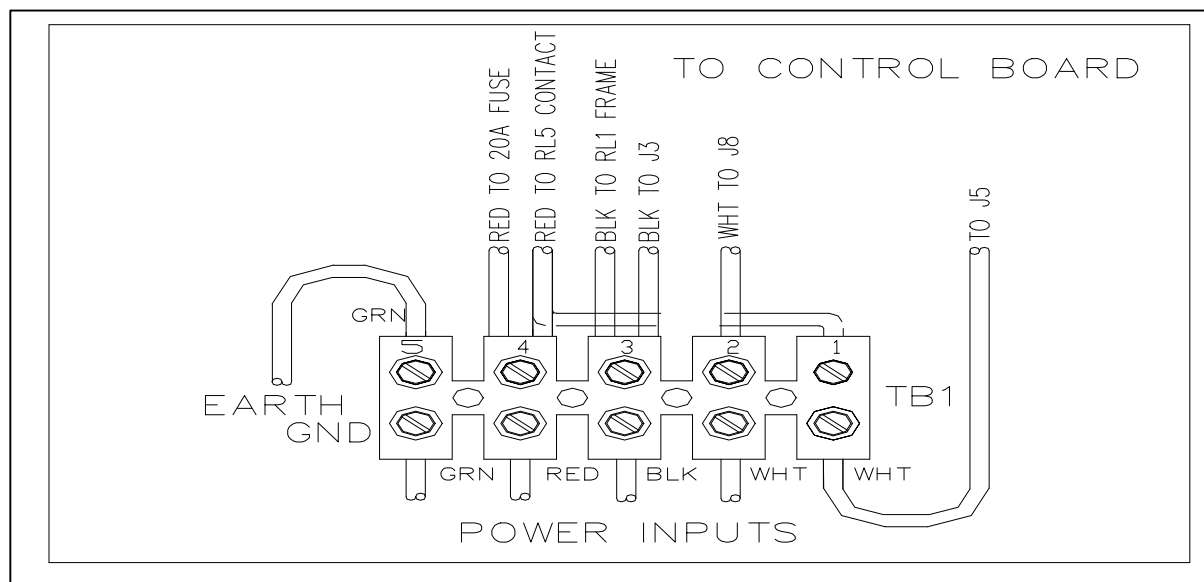
IMPORTANT – BE SURE TO CHANGE THE JUMPERS IN LOCATIONS J-23 ONLY. ALTERING OTHER JUMPER LOCATIONS MAY CAUSE DAMAGE TO THE SPA.

Use a pair of needle-nose pliers to place J-23 from the 20A position to the 30A.

CAUTION: Damage may result if the jumpers are incorrectly set.

The subpanel (38661) containing the GFCI breakers must be installed in series between the spa and the main electrical panel. A LICENSED electrician must install the subpanel.

WARNING! Damage may result if the terminal block or subpanel is incorrectly wired.



MISCELLANEOUS SERVICE TOOLS AND PARTS

TOOLS

DESCRIPTION	PART No.
Thermometer	10066
Meter, Electrical Usage, 230V	70357
Meter, Electrical Usage, 115V	10069
Hose Clamping Pliers	50306
Meter Lead, Needle Black	71129
Meter Lead, Needle Red	71131
Spring Clamp Pliers	50051
3.6" Spanner Wrench	71703
Clear Silicone, DOW 732, 3oz	71525
Leak Detection Kit	73740

SERVICE LITERATURE

DESCRIPTION	PART No.
Oasis Service Manual	61625
Warranty Service Claims (Pack of 25)	60091
ARS UPS LABEL	36071

FOAM AND WINTERIZING BLANKET

DESCRIPTION	PART No.
Foam, Kit — 2 lb. — 1cu.ft.	70293
Foam, Kit — 2 lb — 8.75 cu. ft.	70869
Winterizing Kit For Spas	31399



Electrical Schematics

Watkins Manufacturing Corporation
1280 Park Center Drive
Vista, California 92083

Revised: 08-06

Electrical Schematic Index

Use the following Electrical Schematics to perform a point-to-point electrical testing on all Caldera Export Spa models. Use these schematics when performing a service call with an electrical problem.

Symbol Definition Sheet

YR 2000 Cover Sheet

Oasis

YR 2001 Cover Sheet

Oasis

YR 2002 – 2005 Cover Sheet

Oasis

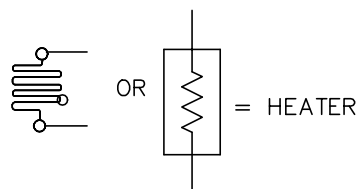
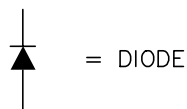
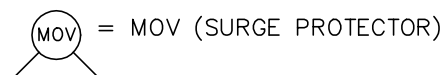
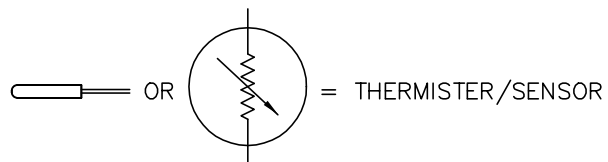
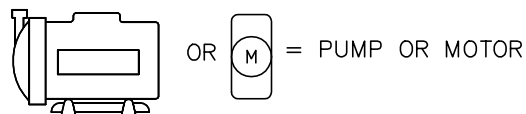
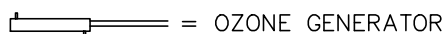
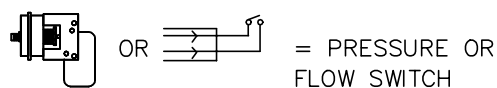
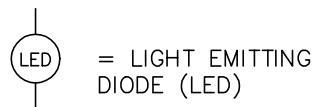
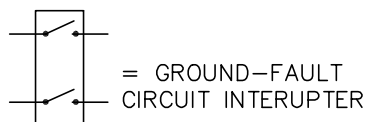
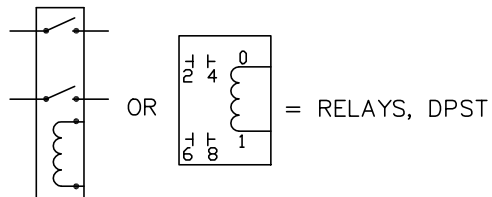
Oasis II

YR 2006 Cover Sheet

Oasis

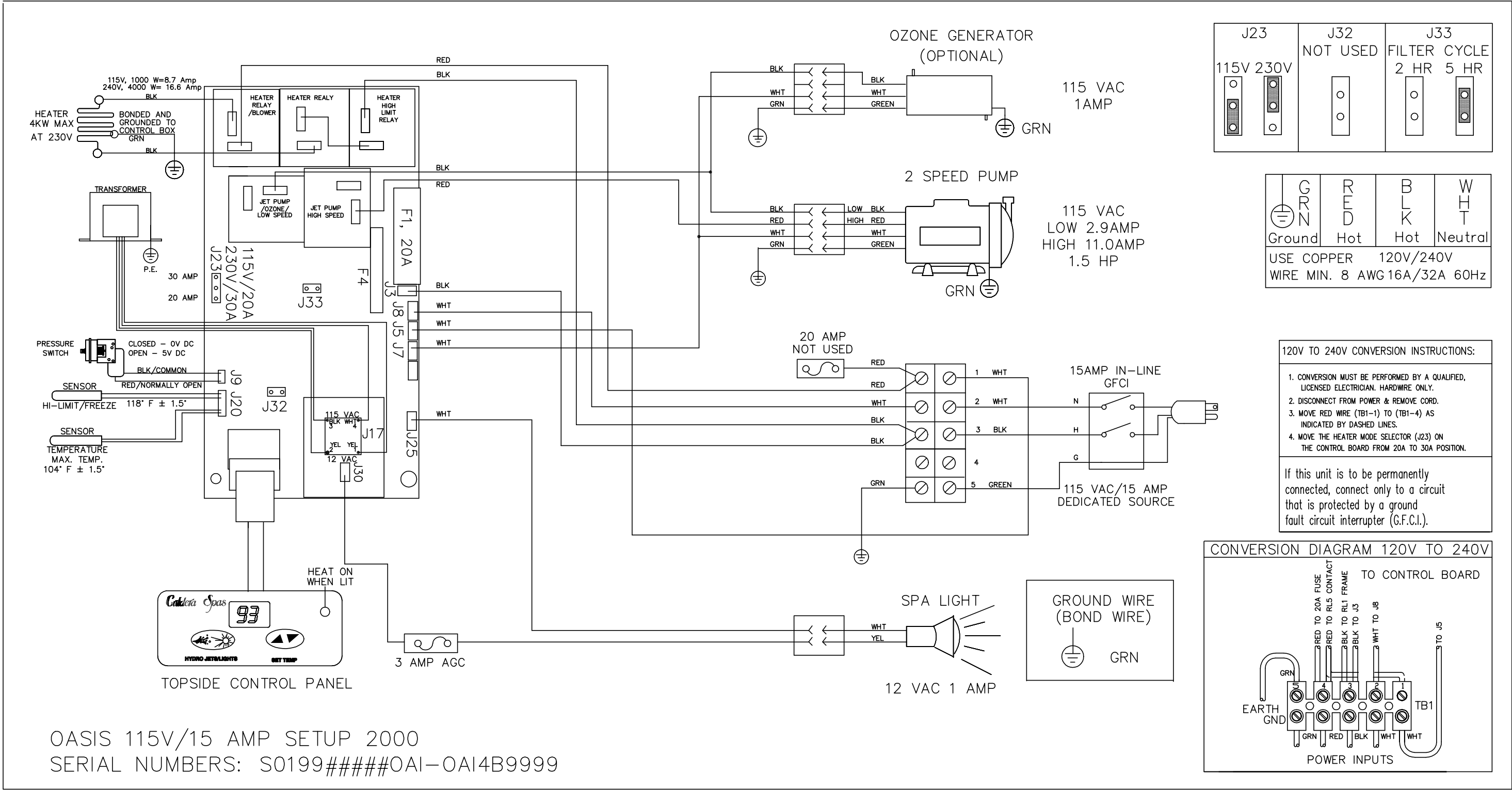
Oasis II

SYMBOL DEFINITION SHEET





Year 2000





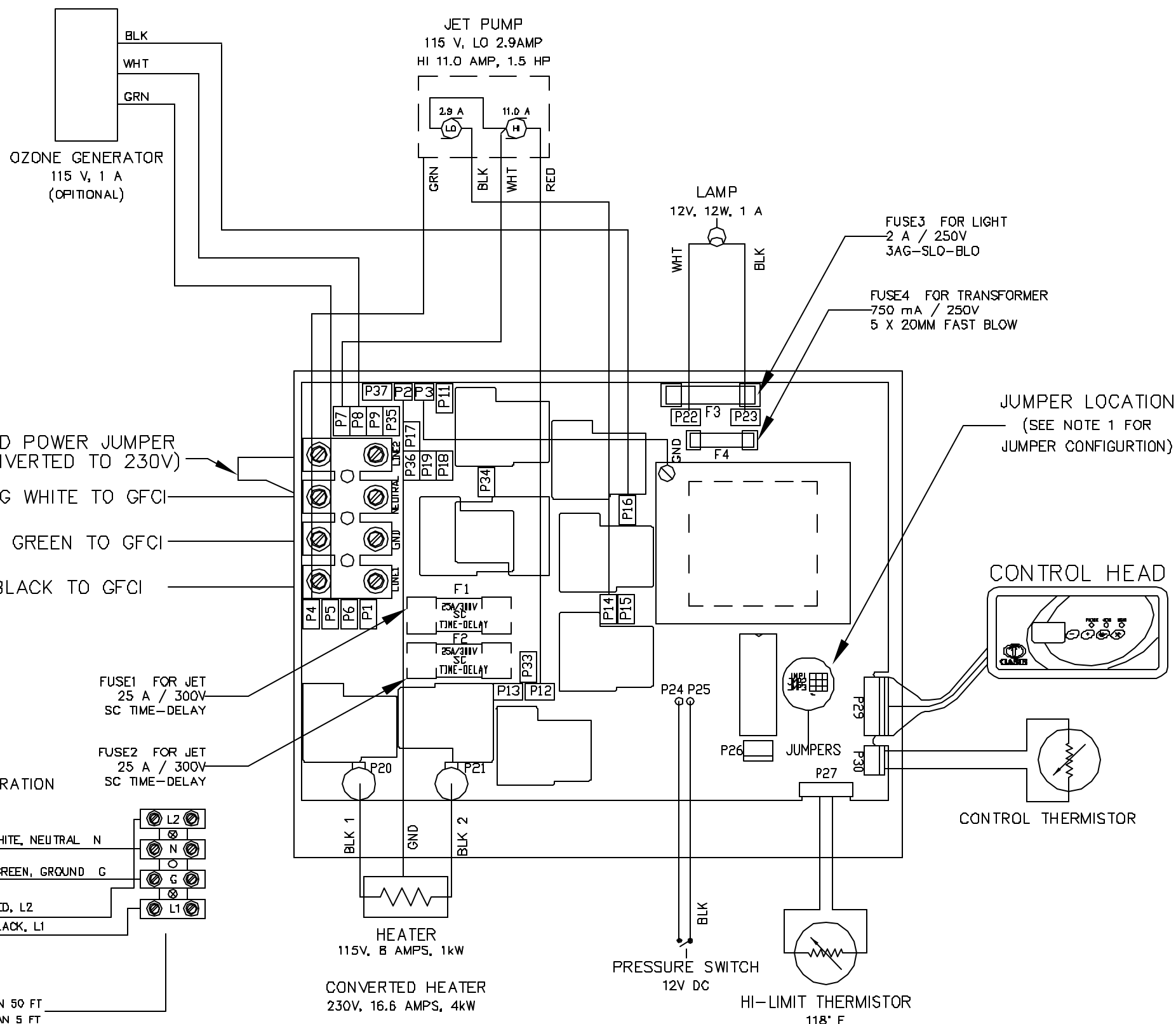
Year 2001



YR2002 – YR2005
Electrical
Schematics

NOTE 1: JUMPER CONFIGURATIONS

PINS 1 2 3		PINS 1 2 3	
	JMP 1		JMP 1
	JMP 2		JMP 2
	JMP 3		JMP 3
STANDARD, 15A, 115V CORD CONNECTED CONFIGURATION		CONVERTED 50A, 230V CONVERTED CONFIGURATION	



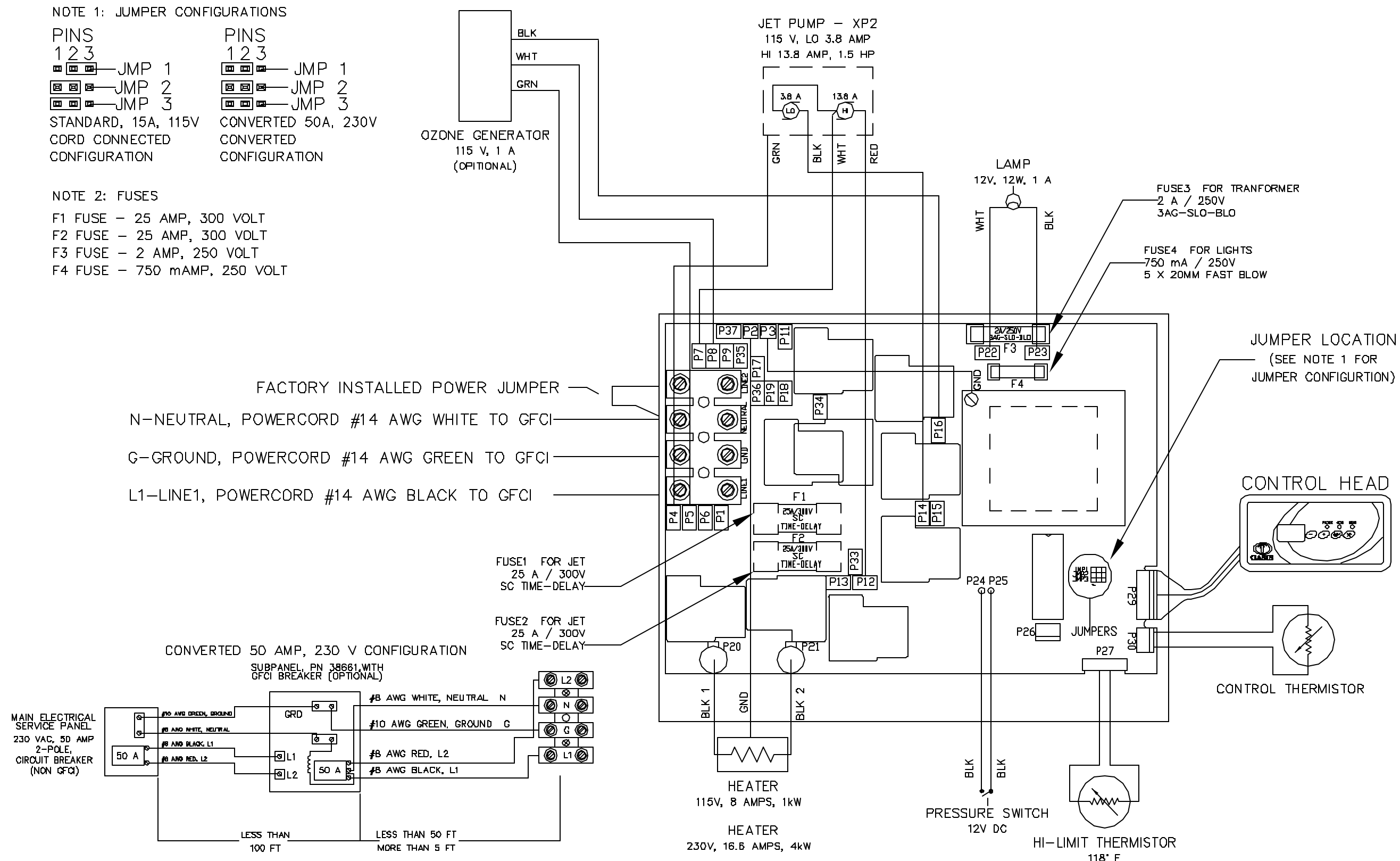
OASIS SERIES 115V/20 AMP SETUP YR2002 – MID2005
SERIAL NUMBERS: OAI11D1001 TO OAI2G1059 (OAI3G3003)

NOTE 1: JUMPER CONFIGURATIONS

PINS 1 2 3		PINS 1 2 3	
	JMP 1		JMP 1
	JMP 2		JMP 2
	JMP 3		JMP 3
STANDARD, 15A, 115V CORD CONNECTED CONFIGURATION		CONVERTED 50A, 230V CONVERTED CONFIGURATION	

NOTE 2: FUSES

F1 FUSE – 25 AMP, 300 VOLT
F2 FUSE – 25 AMP, 300 VOLT
F3 FUSE – 2 AMP, 250 VOLT
F4 FUSE – 750 mA, 250 VOLT



OASIS II 115V/20 AMP SETUP 2002 TO MID2005
SERIAL NUMBERS: OAI21D1001 – OAI22G1071 (OAI23G3003)

Diagram illustrating the wiring configurations for the 15A, 115V standard and the 50A, 230V converted version.

STANDARD, 15A, 115V CORD CONNECTED CONFIGURATION:

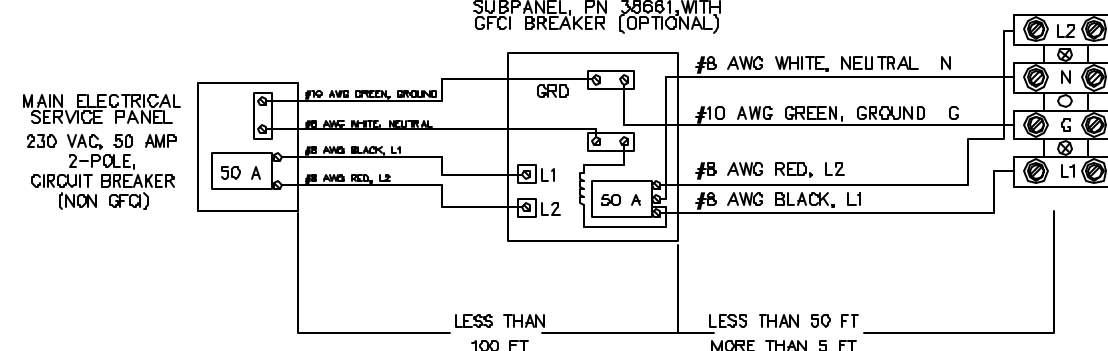
- PINS 1, 2, 3 are connected to three separate jumpers.
- Each jumper is connected to a terminal labeled 1, 2, and 3 respectively.

CONVERTED 50A, 230V CORD CONNECTED CONFIGURATION:

- PINS 1, 2, 3 are connected to three separate jumpers.
- Each jumper is connected to a terminal labeled 1, 2, and 3 respectively.

FACTORY INSTALLED POWER JUMPER
N-NEUTRAL, POWERCORD #14 AWG WHITE TO GFCH
G-GROUND, POWERCORD #14 AWG GREEN TO GFCH
L1-LINE1, POWERCORD #14 AWG BLACK TO GFCH -

SUBPANEL, PN 38681, WITH
GFCI BREAKER (OPTIONAL)



JET PUMP
15 V. LO 2.9AMP
11.0 AMP, 1.5 HP

PUMP #1 CONNECTOR	
VOLTAGE	120V 
GREEN / GROUND	P4
WHITE / COM	P7
BLACK / LOW SPEED	P14
RED / HIGH SPEED	P101

	LIGHT
WHITE / LIGHT	P22
BLACK / 12 VAC	P23

HEATER	
BLACK 1	P20
BLACK 2	P21

VOLTAGE	120V
GREEN / GROUND	P5
WHITE / COM	P8
BLACK / LINE	P16

BLACK	P25
BLACK	P24

FUSE 3 FOR LIGHTS
2 A / 250V
3AG-SLO-BLO

FUSE 4 FOR TRANSFORMER
750 mA / 250V
5 X 2DMM FAST BLOW

JUMPER LOCATION
— (SEE NOTE 1 FOR
JUMPER CONFIGURATION)

CONTROL HEAD

CONTROL THERMISTOR

PRESSURE SWITCH
5V DC, OPEN
0V DC, CLOSED

HI-LIMIT THERMISTOR
118° F

HEATER
120V, 8 AMPS, 1kW

CONVERTED HEATER
230V, 16.6 AMPS, 4kW

OASIS SERIES 115V/20 AMP SETUP MID2005 – 2005
SERIAL NUMBERS: OAI2G1059 (OAI3G3003) TO CURR

NOTE 1: JUMPER CONFIGURATIONS

PINS		PINS	
1 2 3		1 2 3	
— JMP 1		— JMP 1	
— JMP 2		— JMP 2	
— JMP 3		— JMP 3	
STANDARD, 15A, 115V		CONVERTED 50A, 230V	
CORD CONNECTED		CONVERTED	
CONFIGURATION		CONFIGURATION	

NOTE 2: FUSES

- F1 FUSE — 25 AMP, 300 VOLT
- F2 FUSE — 25 AMP, 300 VOLT
- F3 FUSE — 2 AMP, 250 VOLT
- F4 FUSE — 750 mA, 250 VOLT

PUMP #1 CONNECTOR

VOLTAGE	120V
GREEN / GROUND	P4
WHITE / COM	P7
BLACK / LOW SPEED	P14
RED / HIGH SPEED	P101

LIGHT CONNECTOR 12 W MAX.

	LIGHT
WHITE / LIGHT	P22
BLACK / 12 VAC	P23

HEATER

BLACK 1	P20
BLACK 2	P21

OZONE CONNECTOR

VOLTAGE	120V
GREEN / GROUND	P5
WHITE / COM	P8
BLACK / LOW	P16

PRESSURE SWITCH

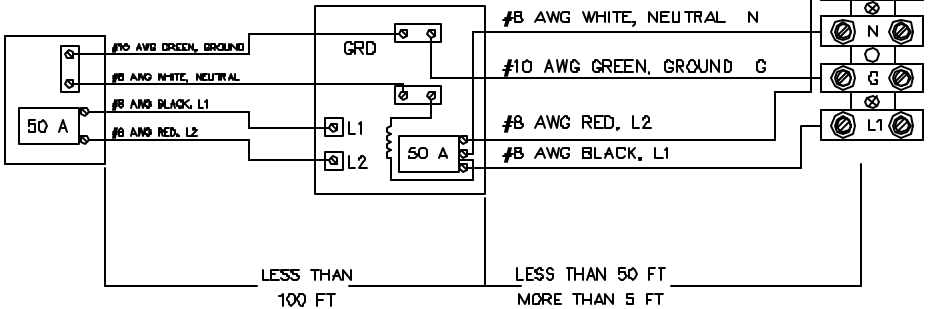
BLACK	P25
BLACK	P24

FACTORY INSTALLED POWER JUMPER
N-NEUTRAL, POWERCORD #14 AWG WHITE TO GFCI
G-GROUND, POWERCORD #14 AWG GREEN TO GFCI
L1-LINE1, POWERCORD #14 AWG BLACK TO GFCI

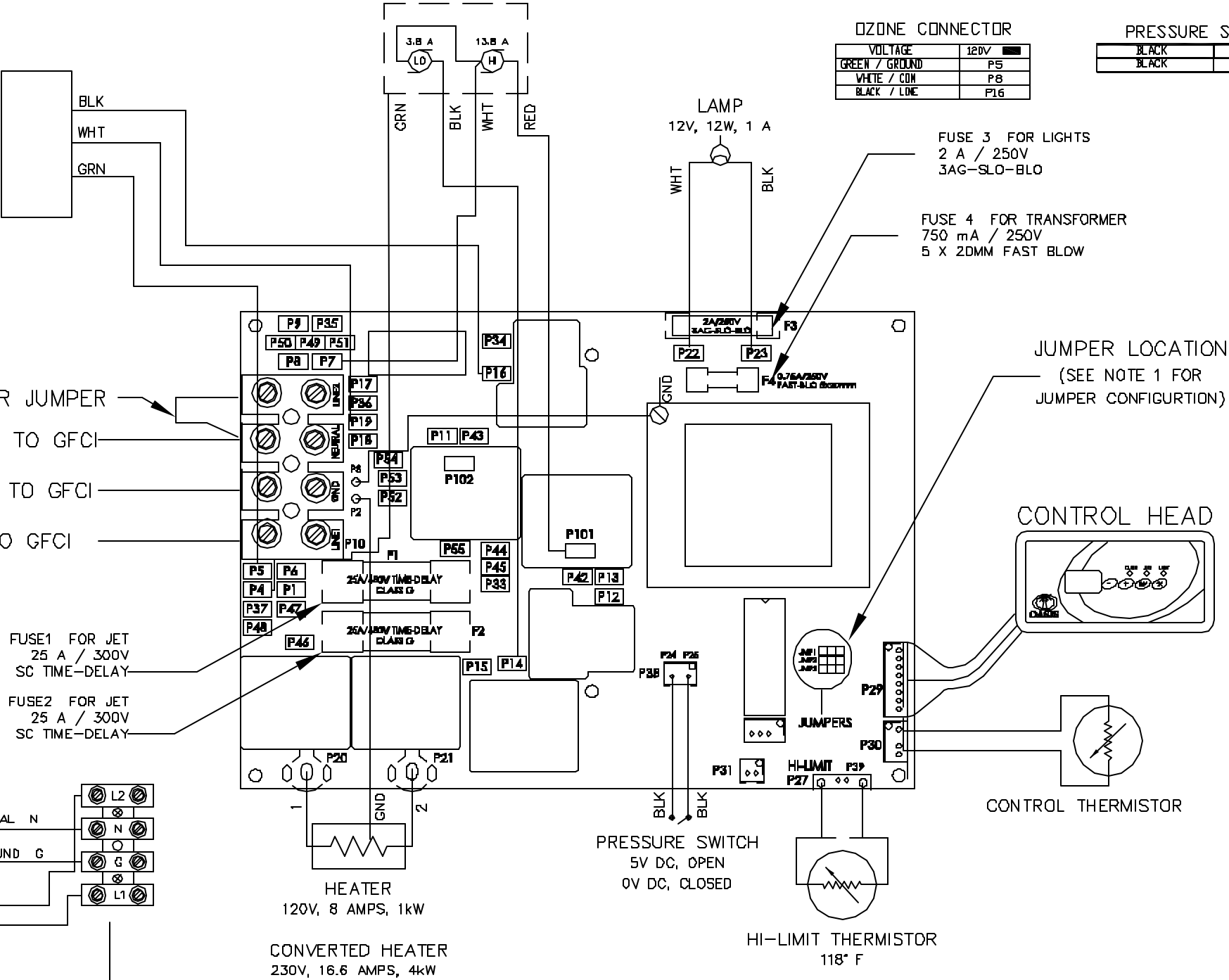
CONVERTED 50 AMP, 230 V CONFIGURATION

SUBPANEL, PN 38661, WITH
GFCI BREAKER (OPTIONAL)

MAIN ELECTRICAL
SERVICE PANEL
230 VAC, 50 AMP
2-POLE,
CIRCUIT BREAKER
(NON GFI)



JET PUMP — XP2
115 V, LO 3.8 AMP
HI 13.8 AMP, 1.5 HP



OASIS II 115V/20 AMP SETUP MID2005 THRU 2005
SERIAL NUMBERS: OAI22G1071 (OAI23G3003) TO OAI24G9999



YR2006
Electrical
Schematics

FACTORY 15A/115V 60 HZ GFCI USE JUMPER CONFIGURATION A

N-NEUTRAL, POWERCORD #14 AWG WHITE TO GFCI
L1-LINE1, POWERCORD #14 AWG BLACK TO GFCI
G-GROUND, POWERCORD #14 AWG GREEN TO GFCI

SEE DETAIL FOR OPTIONAL SUBPANEL CONNECTION

PUMP CONNECTOR
115V, LO 2.9 A, HI 11.0 A, 1.0 HP

RED / LOW SPEED	P28
BLACK / HIGH SPEED	P19
WHITE / COM	P18
GREEN / GROUND	JET PUMP 1

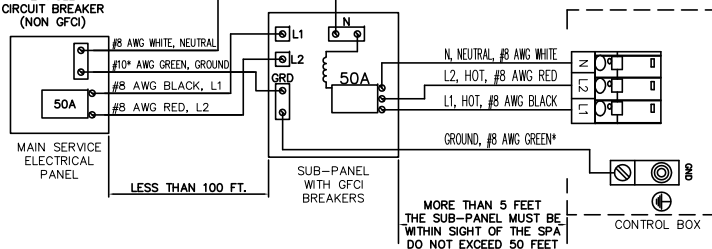
OZONE CONNECTOR
115 V, 1 A

GREEN / GROUND	OZONE
WHITE / COM	P25
BLACK / LINE	P26

OPTIONAL SUBPANEL CONVERSION CONNECTION

230VAC, 50 Amp
2-POLE
CIRCUIT BREAKER
(NON GFCI)

230 VAC, 40A, 60Hz, MODELS



OASIS I 115V/15 AMP SETUP 2006 - 2007
SERIAL NUMBERS: OAI11H3001 TO OAI4J9999



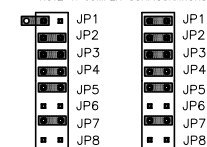
TERMINAL BLOCK
POWER JUMPER
2 POSITION POWER JUMPER
MUST REMAIN ON THE
TERMINAL BLOCK TO
OPERATE 115 VAC
SPA MODELS

NOTE: REMOVE WHEN CONVERTING TO 230V

PUMP RETURN JUMPER (RED WIRE)

BRANCH CIRCUIT
FUSES - F7, F8
25 A / 300V
SC TIME-DELAY

NOTE 1: JUMPER CONFIGURATION



REMOVE POWER JUMPER
FROM TERMINAL BLOCK
TO CONVERT SPA TO
OPERATE AT 230 VAC.

FACTORY, 15A/115V
CORD CONNECTED
CONFIGURATION

OPTIONAL 50A/230V
CONVERTED
CONFIGURATION

CONTROL PANEL 18V/25mA
CONNECTOR P2

BLUE LIGHT 18V/25mA
CONNECTOR P11

CONTROL THERMISTOR
104° F/40° C
CONNECTOR P27

FACTORY WIRING REMOVED FOR CLARITY

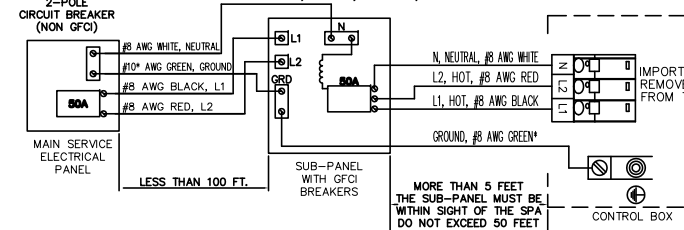
PRESSURE SWITCH
(LOCATED ON HEATER)
5V DC, OPEN
0V DC, CLOSED
CONNECTOR P34

HI-LIMIT THERMISTOR
(LOCATED ON HEATER)
118° F/48° C MAX
CONNECTOR P5

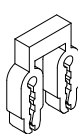
OPTIONAL 50A/230V SUBPANEL CONVERSION CONNECTION USE JUMPER CONFIGURATION B

230VAC, 50 Amp
2-POLE
CIRCUIT BREAKER
(NON GFCI)

230 VAC, 40A, 60Hz, MODELS



MORE THAN 5 FEET
THE SUB-PANEL MUST BE
WITHIN SIGHT OF THE SPA
DO NOT EXCEED 50 FEET

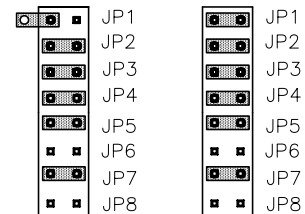


TERMINAL BLOCK
POWER JUMPER
2 POSITION POWER JUMPER
MUST REMAIN ON THE
TERMINAL BLOCK TO
OPERATE 115 VAC
SPA MODELS
NOTE: REMOVE WHEN CONVERTING TO 230V

PUMP RETURN JUMPER (RED WIRE)

BRANCH CIRCUIT
FUSES - F7, F8
25 A / 300V
SC TIME-DELAY

NOTE 1: JUMPER CONFIGURATIONS



A
FACTORY, 15A/115V
CORD CONNECTED
CONFIGURATION

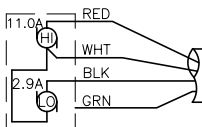
B
OPTIONAL 50A/230V
CONVERTED
CONFIGURATION

REMOVE POWER JUMPER
FROM TERMINAL BLOCK
TO CONVERT SPA TO
OPERATE AT 230 VAC.

N-NEUTRAL, POWERCORD #14 AWG WHITE TO GFCI
L1-LINE1, POWERCORD #14 AWG BLACK TO GFCI
G-GROUND, POWERCORD #14 AWG GREEN TO GFCI
SEE DETAIL FOR OPTIONAL SUBPANEL CONVERSION CONNECTION

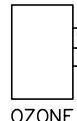
PUMP CONNECTOR
115V, LO 3.8 A, HI 13.8 A, 1.5 HP

RED / LOW SPEED	P28
BLACK / HIGH SPEED	P19
WHITE / COM	P18
GREEN / GROUND	JET PUMP 1



OZONE CONNECTOR
115 V, 1 A

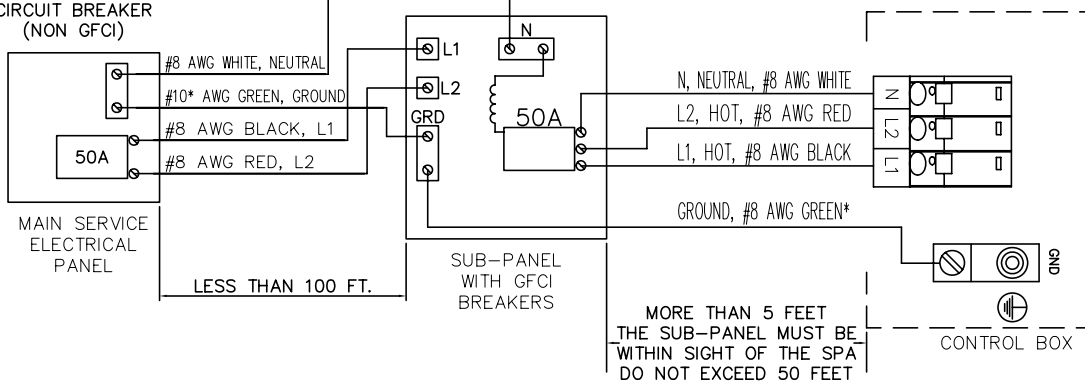
GREEN / GROUND	OZONE
WHITE / COM	P25
BLACK / LINE	P26



OPTIONAL SUBPANEL CONVERSION CONNECTION USE JUMPER CONFIGURATION B

230VAC, 50 Amp
2-POLE
CIRCUIT BREAKER
(NON GFCI)

230 VAC, 40A, 60Hz, MODELS



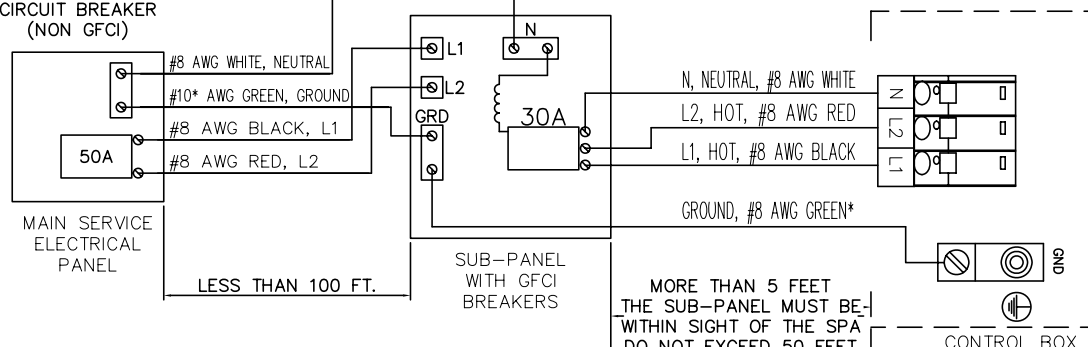
OASIS II 115V/20 AMP SETUP 2006 - CURR
SERIAL NUMBERS: OAI21H3001 TO CURR

FACTORY WIRING REMOVED FOR CLARITY

OPTIONAL SUBPANEL CONVERSION CONNECTION USE JUMPER CONFIGURATION A

230VAC, 50 Amp
2-POLE
CIRCUIT BREAKER
(NON GFCI)

230 VAC, 24A, 60Hz, MODELS



PRESSURE SWITCH
(LOCATED ON HEATER)
5V DC, OPEN
0V DC, CLOSED
CONNECTOR P34

HI-LIMIT THERMISTOR
(LOCATED ON HEATER)
118° F/48° C MAX
CONNECTOR P5

BLUE LIGHT 18V/25mA
CONNECTOR P11

CONTROL PANEL 18V/25mA
CONNECTOR P2

CONTROL THERMISTOR
104° F/40° C
CONNECTOR P27

HEATER
115V, 8 AMPS, 1kW
CONVERTED HEATER
230V, 16.6 AMPS, 4kW

BLACK 1	P1
BLACK 2	P3



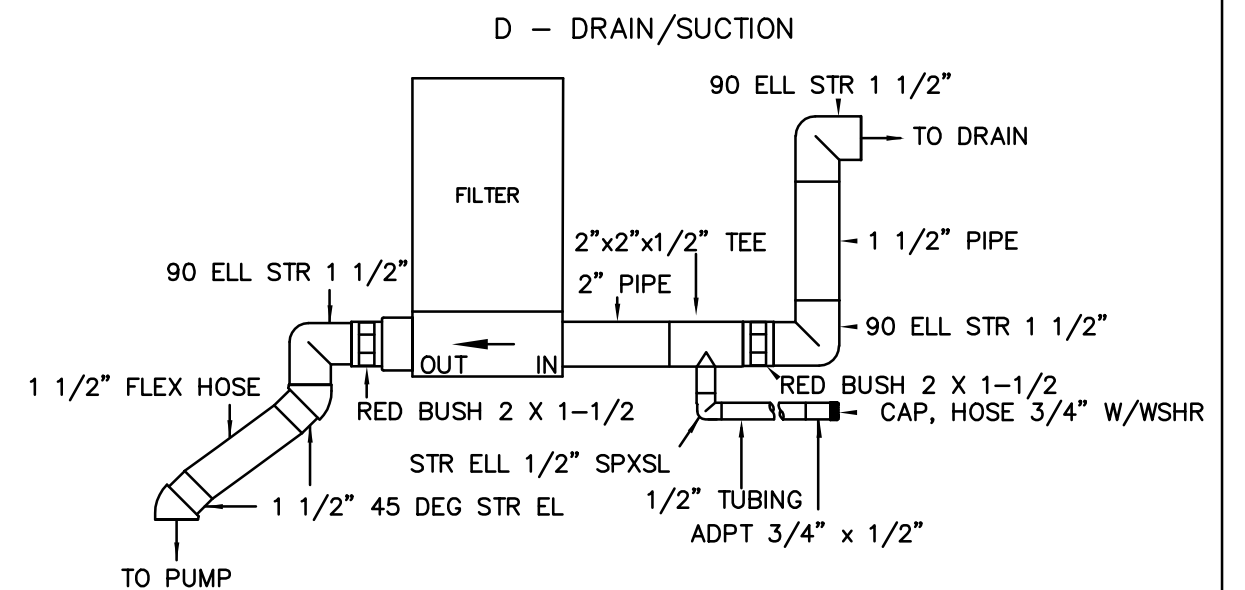
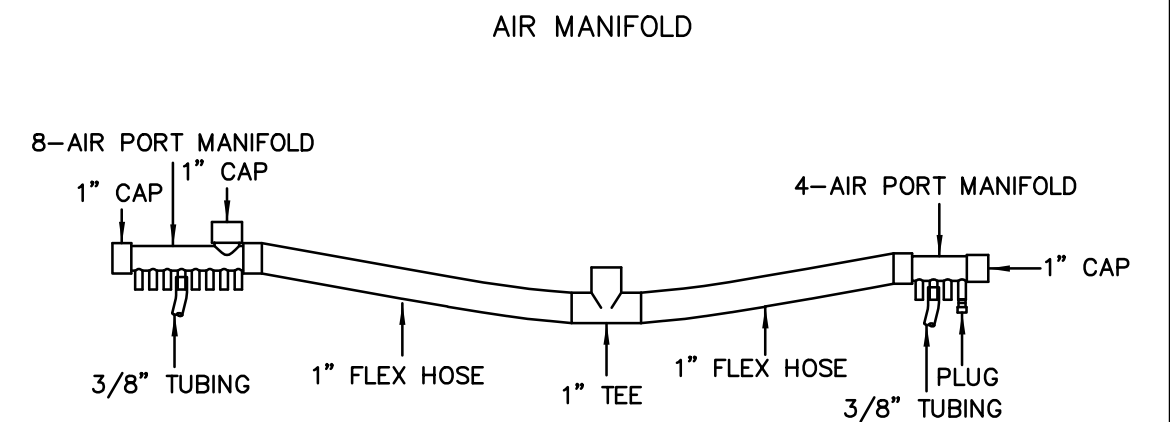
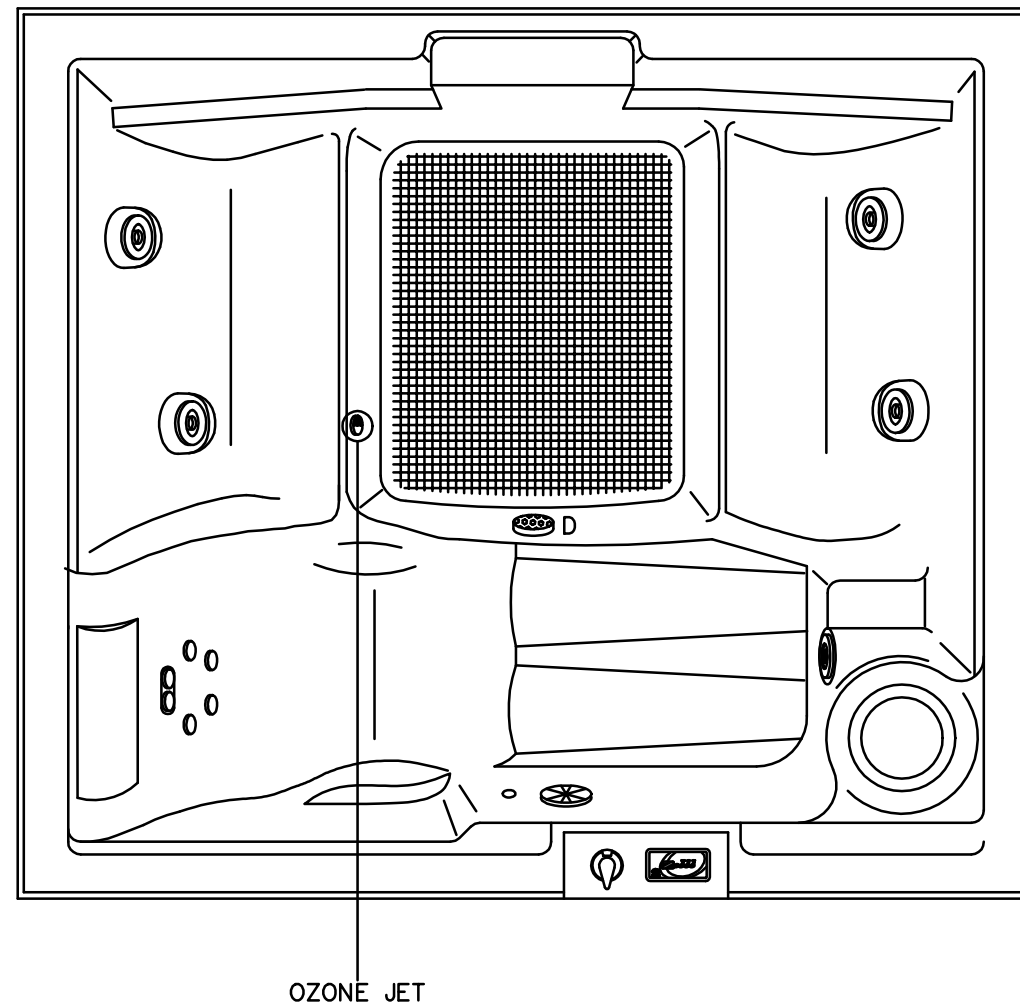
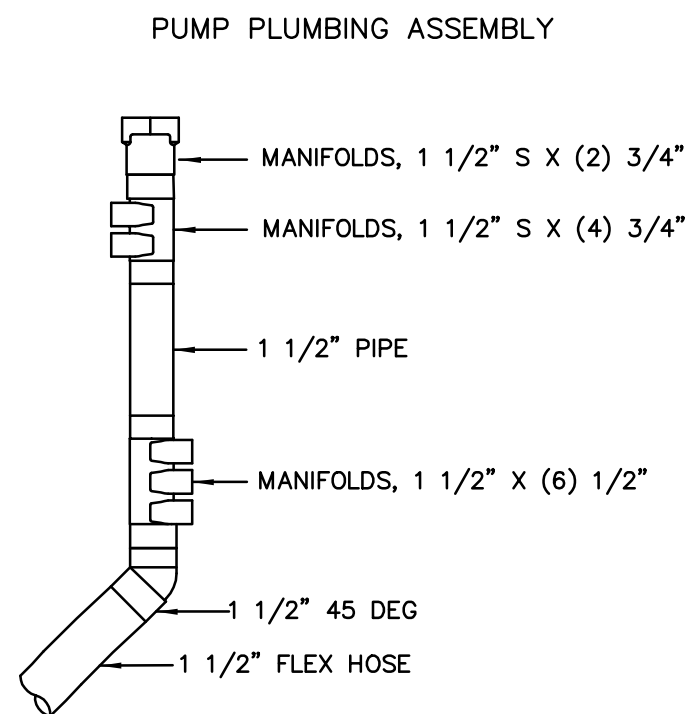
Plumbing Diagrams

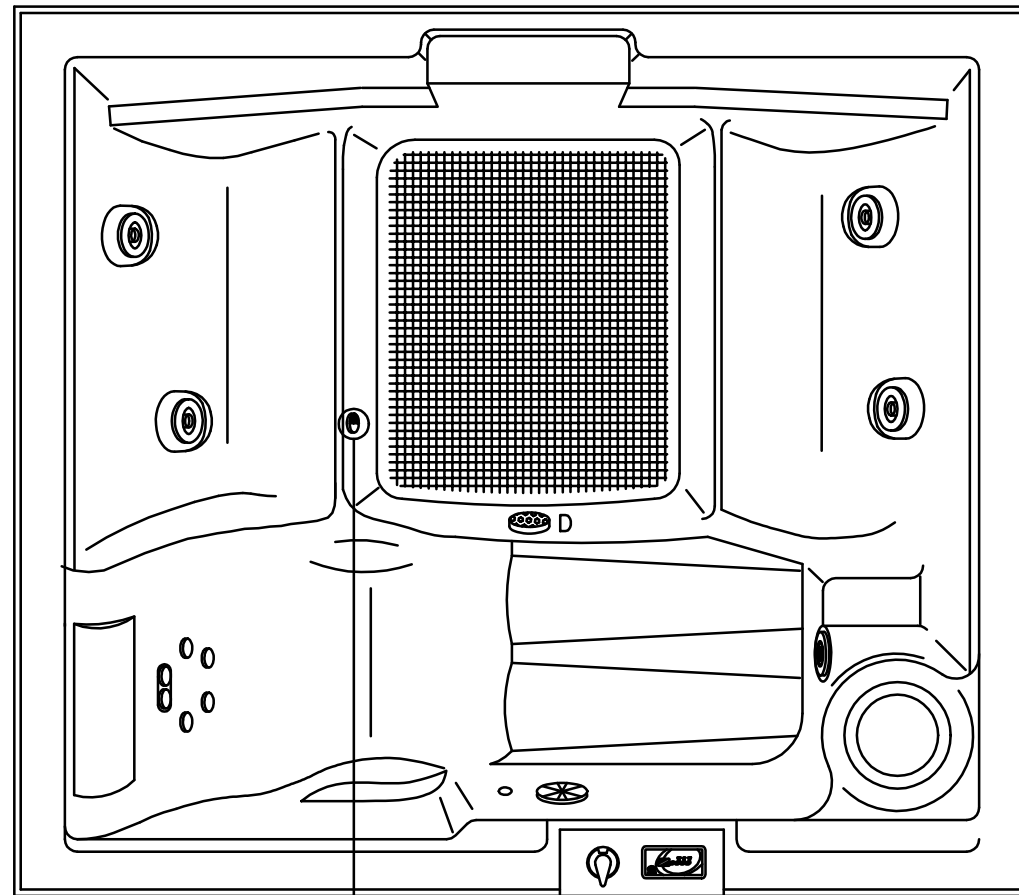
Watkins Manufacturing Corporation
1280 Park Center Drive
Vista, California 92083

Revised: 05-03

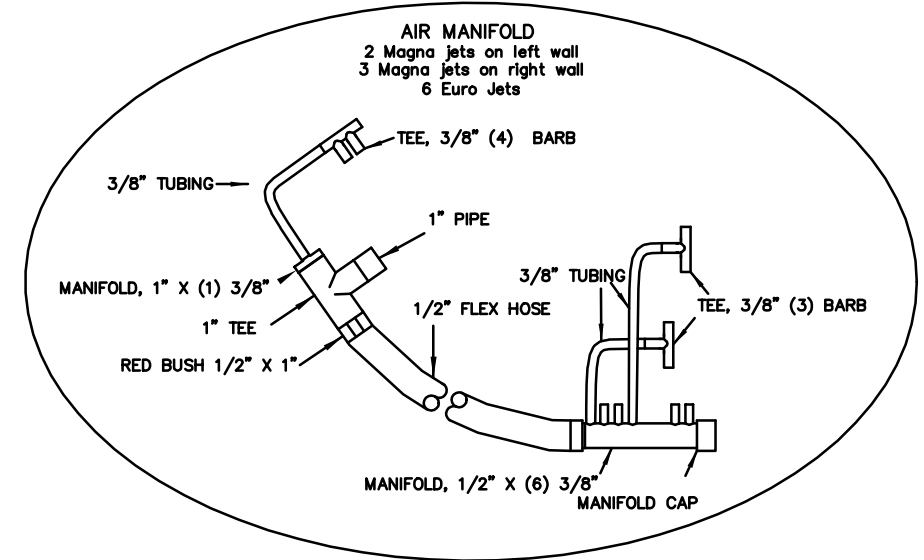
Plumbing Diagram Index

<u>Model</u>	<u>Year</u>	<u>Serial Number</u>
Oasis	2000 – 2002	OAI11B1001 - OAI14D9999
Oasis	2003 – Curr	OAI11E1001 - Curr
Oasis II	2002 – Curr	OAI21D1001 - Curr

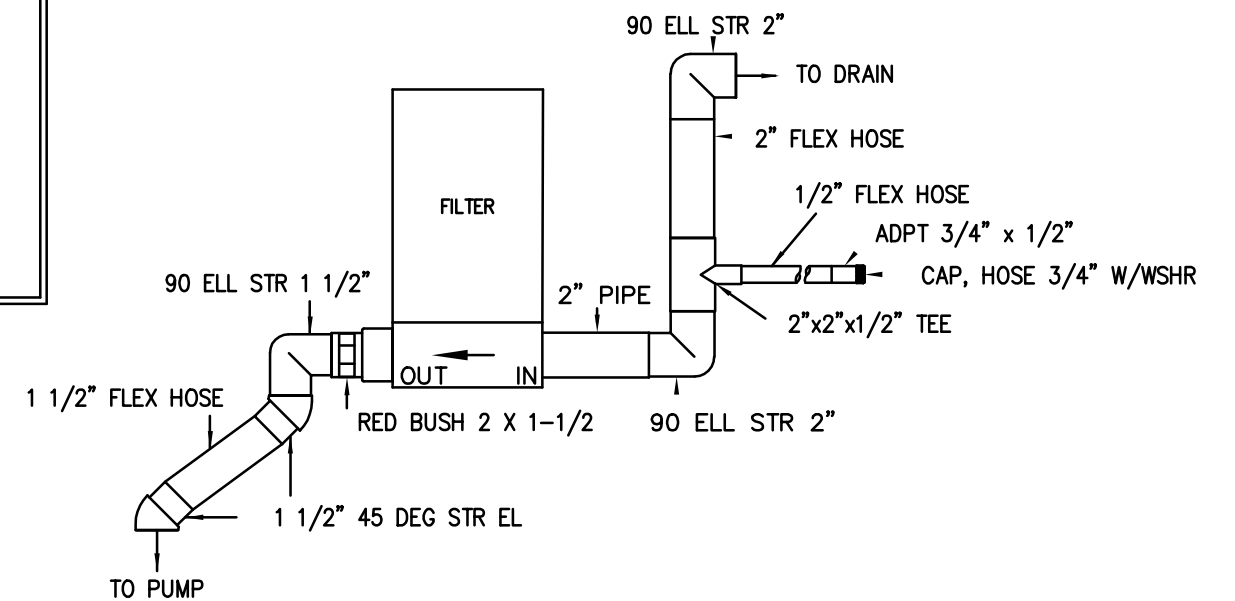




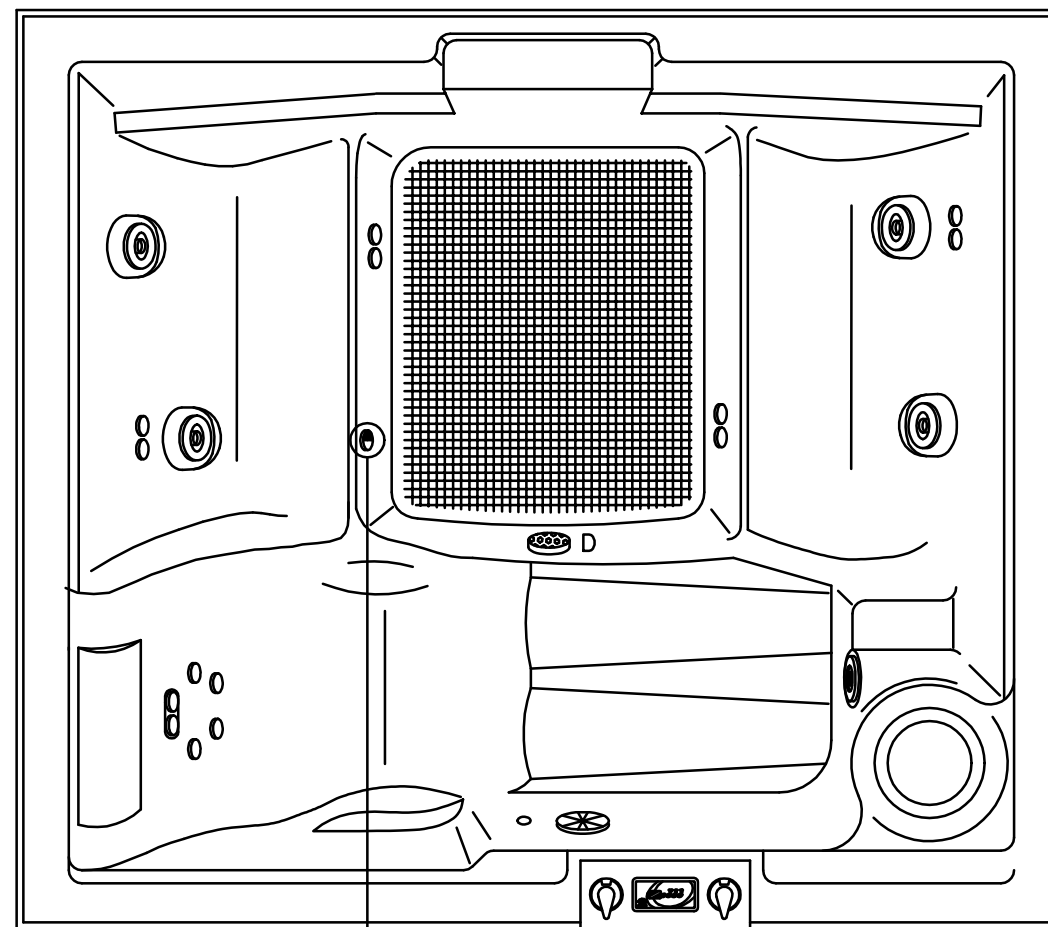
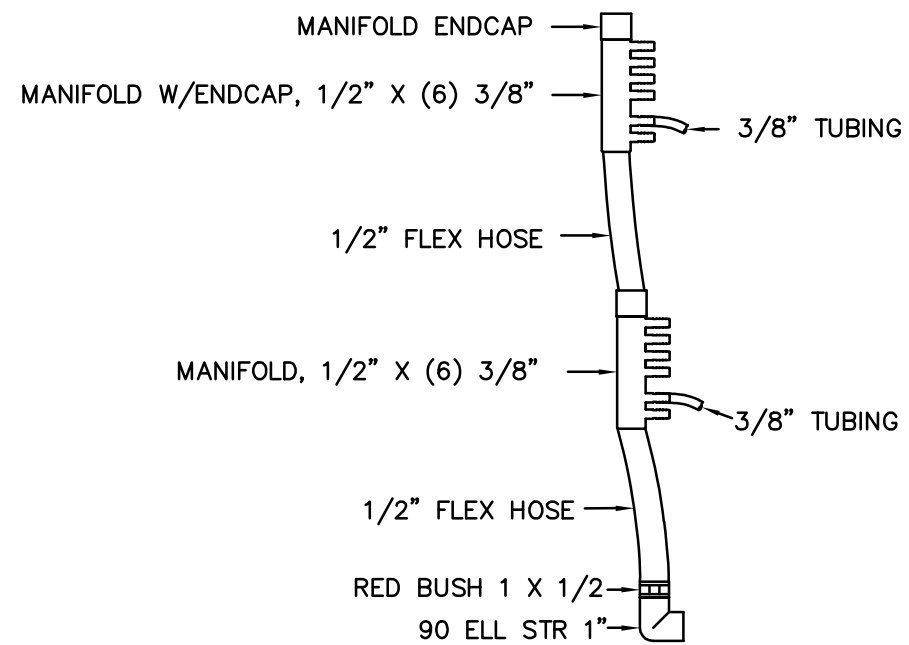
OZONE JET



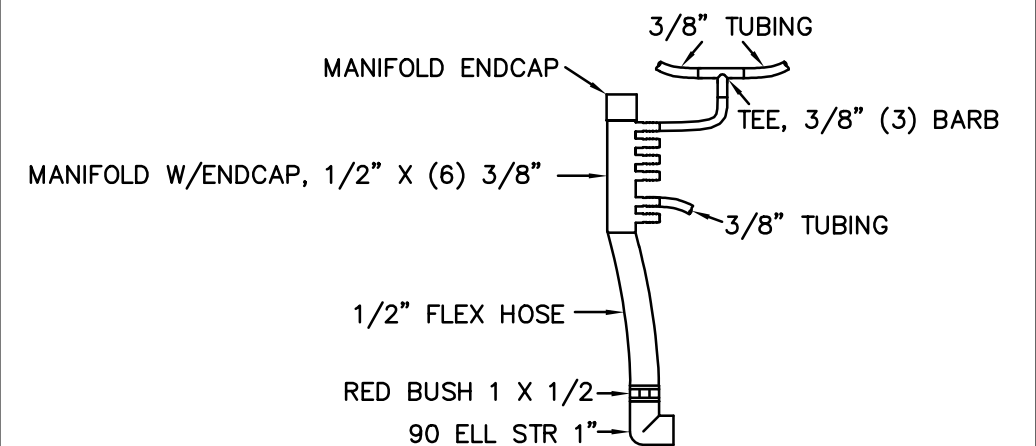
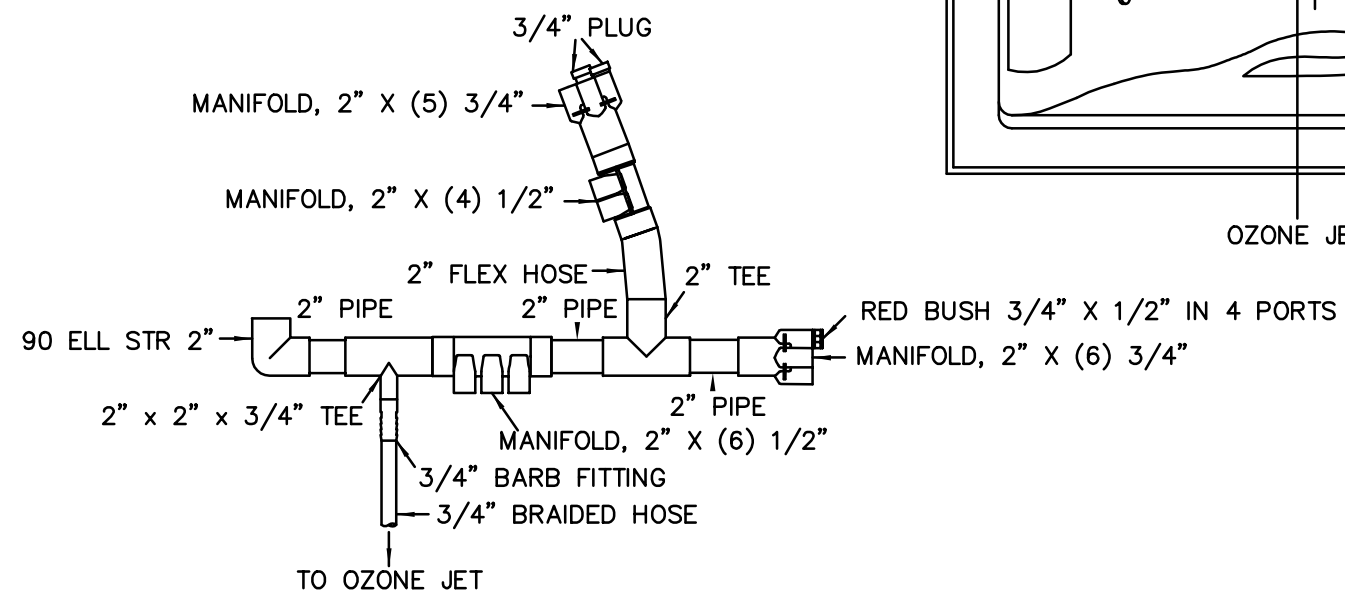
D - DRAIN/SUCTION



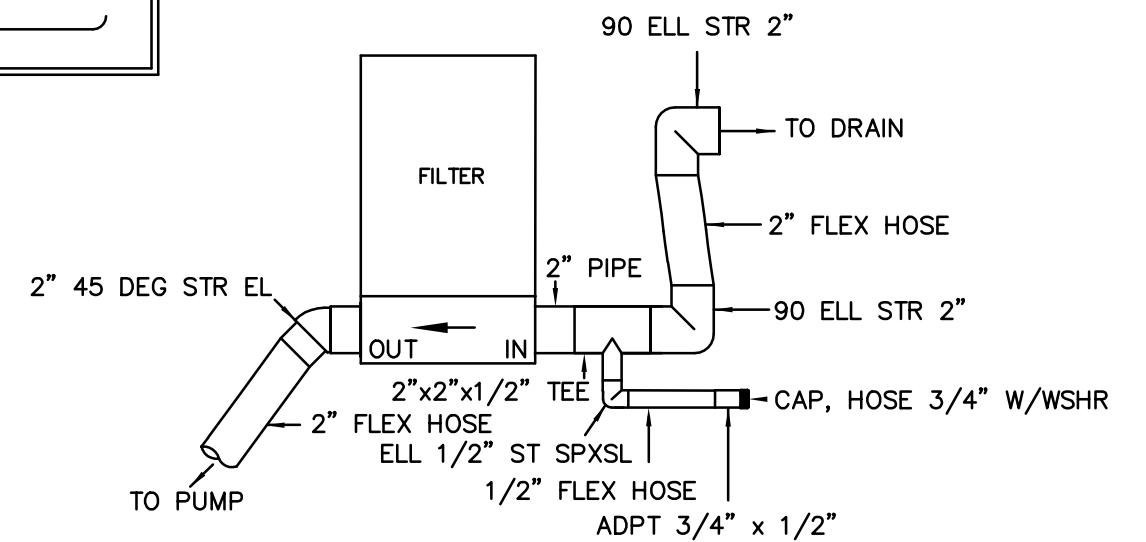
PLUMBING, OASIS YR 2003 - CURR: OAI11E1001 - CURR



OZONE JET



D- DRAIN/SUCTION



PLUMBING, OASIS II YR 2002: OAI21D1001 - CURR



Trouble Shooting
Guide is under
development and will
be released at a later
date



Service Bulletin

DEALER MEMO

TO: All Oasis Service Centers
SUBJECT: Service Bulletins/Fast
Facts/Service Updates

DATE: July 1, 2001
FROM: Technical Service

The Service Bulletins/Fast Facts/Service Updates is used as a vital communication line between the Technical Service department and the service technician in the field. It serves as a means to update the service manual with the latest information regarding retrofits, repair procedure, recalls, and/or the availability of special service tools.

The Service Bulletins/Fast Facts/Service Updates is distributed to the service technician via the monthly dealer mailing. Each month the Technical Writer distributes a memo detailing the various deadlines associated with the monthly dealer mailing.

Determine the Service Bulletins/Fast Facts/Service Updates number as indicated (as indicated by XXXX). Bulletins are organized by numerical code, 1000-9000 series by content as follows. The following is a list of the categories for the Service Bulletins:

Service Bulletin No.	Category
1000	Surface
3000	General Maintenance
5000	Electrical
7000	Plumbing
9000	Accessories/Miscellaneous

If you have any questions, do not hesitate to contact the Technical Support Department at (800) 669-1881 @ extension 115.



Fast Fax/Service Update/Service Bulletin Index

Section 1000: Surface

Document Title

*Doc Ref.
(Date/SB No.)*

Section 3000: General Maintenance

Document Title

*Doc Ref.
(Date/SB No.)*

Section 5000: Electrical

Document Title

*Doc Ref.
(Date/SB No.)*

Section 7000: Plumbing

Document Title

*Doc Ref.
(Date/SB No.)*

Section 9000: Accessories/Miscellaneous

Document Title

*Doc Ref.
(Date/SB No.)*



Section 1000: Surface

Document Title

Doc Ref.
(Date/SB No.)



Section 3000: General Maintenance

Document Title

Doc Ref.
(Date/SB No.)



Section 5000: Electrical

Document Title

Doc Ref.
(Date/SB No.)



Section 7000: Plumbing

Document Title

Doc Ref.
(Date/SB No.)



Section 9000: Accessories/Miscellaneous

Document Title

Doc Ref.
(Date/SB No.)



Section 1 : Ground Fault Circuit Interrupter

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PEAK LOAD CHECK	11

GFCIs/GFCI BREAKERS

PART I: CURRENT LEAKAGE

Ground Fault Circuit Interrupters (GFCI) serve an important purpose on OASIS Spas or any electrical circuit. They provide protection against current leakage to ground, which can cause injury or, in extreme cases, death. GFCI breakers, along with providing current leakage to ground protection, protect the spa's equipment and components from overloads and short circuits (excessive current draw). This is the reason GFCIs must always be installed properly and never be removed or altered. 115-volt model spas are protected by a Class A GFCI, which is included with the spas; 230-volt (optional model spas must have GFCI breakers, which are included in a subpanel. Both the Class A GFCI and GFCI breakers must be installed by a qualified electrician. It is important that ALL installation instructions be followed.

Monthly testing must be stressed to spa owners, both for your and their protection. When a GFCI trips, it normally indicates current leakage to ground caused by faulty insulation, wet wiring or wet equipment connected to the circuit. Tripping of GFCI breakers may also indicate an excessive current draw on the circuit.

There is no known device that can offer complete protection against the hazards of an electrical accident under all conceivable conditions. For example, the GFCI does not protect a person who comes in contact with both hot wires on 230-volt models, or in contact with either one of the hot wires and the neutral wire on 115-volt models. There is also no protection from voltage introduced from an exterior source (radio, hair dryer, etc.).

Even with a GFCI installed, an electrical shock may be felt, but such a shock will be less than what is normally considered dangerous. However, individuals with acute heart problems or other conditions that may make someone susceptible to electrical shock may still be seriously injured. Remember, electricity, when handled carelessly or misused, is dangerous.

DIAGNOSIS AND TESTING

TESTING GFCIS

To test a GFCI, push the test button. The GFCI should trip (current will not flow). Then push the reset button. This should activate the GFCI, which will activate that particular circuit. If either of these tests fail, replace the GFCI.

CAUTION: A GFCI breaker provides protection only to the circuit to which it is connected. It does NOT provide protection to other circuits.

MEASURING FOR CURRENT LEAKAGE

IMPORTANT: Always perform a current leakage test when experiencing GFCI tripping.

IMPORTANT: Be sure that any additional ground or bond wires are disconnected. If there are two paths to ground the current leakage. Will be split, which will give inaccurate readings.

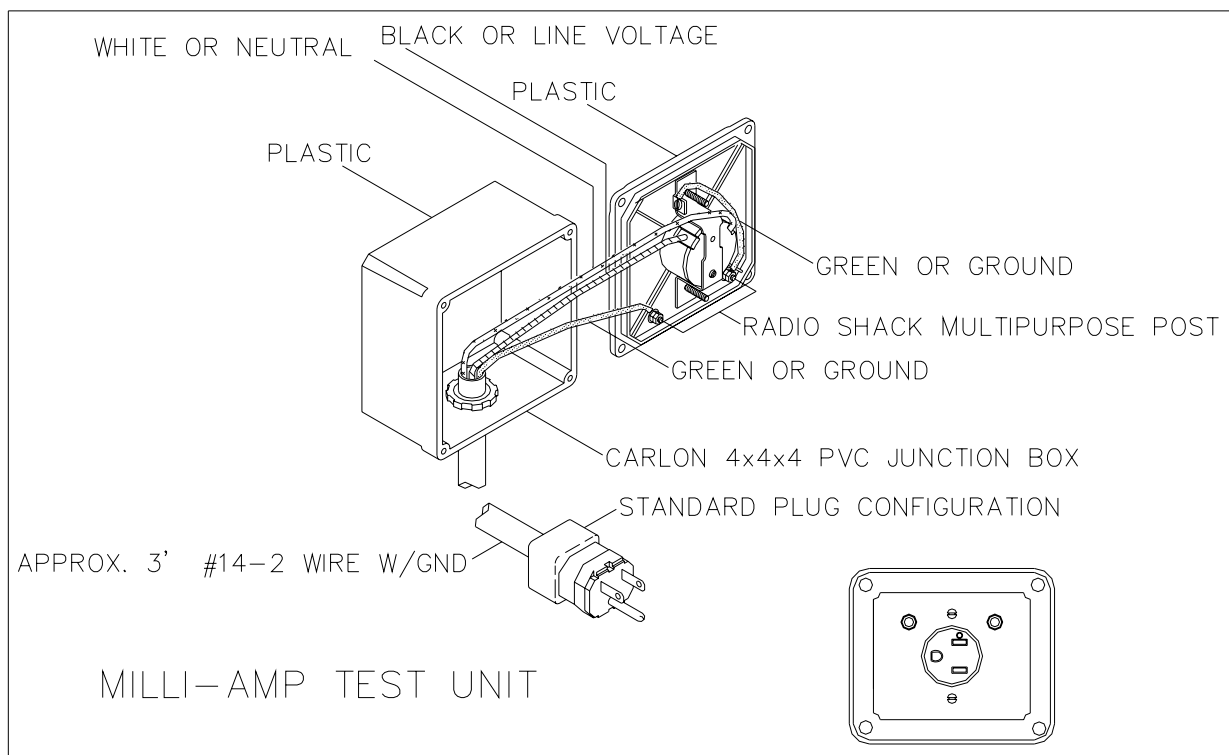
Note: GFCIs only test the load side of the circuit they are attached to.

These instructions are for use a meter that is capable of reading AC milliamps.

Refer to the following instructions to test for current leakage:

1. Set the volt/ohm meter to the AC milliamp function (20m on the meter).
2. Plug the black test lead into the "COM" receptacle on the meter. Plug the red test lead into the appropriate mA receptacle on the meter.
3. To test for AC milliamp leakage, break the ground connection of the circuit being tested and use the test leads with alligator clips to complete the circuit. On a 230-volt spa, testing is done at the subpanel grounding bar. On a 115-volt model spa, testing is done in the control box on the main grounding lug.

Note: To simplify testing current leakage on 115-volt spa models, build the test box illustrated below.



Determine current leakage from each of the spa's components (heater, jet pump, lamp and circulating pump) by isolating the individual components. Do this by operating each individual component alone (nothing else in the spa in operation). Check the meter for milliamp reading.

Any component registering more than .5 milliamps (mA) of current leakage is considered defective and should be replaced.

GFCI SPECIFICATIONS

Breakers

- Trips at 5 mA (+/- 1 mA) and within 1/4 of the sine wave or 8 milliseconds.
- Determines maximum current draw of a circuit, for example 15, 20 or 30 amps.

All Others

- All Class A GFCIs trip at 5 mA (+/- 1 mA) and within 1/4 of the sine wave or 8 milliseconds.

There is an industrial-grade GFCI available that is intended to protect equipment only. The trip point is 30 mA, which is beyond the safe level to protect people.

GFCI INSTALLATION

Receptacles

A receptacle is wired in series with the circuit it is protecting. The GFCI is generally marked "line or source" for the incoming wire and "load" for the wires going to the circuit. Both black and white wires, along with the ground wire need to be connected.

Note: Proper GFCI installation is imperative for the spa owner's safety and spa's protection.

Breakers

All wires of the circuit (hot and neutral wires for a 115-volt circuit; both hot wires for a 230-volt circuit) are to be attached to the breaker. The breaker is generally marked "load power" on the black wire(s) (power) and "load neutral" on the white wire (neutral).

Note: Failure to connect the neutral wire to the GFCI breaker will cause the breaker to trip immediately when power is applied.

230 Breakers - Additionally, it is important to connect the "pigtail" wire (white) on the breaker to the neutral buss bar in the electrical subpanel. This wire acts as a path to ground for the neutral wire in a 230-volt circuit and as a path for the trip circuit on the breaker.

Note: On 230-volt breakers if the pigtail is not installed to the neutral buss bar, the breaker will not manually test, but it will protect the circuit.

GFCI OPERATION

A GFCI measures the differential of current in an electrical circuit. When the differential of current across the load is greater than the trip value of the GFCI (5 mA, +/- 1 mA), the GFCI will trip. When tripped, the GFCI internally opens a set of switches, which will not allow current to flow through the incoming wires of the circuit. This is particularly important in a 115-volt circuit where reversed polarity may exist due to the circuit being improperly wired.

It is also important to understand that a GFCI will only test the load side of the attached circuit. It will not test the source side connected to it.

Note: A spa being protected by a GFCI must have a path to ground for the GFCI to operate. Complaints of spa owners receiving a low current shock is usually the sign of improperly grounded piece of equipment.

COMMON REASONS FOR GFCI TRIPPING

- Excessive current leakage from an electrical part.
- Moisture on electrical component(s).
- Incorrect wire installation to the GFCI.

OTHER MINOR REASONS FOR GFCI TRIPPING

- Low voltage. Some designs are more immune to this than others since the internal tripping circuit is not mechanically held.
- Radio frequency (RF).
 - A. Radiated RF (traveling through air)
 - B. Conducted RF (traveling through wires), such as that generated by chattering relays and contactors.
 - Excessive voltage spikes.
 - Scraping off wire insulation when installing wire into conduit.
 - Lightning strikes and high-energy utility switch noises.

G.F.C.I. TROUBLESHOOTING

IMPORTANT! Keep in mind that a majority of G.F.C.I. tripping problems can be attributed to incorrect wiring. G.F.C.I. troubleshooting usually finds the problem.

If Correct Wiring Is Verified

- Check to see if the proper G.F.C.I. is installed.
- Check the label in the system box near TB1 to determine the maximum amperage draw for the system.
- Be sure the G.F.C.I. is rated for more amperage than the system will draw.
- For a 230 V dedicated system, a 2-pole G.F.C.I. with a neutral.
- For a 120/230 V system, the G.F.C.I. must include a load neutral out.
- If the load neutral wire is routed from the sub-panel neutral bar directly to TB1 in the system box, then the G.F.C.I. will trip when a 120 V device is activated.
- If the wiring is correct and the G.F.C.I. will not reset, then unplug the pump and try to reset the G.F.C.I.
- If the G.F.C.I. trips again, then unplug the blower and push the reset button. If the G.F.C.I. continues to trip, then do the same procedure for the ozone generator.
- If the G.F.C.I. stops tripping after you unplugged one of the spa's components, turn off the power to the spa then plug in each component except the one that tripped the G.F.C.I.
- Power up the system. If the G.F.C.I. no longer trips, then you have correctly identified the problem.
- Repair or replace the component as instructed by the spa manufacturer.
- If you have unplugged all of the spa's components and the G.F.C.I. still doesn't reset then the problem is most likely a ground fault in the heater.

To Disconnect the Heater

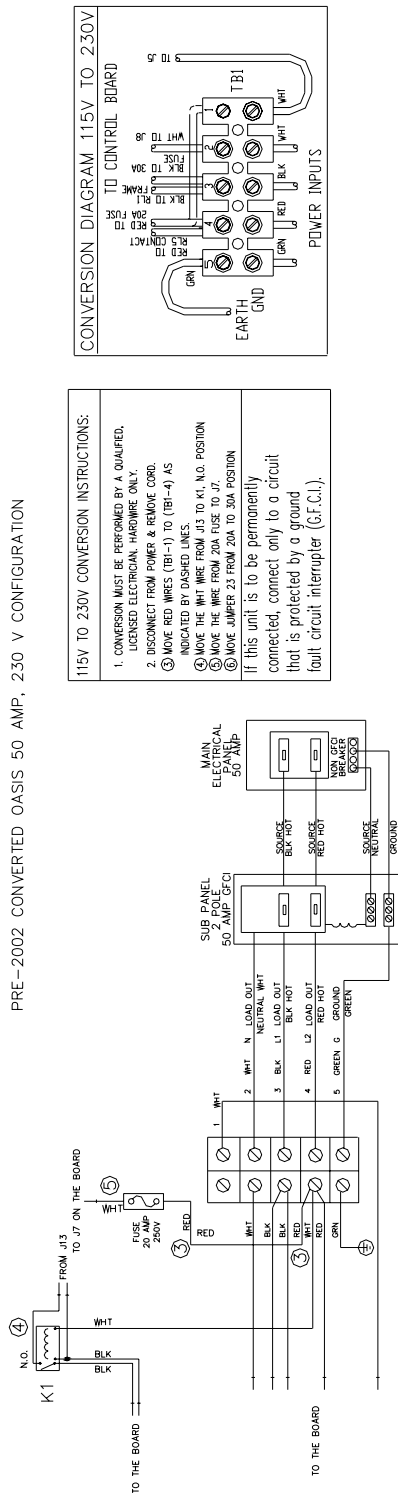
- First, turn off the main circuit breaker,
- For Pre-2002 Oasis, remove either heater straps or wires from the terminal block, not the heater itself. For 2002 Oasis, disconnect the green ground wire from the heater.
- After restoring the power, try to reset the G.F.C.I., again. If it no longer trips after the system calls for heat, then replace the heater.
- If the G.F.C.I. still trips, look for pinched or shorted wires at the transformer. Make sure that the screws that attach the transformer to the system box have not pinched or damaged the insulation of the transformer wires.
- If the transformer wires are undamaged, check for any other pinched wires. Refer to the wiring diagram to verify the correct wiring of the control system.
- If everything looks to be in perfect working order, then the G.F.C.I. may be defective.

IMPORTANT: More than one GFCI in series will not cause the GFCI to trip. A GFCI measures the differential of current across each side of an electrical circuit and in no way could contribute to the tripping of another GFCI.

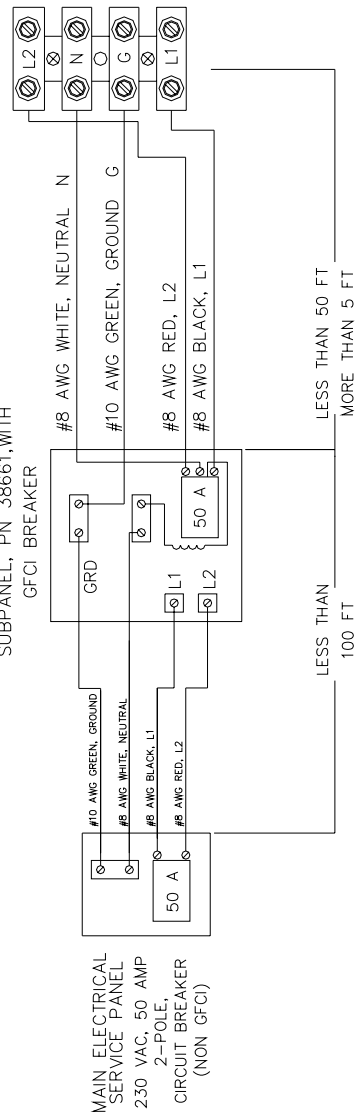
For example, if there is a GFCI breaker in the service panel and a GFCI cord connection on the spa it is possible that the GFCI breaker will trip first should a component develop current leakage. Keep in mind GFCI specifications - five (5) milliamps (+/- one (1) milliamp). Should this happen it is probably because the GFCI breaker is calibrated lower than the cord GFCI and not due to the proximity of the two GFCIs.

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PRE-2002 CONVERTED OASIS 50 AMP, 230 V CONFIGURATION

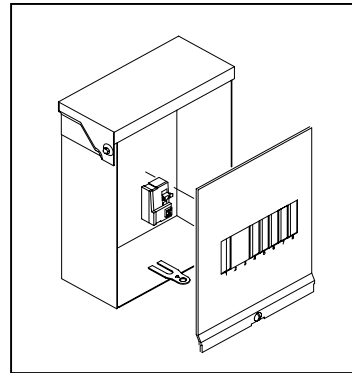


YR2002 – CURRENT CONVERTED OASIS 50 AMP, 230 V CONFIGURATION
SUBPANEL, PN 38661, WITH



WIRING CHECK FOR G.F.C.I./SERVICE DISCONNECT

The National Electrical Code states that a service disconnect breaker box (a G.F.C.I. can be used for this purpose) must be located at least five feet away from the spa and should be conveniently located near the equipment bay. If it is not in plain sight, keep the disconnect padlocked when in the off position.



Precautions - Effective January 1994, G.F.C.I.'s are required for spa installations. Prior to that, G.F.C.I.'s had been recommended for spa installations, but were not mandatory. If the spa you are servicing was not installed with a G.F.C.I., strongly urge your customer to improve safety and comply with current standards by installing one.

IMPORTANT! Remember, high voltage is still accessible in the house breaker box even though you have turned off the spa breaker.

PART II: VOLTAGE TESTING

G.F.C.I. LINE-IN WIRING CHECK

Locate the proper circuit breaker and turn it off.

Remove the cover from the house breaker box.

Check the main service amperage rating to the breaker box.

Note: Typically, a house circuit will require at least a 100 Amp service when a spa is installed.

From the circuit breaker, locate the red load wire and the black load wire.

From the neutral bar the green ground wire.

Be sure there are no other appliances on the spa circuit.

If there are, service must be re-wired to supply the spa only.

Make sure all four wires exit the house breaker box via conduit, routed to the G.F.C.I. breaker box.

The black should be connected to the G.F.C.I. line-in 1. The red should be connected to the line-in 2.

The white load neutral connects to the neutral bar. The green ground wire should be attached securely to the ground lug inside the G.F.C.I. box.

G.F.C.I. LINE-OUT WIRING CHECK

If the spa is equipped with a 115/230 V control system, an additional white load neutral wire must connect to the load neutral out. This wire runs the others to the system box. Proper placement of this neutral wire is essential. If miswired, the G.F.C.I. will trip when a 115 V device is activated.

Once you have found all wiring correctly installed, begin to check for proper voltage.

VOLTAGE CHECKS - BREAKER BOX/G.F.C.I.

When checking for proper voltage, please keep in mind that the acceptable voltage range is $\pm 10\%$ of the recommended voltage. Acceptable voltage when 115 V is specified as the desired voltage, is between 103.5 and 126.5 V. Acceptable voltage when 230 V is specified as the desired voltage, is between 207 and 253 V.

IMPORTANT: This service must be single phase. Any abnormal voltage reading requires an electrician. Do not attempt to fix problems yourself. High voltage can seriously injure or kill.

BREAKER BOX VOLTAGE CHECK

Set your multi-meter or voltmeter for AC Volts.

Make sure the G.F.C.I. is off.

Carefully turn on the spa circuit breaker at the sub-panel.

At the house breaker box, probe the spa circuit breaker between the black and red wires. Your meter should read 230 V.

Probe the black and green ground wires. You should see 115 V.

Probe between the red and green ground wires. You should also see 115 V.

The voltage between the white load neutral and the green ground wire. You should see 0 volts.

G.F.C.I. LINE-IN VOLTAGE CHECK

230 V Dedicated System:

- Be sure the spa circuit breaker (located in the house breaker box) is on.
- Make sure the G.F.C.I. is off.
- Probe the black and red wires. The meter should read 230 V.
- Probe the black and green ground wires. The meter should read 115 V.
- Probe the red and green ground wires. This should also read 115 V.
- The voltage between the white load neutral and the G.F.C.I. box ground lug should read 0 V.
- Turn on the G.F.C.I. breaker before continuing to the system box.

115/230 V Convertible System:

- Be sure the spa circuit breaker (located in the house breaker box) is on.
- Make sure the G.F.C.I. is off.
- Probe the black and red wires. The voltage should be 230 V.
- Probe the black and white load neutral wires. The voltage should read 115 V.
- Probe the red and the white load neutral wires. The voltage should read 115 V.
- Turn on the G.F.C.I. breaker before continuing to the system box.

VOLTAGE CHECKS - SYSTEM BOX

G.F.C.I. LOAD OUT VOLTAGE CHECK

230 V Dedicated System:

- Be sure that the house breaker is on.
- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires at G.F.C.I. load out L1 and L2. The voltage should be 230 V.
- Probe the black wire and the G.F.C.I. neutral bar. The meter should read 115 V.

- Probe the red wire and G.F.C.I. neutral bar for 115 V.
- Probe the black load out wire and the box chassis ground lug. Your meter should read 115 V.
- Probe the red load out wire and the ground lug. The voltage should read 115 V.
- Recheck voltage under peak load conditions.*

115/230 V Convertible System Check:

- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires at G.F.C.I. load out L1 and L2. Look for 230 V.
- Probe the black and white load neutral wires. The meter should read 230 V.
- Probe the red load out and box chassis ground lug for 115 V.
- Probe the black load out wire and ground lug for 115 V.
- Recheck voltage under peak load conditions.*

*Peak Load Check - It is important to check the voltage again under peak load conditions. To reach peak load, turn on the blower, heater, lights and all pumps.

SYSTEM BOX CHECK (AT TERMINAL BLOCK)

230 V Dedicated System Check:

- Be sure the G.F.C.I. breaker is on.
- Probe the black and red wires. The voltage should be 230 V.
- Probe the black wire and green ground wire. The meter should read 115 V.
- Probe the red wire and green ground wire. The voltage should read 115 V.
- Probe the white neutral wire and the green ground wire. This should be 0V.
- Recheck voltage under peak load conditions.*

115/230 V Convertible System Check:

- Be sure the G.F.C.I. breaker is on.
- Begin the voltage check at TB1.
- Probe the black and red wires. Your meter should read 230 V.
- Probe the black and white wires. Look for 115 V.
- Probe the black and green ground wires also 115 V.
- Probe the red and green ground wires again 115 V.
- Probe the red and white load neutral wires. This should read 115 V.
- Recheck voltage under peak load conditions.*

*Peak Load Check - It is important to check the voltage again under peak load conditions. To reach peak load, turn on the blower, heater, lights and all pumps.

PEAK LOAD CHECK

Peak Load Check for 230 V System:

- Check the voltage between the black and red wires. The acceptable voltage range is between 216 and 264 V. Probe the black and white neutral wires. This voltage must be between 108 and 132 V.

Peak Load Check for 115/230 V Convertible System:

- Check the voltage between the black and red wires. Acceptable range is between 216 and 264 V.
- Probe the black and the white load neutral wires. The voltage must be between 108 and 132 V.
- Probe the red and the white load neutral wires. The voltage must be between 108 and 132 V.

IMPORTANT - If the voltage is not within the acceptable range, call an electrician or the local electric company to diagnose the problem.

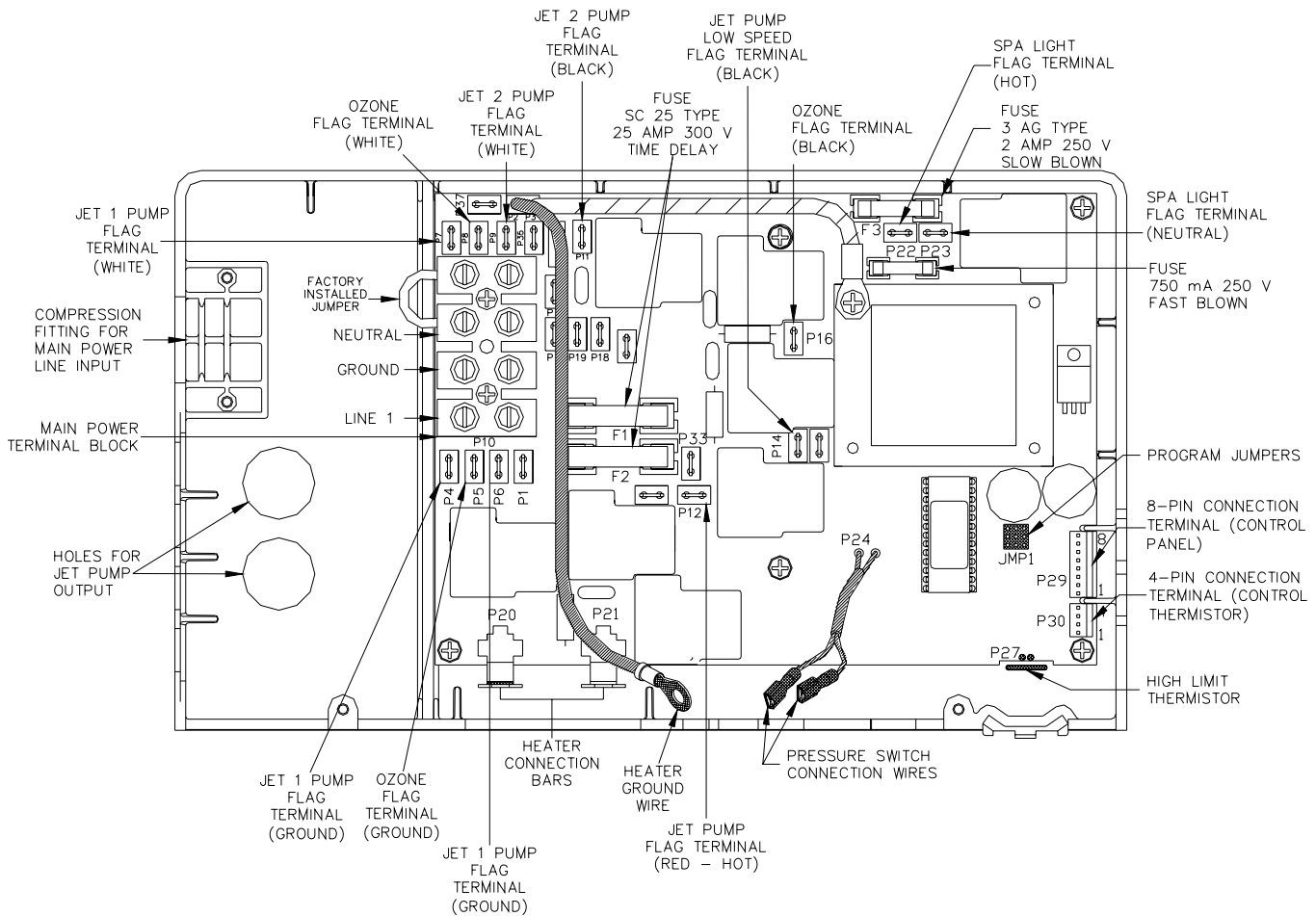


Section 2: Spa Control System

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Section 2: Spa Control System CONTROL SYSTEM – OASIS 2002 – CURRENT



Compression fitting - Main Power Line – secures the main power line to the control box.

Output Holes – Jet Pump – where the jet pump power cord exits the control box.

Terminal Block – receives the main power line and passes source voltage on to the circuit board.

Heater Connection Bars (2) – supplies power to the heater.

Heater Ground Wire – attaches to the grounding terminal on the heater.

Pressure Switch Connection Wires (2) – attaches to the pressure switch.

High Limit Thermistor – measures the temperature of the heater.

Control Thermistor 4-pin Connection Terminal – seats the control thermistor 4-pin connection wire.

Control Panel 8-pin Connection Terminal – seats the control panel 8-pin connection wire.

Program Jumpers (3) – control the spa settings.

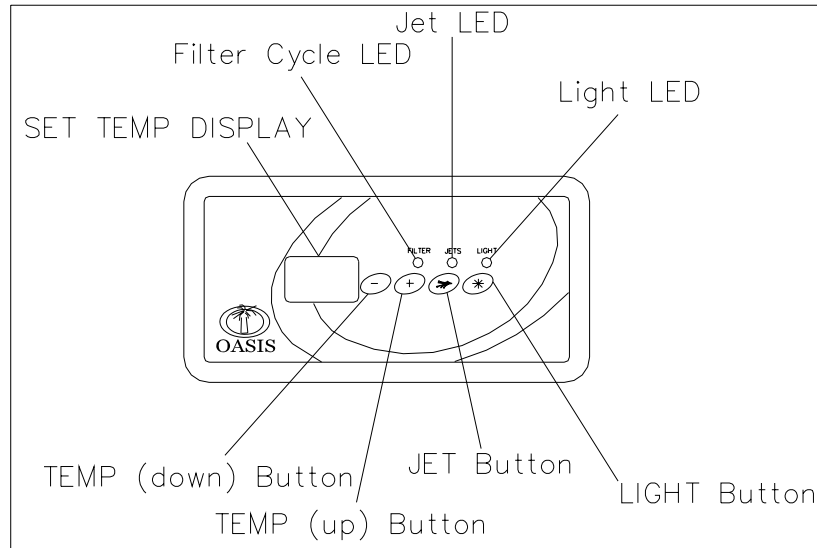
Fuses (4) – two are replaceable, and two are not.

Flag Terminals/Pin Connectors(12) – supply power to components such as the spa light and jet pump.

Section 2: Spa Control System

OASIS CONTROL PANEL – 2002 – CURRENT

The control panel is a complete assembly that contains the control head (electronics portion) and control panel (touch pads, display window, and indicators) as one unit.



SET TEMP DISPLAY – displays the spa’s actual water temperature. In certain situations, it will show error messages.

TEMP (up) Button - raises the thermostat setting (the maximum temperature setting is 104° F / 40° C). **NOTE: Factory setting of the spa is 102° F/ 38° C**

TEMP (down) Button - lowers the thermostat setting (the minimum temperature setting is 80° F / 27° C). **NOTE: Factory setting of the spa is 102° F/ 38° C**

JET Button – Activates and deactivates Jet Pump.

Light Button – Activates and deactivates the lighting.

Filter Cycle LED – Activates when the filter cycle is running.

Jets LED – Activates when the jet pumps are in use.

Light LED – Activates when the light is in use.

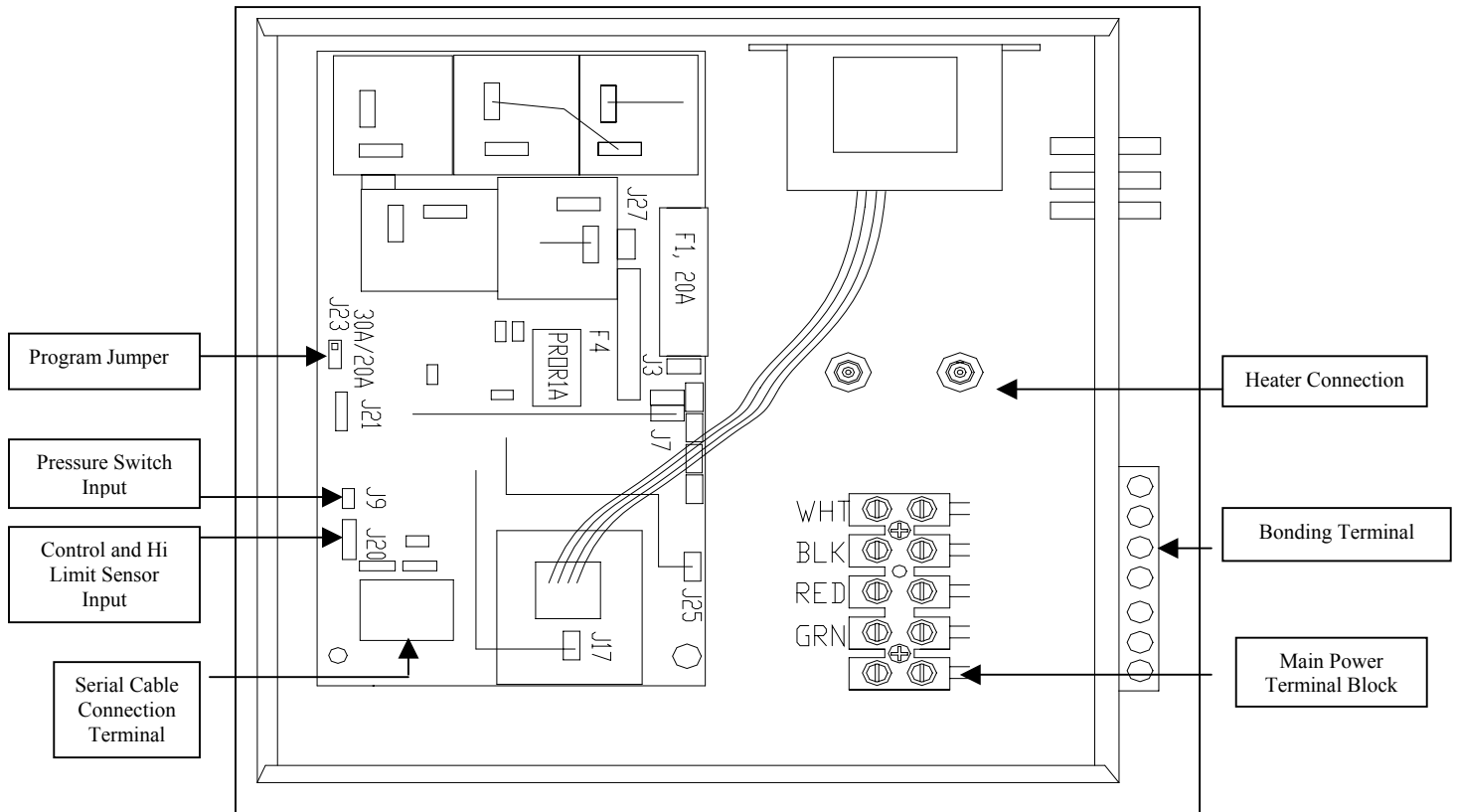
Section 2: Spa Control System

CONTROL SYSTEM

PRE-2002

The PROP1A control panel, used in the OASIS spa, allows the spa user to regulate and view the status of the spa functions. These functions are regulated by the topside control panel and include the temperature control, light, hydro and air jets. The topside control panel also displays the set temperature. The PROP1A topside control panel consists of two items: the digital display and control touch pads. It is connected to the control system, via a serial cable.

There is one control box, PROP1A, used for both series:



The control box is designed to control the spa functions (jets, light, set temperature) and monitor the status of the control system. Since the same box can be used in different models with different functions, the Program Jumpers must be set to “program” the control box for the spa model it is installed in. Additionally, the control box is installed for a 115 VAC model.

- **Serial Cable Connection Terminal (Topside Control Panel)** – seats the control panel’s serial cable.
- **Program Jumpers** – control the spa functions and settings (see *Jumper Configuration*).
- **Input (Control Sensor)** – receives the control sensor connection wire.
- **Input (High Limit Sensor)** – receives the high limit sensor connection wire.
- **Main Power Terminal Block** – receives the main power line and passes source voltage on to the circuit board and heater relays.
- **Power Connection** – supply power to the spa light, jet pump, and ozonator.
- **Heater Connector** – attaches to the via the hold straps.
- **Bonding Terminal** – connects all of the solid copper bonding wires to the main ground line.

All metal permanently placed near the spa must be bonded to the grounding lug as described in the Pre-Delivery instructions and the Homeowners Manual. Please refer to these documents for more complete information.

Power Connections For The Equipment Compartment Components

The jet pump and blower are wired via flag terminals to the appropriate relays.

Terminal Block (TB-1)

The terminal block (TB-1) receives the source voltage from the 115 VAC dedicated source and delivers voltage to the various circuits and electronic components within the control box.

Grounding (bonding) Lug

The minimum size bonding wire is #8 AWG (solid copper). The ozonator (if installed) must be bonded to the control box grounding lug.

Heater Ground

The heater is grounded to the box via the holding straps.

Program Jumpers

The setting of the program jumpers determines the operation of the spa functions. When replacing a control box, the program jumpers must be set for the specific model, according to the label on the control box cover or the instructions in this manual.

Serial Cable Input

The serial cable input is the interface between the topside control panel and the control box. The spa will be non-operational if the serial cable is broken or disconnected from the terminal.

Control Sensor Input

The control sensor is electronically connected to the control box circuitry at the J20 input terminal.

Control Sensor

The control sensor is an electronic component that senses the temperature of the spa water by changing its electrical resistance value (ohms, Ω). The control box compares the temperature reading (in ohms) of the control sensor to the setting of the thermostat, and heat is turned on or off.

Hi-Limit Sensor Input

The hi-limit sensor is electronically connected to the control box circuitry at the J20 input terminal.

Hi-Limit Sensor

The hi-limit sensor is an electronic component that senses the temperature of the spa water by changing its electrical resistance value (ohms, Ω). Within the safety features of the spa, the temperature reading (in ohms) of the hi-limit sensor is compared to the permanent value of the hi-limit (the hottest temperature which the spa can reach without shutting off).

If the spa water temperature exceeds the hi-limit value of 118° F, or the high limit sensor is faulty or unplugged, the high limit will trip and the spa functions will not work. “OH” is displayed as an indication that a high limit problem exists.

Jet Relay 1 / Jet Relay 2

There are two (2) relays available on the board for low and high jet pump operation.

Lamp Assembly

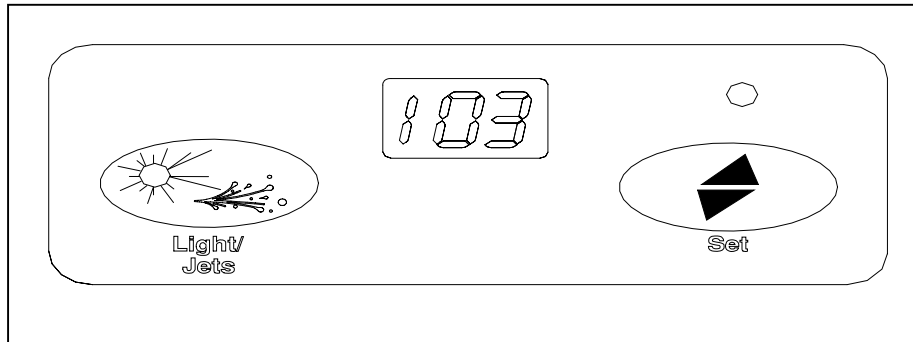
The lamp assembly consists of the wire harness, reflector, bulb light, and lens which is plugged into the control box via a four-pin connector that is wired to the main circuit board via the J5 and J20 terminal.

Section 2: Spa Control System TOPSIDE CONTROL PANEL PRE-2002

The topside control panel is a complete assembly that contains the control head (electronics portion) and control panel face (touch pads, display, and indicator) as one unit. The panels are used in conjunction with overlays. The control panel is attached to the spa with four screws.

Overlay

Topside Control



Light and Pump Touch Pad – activates and deactivates the spa light and the pumps.

Temperature Control Touch Pad – raises and lower the thermostat setting (the maximum temperature setting is 104° F / 40° C; the lowest temperature setting is 80° F / 27° C).

Main Display - shows the set temperature value and LED messages.

Serial cable - the serial cable is the connection between the topside control panel and the control box. The cable is removable and replaceable.



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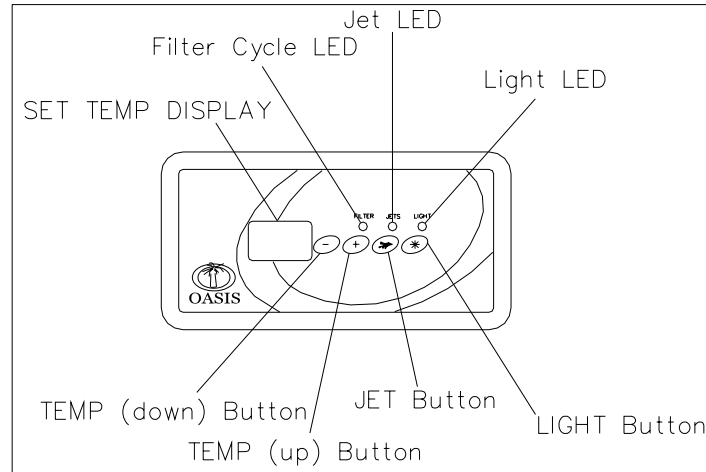
Section 3: Operation

Control Panel – Oasis

2002-CURRENT

Control Panel Display and Spa Functions

The control panel consists of an LED display and touch pads that allow the water temperature to be set, adjust the filter cycles, and to control the jets and the light. The LED displays the water temperature and certain trouble-shooting messages.



Light Control

Push the LIGHT button to turn the light on. Push it a second time to turn it off.

Jet Control

Pressing the Jet button changes the jet pump functions, depending on the number of presses:

- 1st press: Two-speed jet pump runs on low speed
- 2nd press: Two-speed jet pump runs on high speed
- 3rd press: Jet pump off

During filter cycle or when the thermostat is calling for heat, the pump's low speed operates automatically and cannot be turned off by pressing the Temp Plus (+) button or the Temp Minus (-) button. When the jets are active, the JETS LED on the control panel will illuminate.

The pump on high speed automatically turns off after operating for one hour. The pump on low speed automatically turns off after operating for one hour.

In the HEAT mode, the jet pump's low speed will run until the spa has reached the set temperature.

Temperature Control

The set temperature range is from 50° F to 104° F. The factory-set temperature is 102° F. Whenever power is disconnected and then reconnected to the spa, the water temperature will automatically seek this temperature, unless you set it otherwise using the temperature controls.

Pressing the Temp Plus (+) button displays the spa's set temperature. To increase the set temperature, press the Temp Plus (+) button again, once for each degree of increased temperature desired. If the button is held down, the set temperature will increase by one degree every half a second.

Pressing the Temp Minus (-) button display the spa's set temperature. To decrease the set temperature, press the Temp Minus (-) button again, once for each degree of decreased temperature desired. If the button is held down, the set temperature will decrease by one degree every half second.

Filter Cycle

When the spa is first powered up, the filter cycle will automatically start 12 hours later. A six hour filter cycle using the low speed of the jet pump is the programmed default setting. The filter cycle occurs once every 12 hours or twice per day.

To modify the length of the cycle:

- Press and hold the "LIGHT" button for approximately five seconds until the display shows "xx" where xx is a number from 0-12. This is the length of the filter time.
- Using the Temp Plus (+) or Temp Minus (-) button, adjust the value as desired.
- Pressing the LIGHT button again confirms the setting and starts a cycle immediately.
- If the LIGHT button is not pressed again, the new filter time is stored but the cycle is started at the next scheduled time.

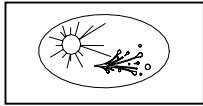
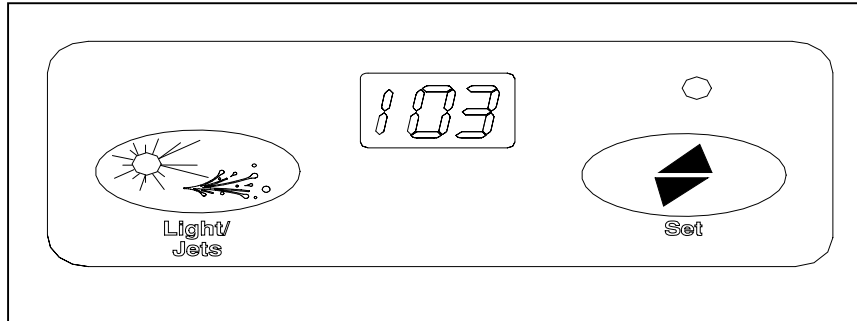
The filter cycle will activate the jet pump on low speed for the entire cycle. When activated, the FILTER CYCLE LED on the display is turned on. If the temperature exceeds the set point by three degrees F for more than three hours, the system will cancel the filter cycle and FILTER CYCLE LED will blink for the remaining filter cycle time. Should the temperature cool down to two degrees F above the set point before the end of the cycle, the cycle will resume for the time. Should the spa be used during the filter cycle by placing the jet pump on high speed, the FILTER CYCLE LED will blink. Once the jet pump has been shut off, a ten (10) minute tranquil mode begins (FILTER CYCLE LED will be blinking during the tranquil mode) before the filter cycle will resume. The blinking FILTER CYCLE LED indicates that the filter cycle timer is still counting but the cycle has been temporarily interrupted.

NOTE: The interrupted time of a filter cycle is not extended after the filter cycle resumes. Power interruptions of the spa will reset the filter cycle to the default six hour cycle time and begin 12 hours after the spa regains power.

Section 3: Operation Topside Control Panel Pre-2002

Topside Control Panel Display and Spa Functions

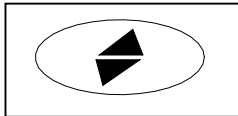
The topside control panel uses a display to show the temperature and other LED messages. When there is a demand for heat, the indicator will light up to show the call for heat.



LIGHT/JETS

Pressing the LIGHT/JET touch pad sequences the light and pump through the controls as follows:

- | | | |
|---|-----------------|-----|
| 1 | Light | On |
| 2 | Low Speed Pump | On |
| 3 | High Speed Pump | On |
| 4 | Light | Off |
| 5 | Pump | Off |



TEMPERATURE CONTROL

To adjust the temperature, press the SET touch pad again while the temperature is displayed. This will increase or decrease the temperature, depending on which direction was chosen last. Each successive press will change the set temperature by one degree in the same direction.

If the temperature is to be adjusted in the opposite direction, wait four (4) seconds to allow the display to revert to the actual water temperature. Press the SET touch pad again until to adjust in the desired directions.

MODE

Standard Mode – the heater and pump's low speed turn on as needed to reach and maintain the set temperature.

Economy Mode – the spa heats only during the filtration cycles.

To switch from the Standard mode to the Economy mode, press the SET touch pad then, while the temperature is displayed, press the LIGHT/JET touch pad. EC will flash alternately with the temperature signifying the Economy mode.



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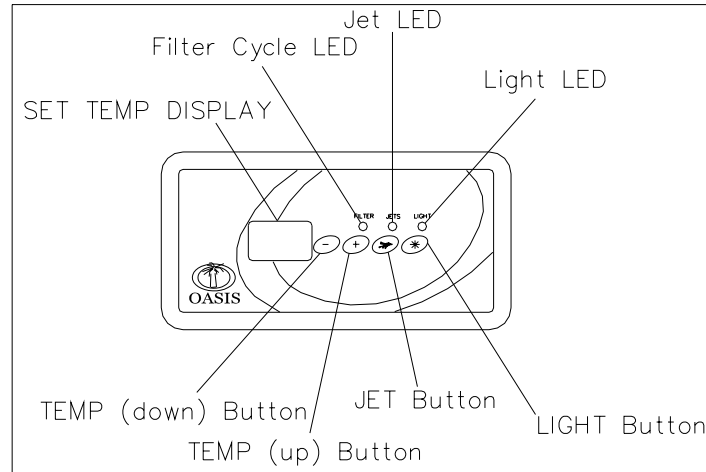
Section 3: Operation

Control Panel – Oasis

2002-CURRENT

Control Panel Display and Spa Functions

The control panel consists of an LED display and touch pads that allow the water temperature to be set, adjust the filter cycles, and to control the jets and the light. The LED displays the water temperature and certain trouble-shooting messages.



Light Control

Push the LIGHT button to turn the light on. Push it a second time to turn it off.

Jet Control

Pressing the Jet button changes the jet pump functions, depending on the number of presses:

- 1st press: Two-speed jet pump runs on low speed
- 2nd press: Two-speed jet pump runs on high speed
- 3rd press: Jet pump off

During filter cycle or when the thermostat is calling for heat, the pump's low speed operates automatically and cannot be turned off by pressing the Temp Plus (+) button or the Temp Minus (-) button. When the jets are active, the JETS LED on the control panel will illuminate.

The pump on high speed automatically turns off after operating for one hour. The pump on low speed automatically turns off after operating for one hour.

In the HEAT mode, the jet pump's low speed will run until the spa has reached the set temperature.

Temperature Control

The set temperature range is from 50° F to 104° F. The factory-set temperature is 102° F. Whenever power is disconnected and then reconnected to the spa, the water temperature will automatically seek this temperature, unless you set it otherwise using the temperature controls.

Pressing the Temp Plus (+) button displays the spa's set temperature. To increase the set temperature, press the Temp Plus (+) button again, once for each degree of increased temperature desired. If the button is held down, the set temperature will increase by one degree every half a second.

Pressing the Temp Minus (-) button display the spa's set temperature. To decrease the set temperature, press the Temp Minus (-) button again, once for each degree of decreased temperature desired. If the button is held down, the set temperature will decrease by one degree every half second.

Filter Cycle

When the spa is first powered up, the filter cycle will automatically start 12 hours later. A six hour filter cycle using the low speed of the jet pump is the programmed default setting. The filter cycle occurs once every 12 hours or twice per day.

To modify the length of the cycle:

- Press and hold the "LIGHT" button for approximately five seconds until the display shows "xx" where xx is a number from 0-12. This is the length of the filter time.
- Using the Temp Plus (+) or Temp Minus (-) button, adjust the value as desired.
- Pressing the LIGHT button again confirms the setting and starts a cycle immediately.
- If the LIGHT button is not pressed again, the new filter time is stored but the cycle is started at the next scheduled time.

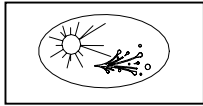
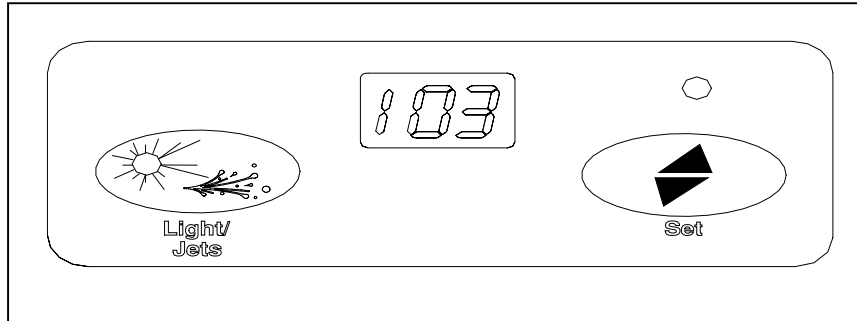
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Section 3: Operation Topside Control Panel Pre-2002

Topside Control Panel Display and Spa Functions

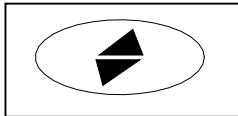
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Economy Mode – the spa heats only during the filtration cycles.

To switch from the Standard mode to the Economy mode, press the SET touch pad then, while the temperature is displayed, press the LIGHT/JET touch pad. EC will flash alternately with the temperature signifying the Economy mode.



Section 4: Troubleshooting

IMPORTANT TROUBLESHOOTING TIP

Before beginning a troubleshooting procedure, if the control system has never operated properly (either from the factory or after a repair), check the setting of the JUMPERS. If the JUMPERS are set improperly, the spa will malfunction.

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Section 4: Trouble Shooting

MAIN PANEL MESSAGES – 2002 – Current

Panel messages are a quick clue toward solving a variety of problems. Here are the most common messages and what they mean.

“●●●” - if the control panel displays three dots, there is a problem with the pressure switch or the hi limit thermostat.

MAIN PANEL MESSAGES – PRE-2002

Panel messages are a quick clue toward solving a variety of problems. Here are the most common messages and what they mean.

OH

The Panel Display is Flashing OH or the Status/Heat LED is Flashing (1 second on, 1 second off) - This means that the control system is at a high-limit condition and the spa is deactivated.

Note: Overheating may occur if the low-speed pump is set to operate for extended periods of time, or if the incorrect pump is installed.

The following is a list of most probable causes of this message. Inspect these first:

- Check ball valves. Make sure that they are open.
- Make sure the correct pump and pressure switch are installed.
- Clean the filter/skimmer if there is any blockage.
- In extremely hot weather, check for proper cabinet ventilation.
- Make sure the temperature sensor is fully inserted into the sensor mount and that the sensor mount is properly insulated with foam.

Note: Improper insulation is one of the most common causes of spa overheating. Aerosol foam insulation is available at most hardware stores.

- Check for excessive filter duration.
- Check the water level.
- Check the water temperature with an accurate temperature thermometer:
- Remove the spa cover and allow the water to cool to below 110 F.
- Adding cool water may be necessary.
- Touch any button to reset the system.
- If the Problem Reoccurs, Test the Sensor Set:

Check sensor wires for cracks or damage that may indicate the presence of a rodent.

Inspect the connections of both sensors on the circuit board. Verify that the plugs are clean by unplugging the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test. If the problem is solved, replace the sensor set. If the problem is not solved, do not replace the sensor set. Plug in the original sensor set to verify that there is not a connection problem. If the problem continues after following the above steps, then replace the circuit board.



The Panel Display Shows Sn1

When this is shown it means that the high-limit or water temperature sensor has malfunctioned.

For the High-limit sensor:

- Check sensor wires for cracks or damage that may indicate the presence of a rodent.
- Inspect the connections of both sensors on the circuit board. Verify that the plugs are clean by unplugging the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test.
- If the problem is solved, replace the sensor set.
- If the problem is not solved, do not replace the sensor set.
- Plug in the original sensor set to verify that there is not a connection problem.
- If the problem continues after following the above steps, then replace the circuit board.

For the water temperature sensor:

- Unplug the sensor set (at the temp/high-limit jumper, J-20) and plug in the test sensor set. Hang the test sensors over the spa side and into the water. Keep in mind there is no high-limit sensor protecting the heater during this test.
- If the problem is solved, replace the sensor set. If the problem is not solved, do not replace the sensor set.
- Plug in the original sensor set to verify that there is not a connection problem.
- If the problem continues after following the above steps, then replace the circuit board



The Display Continually Shows FL

When this is happening, the software is detecting no pressure at the pressure switch with the pump. This may indicate a flow restriction, such as a dirty filter. Another cause of the flashing FL message could be a faulty or malfunctioned pressure switch

- Check the water level.
- Remove the filter cartridge, then run the spa. If the message disappears, then clean or replace the filter cartridge.
- Make sure the ball valves are open. If any valves are closed, opening them may remove the error message from the display and the spa should operate normally.
- Make sure the pump motor is functioning properly. If not, use a multi-meter to verify system box output to the pump.
- Check the floor/suction grills for obstructions.
- Check the pump to be sure it is primed and pumping properly.
- Check the pressure switch wire for cuts.
- Inspect the connection on the circuit board. Verify that the plug is clean by unplugging it (at jumper, J-9).

- If the spa does not operate normally, disconnect the wires from the pressure switch.
Disconnect the pressure switch from the circuit board at J-9
Set your voltmeter to the continuity setting.
Place the meter leads onto the two wires of the pressure switch.
Turn on the low-speed pump.
The meter should register continuity.
Turn the pump on from low-speed to high-speed if available and run for ten seconds.
After ten seconds has passed, turn off all pumps.
After four seconds, check the voltmeter and verify that there is no continuity registering.

Note: The most common pressure switch failure is poor water chemistry. Instruct the homeowner in proper water maintenance.

- If the spa operates normally, replace the switch.
- Clean the area where the pressure switch is located with a metal brush. Use silicone tape or sealant when installing a new pressure switch.
- If the problem continues, then replace the control.

MISCELLANEOUS MESSAGES

The Panel Display shows Unexplained Error Messages

Ec on - In Economy mode => heater will only operate during filtration cycles. Heater will operate if water temp is 20°F or more below set temp

St An - In Standard mode => heater will operate anytime the thermostat demands heat => water temp is always within 1°F of set temp

Section 4: Trouble Shooting

DIAGNOSIS

The center column (Diagnosis) is arranged so that the most likely cause of the malfunction is listed first, so the corrective actions should be performed in order from top to bottom until the problem is solved.

PROBLEM	DIAGNOSIS	CORRECTIVE ACTION
ENTIRE SPA INOPERATIVE The control panel does not activate any of the spa functions or components.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control box may not be receiving the proper source voltage. Many problems are associated with electronic control systems are caused by low voltage.	Check the GFCI. If the GFCI has tripped, perform the Leakage Current Test. If the GFCI has not tripped, Perform the Source Voltage Test.
	The control panel may be defective.	Perform the Control Panel Test.
	The spa light, heater, and/or jet pump may not be receiving the proper output voltage.	Perform the Output Voltage Test for the spa light, heater, and jet pump.
SPA LIGHT INOPERATIVE The spa light cannot be operated with the control panel.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The light bulb may be burned out.	Check the light bulb, and replace it if necessary.
	The control panel may be defective.	Perform the Control Panel Test.
	If the spa light will not turn ON, the spa light may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Spa light.
	If the spa light cannot be turned OFF from the control panel, and the control panel is not defective then the control box is defective.	

JET PUMP INOPERATIVE The jet pump cannot be operated with the control panel.	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control panel may be defective.	Perform the Control Panel Test.
	If the jet pump will not turn ON (high-speed or low-speed), the jet pump may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Jet Pump.
	If the jet pump cannot be turned OFF from the control panel, and the control panel is not defective then the control box is defective.	
	If there is output voltage from the control box and the jet pump cannot be turned on, it is defective.	
HEATER INOPERATIVE The heater cannot be operated with the control panel.	If there is a message displayed in the control panel.	Refer to the Panel Message section.
	The set point may not be more than 2° F above the current water temperature displayed on the control panel.	Use the Temp button on the control panel to raise the set point more than 2° above the current water temperature displayed in the control panel.
	The electrical connections in the control box may be bad, the jumpers may not be properly configured, and/or one or more of the fuses may be blown.	Check the electrical connections, program jumpers, and fuses.
	The control panel may be defective.	Perform the Control Panel Test.
	The jet pump may not be operative.	Perform the Output Voltage Test, Jet Pump.
	The jet pump may not be producing adequate flow.	Perform the Flow Test (restricted flow is usually the result of dirty or clogged filters).

HEATER INOPERATIVE – cont.	The pressure switch may be defective.	Perform the Pressure Switch Test.
	The sensors may be defective.	Perform the Sensor Test, Control, and the Sensor Test, High Limit.
	The heater may not be receiving the proper output voltage at the control box.	Perform the Output Voltage Test, Heater.
	If there is output voltage from the control box and the heater cannot be turned on, it is defective.	
	If the heater cannot be turned OFF, even though the set point is more than 2° F below the current water temperature displayed on the control panel and the control sensor is not defective then the control box is defective.	
	If there is output voltage from the control box and the heater cannot be turned on, it is defective.	
GFCI TRIPPING	The electrical connections in the control box may be bad.	Check the electrical connections.
	There may be a leak in the equipment compartment and/or excessive moisture on any of the electrical components	Check for a leak in the equipment compartment and/or excessive moisture on any of the electrical components, and repair the leak and/or dry the equipment compartment and electrical components.
	There may be a ground fault.	Perform the Leakage Current Test.

HIGH LIMIT TRIPPING

Immediately after the high-limit trips, it must be determined whether it was just the heater that overheated, the entire spa, or neither.

If the spa water temperature is equal to or below the temperature set point only the heater overheated.	Flow is restricted in conjunction with a defective control box and possibly a defective pressure switch.	Perform the Flow Test (restricted flow is usually the result of dirty or clogged filters). Then perform the Pressure Switch Test.
If the water temperature is above the maximum temperature set point (104° F) the entire spa overheated.	The control sensor may be defective.	Perform the Sensor Test, Control.
	The jet pump and/or heater may not be able to be switched OFF, due to a defective control box.	Verify that the heater and jet pump function normally (refer to the Heater Inoperative and Jet Pump Inoperative sections).
If neither the heater, nor the entire spa overheated.	The high limit sensor may be defective.	Perform the Sensor Test, High Limit.

JET PUMP LEAKING

There are two places where the source of the leak may be located, the wet end and the plumbing union.

Wet End	Damaged volute and/or loose or damaged volute cover.	Tighten volute cover and/or replace volute and/or volute cover.*
	Damaged shaft seal.	Replace the shaft seal.*
	Loose or damaged freeze drain plug and/or o'ring.	Tighten or replace the freeze drain plug and/or o'ring.*
	Damaged volute o'ring.	Replace the volute o'ring.*
Plumbing union, Inlet or Outlet	Damaged volute and/or volute cover.	Replace volute and/or volute cover.*
	Loose or damaged compression fitting.	Tighten or replace the compression fitting.
	Damaged compression fitting o'ring.	Replace the o'ring.

*Replace the entire pump if servicing its components is prohibited by the warranty, or if replacement components are not available.

HEATER LEAKING

There are two places where the source of the leak may be located, the heater body and the plumbing union.

Heater Body	Damaged heater body.	Replace the heater.
	Loose or damage pressure switch.	Remove the pressure switch and reinstall or replace.
	Loose or damaged compression fitting.	Tighten or replace the compression fitting.
	Damaged compression fitting o'ring.	Replace the o'ring.
	Bulk head fitting is leaking.	Replace the bulk head fitting.

FLOW RESTRICTED

The jet pump is weak or sporadic, and/or no air is mixing at the jets, and/or the pressure switch will not close when the jet pump is switched on.

	The filter may be dirty or clogged.	Perform the Flow Test.
	The air control valve may be closed.	Open the air control valve.
	There may be a leak in one of plumbing lines, or the jet pump or heater may be leaking.	Repair any leaks (refer to the Jet Pump Leaking or Heater Leaking sections of this manual).
	One of the air or plumbing lines may be obstructed.	Remove the obstruction from the air or plumbing line.

Section 4: Trouble Shooting

TEST

SOURCE VOLTAGE TEST - 15 or 20 AMP 115 VAC SYSTEMS W/TWO POLE INLINE GFCI

Note: The following tests apply to the Oasis 115 VAC, 15 or 20 Amp, and the Oasis II, 115VAC, 20 Amp.

1. Verify that there is $115 \pm 10\%$ VAC between the neutral wire and the hot wire in the in-line GFCI. If there is not 103.5 to 126.5 VAC between the neutral wire and the hot wire on the input side of the breaker, call an electrician. If the breaker is closed and there is not $115 \pm 10\%$ VAC between the neutral wire and the hot wire on the output side of the breaker, then the breaker is defective.
2. Check the GFCI test button, and reset the breakers. Perform the **Leakage Current Test** if the GFCI breaker test button is tripped.

Note: Every time power is disconnected from the spa (even when resetting the breakers), power should not be reconnected for at least 30 seconds.

3. Measure the voltage on the main power terminal block between the hot wire (black) and the neutral wire (white). There should be $115 \pm 10\%$ VAC present at the terminal block. If the terminal block is not receiving from 103.5 to 126.5 VAC between the neutral wire and the hot wire, then there is a problem with the wiring somewhere between the in-line GFCI and the terminal block. Inspect for damaged wiring or components and check the terminal block, main power cord, and in-line GFCI for loose wiring.

LEAKAGE CURRENT TEST

Note: Every time power is disconnected from the spa, power should not be reconnected for, at least, 30 seconds. If power is prematurely reconnected to the spa, then disconnect power, wait 30 seconds, and then reconnect power.

DANGER! RISK OF ELECTRICAL SHOCK – THE SPA’S GROUND WILL BE INTERRUPTED AND THE TEST METER WILL BE PLACED IN SERIES WITH THE SPA’S GROUND WHEN PERFORMING THIS TEST.

1. Verify that the program jumpers and electrical connections are properly configured, and that none of the fuses are blown.
2. Unplug the spa’s power cord (15 or 20 amp spas) or turn off the breakers at the sub-panel (50 amp spas).
3. Disconnect all of the solid copper wires attached to any bonding terminals, and any additional ground or bonding wires to eliminate all paths to ground but the main power source ground wire.

DANGER! RISK OF ELECTRICAL SHOCK – DISCONNECTING THE GROUND WIRES ELIMINATES ALTERNATE PATHS TO GROUND, USE EXTREME CAUTION WHEN TESTING.

4. Disconnect the heater’s power wires on the appropriate models as per follows:
Remove the heater wires from relays RL5 and RL4 for Pre-2002 boxes or P20 and P21 in the 2002 – Current boxes.
5. Unplug the jet pump from the right side of the control box in a Pre-2002 box. Disconnect the bonding wires from the jet pump(s). All connections should be removed at the bonding terminal block on the right side of the control box.
For 2002 – Current, disconnect the terminals from the appropriate flag terminals – refer to the electrical schematic or the label on the control box.
6. If the spa is equipped with an ozonator, unplug it from the box. Additionally, disconnect its bonding wire.
7. Disconnect all main power lines or cord wires from the main power terminal block.

8. Test for continuity between the main source power coming from the G.F.C.I. to the spa by testing between the black (hot) wire and the ground (green) wire in a 115V setup. In any test situation, if there is any continuity, there is a ground fault condition and either the source power line or cord are defective.

WARNING -The power line or cord must NOT be receiving any voltage when performing this test.

9. Test for continuity between the main source power coming from the G.F.C.I. to the spa by testing between the white (neutral) wire and the ground (green) wire in a 115V setup.

WARNING - If there is ANY continuity at all between the neutral and ground wire, there is a ground fault condition. Either the power cord or the G.F.C.I. is defective

WARNING - This SHOULD show continuity. If there is NO continuity, trace both the neutral (white) and ground (green) wires for a broken or loose wire.

WARNING - The power line or cord must NOT be receiving any voltage when performing this test.

Main Power Cord

10. Disconnect the main power cord hot, ground, and neutral wires from the main power terminal block.
11. Test for continuity between the main power cord hot and ground wires. If there is any continuity at all, there is a ground fault, and the power cord is defective.
12. **WARNING** - The power cord must NOT be receiving any voltage when performing this test.
13. Test for continuity between the main power cord neutral and ground wires. If there is any continuity at all, there is a ground fault, and the power cord is defective.
14. **WARNING** - The power cord must NOT be receiving any voltage when performing this test.

Control Box

15. Reconnect the main hot and neutral wires to the terminal block, and connect the test meter's leads to the ground wire and the terminal block
16. Reconnect power to the spa. If the GFCI trips and there is no ground fault in the power cord, the control box is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage #1 (this value is the current leaking to ground from the control box). The control box is defective if current leakage #1 is equal to or greater than 0.5 AC mA.
17. Disconnect power to the spa.

Jet Pump (High and Low Speed)

18. Disconnect the power leads for the heater.
19. Plug in the jet pump. If this is a two-pump spa, repeat the same procedure on second pump, treating it as a separate pump.
20. Set the meter to read AC milliamps (0.5 milliamps MAX). For a pre-2002, place the meter in-line with the incoming ground wire for the spa and TB1, terminal 5. For the 2002 – current, place the meter leads on the ground terminal and the main terminal block.
21. Reconnect power to the spa and switch the jet pump to low speed. If the GFCI trips, the jet pump is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage #2 (this value is the current leaking from the jet pump when on low speed). The jet pump is defective if the value of current leakage #2 is equal to or greater than 0.5 AC mA.
22. Switch the jet pump to high speed. If the GFCI trips, the jet pump is defective. If the GFCI does not trip, write down how much current is leaking to ground, and label this value current leakage

- #3 (this value is the current leaking from the jet pump when on high speed). The jet pump is defective if the value of current leakage #3 is equal to or greater than 0.5 mA.
23. Disconnect the power to the spa, disconnect the high and low speed jet pump hot and neutral wires, then disconnect the test meter from the ground wire and the flag terminal. Reconnect the pump upon the conclusion of the test.

Heater

24. Reconnect the heater load power connection to the proper heater terminals in the control box.
25. Reconnect power to the spa, raise the set point to the highest setting and verify that the set point is at least 2° F above the actual water temperature. If the GFCI trips, the heater is defective.
26. Disconnect the power to the spa, reconnect the spa light and jet pump wires. Then reconnect all of the No. 8(Pre-2002) or 10(2002-current) AWG solid copper wires that were previously attached to the bonding terminal, and any other ground wires that you disconnected.

Verify that the GFCI device is functioning properly

27. Add the current leakage values #1 (control box) and #2(jet pump), label this value current leakage B (this value is the current leaking to ground from the entire spa when the heater is on). If the value of current leakage B is 4 to 6 mA and the GFCI does not trip, then the GFCI device is defective.
28. Add the current leakage values #2 (jet pump) and #3 (this value is the current leaking from the jet pump when on high speed), label this value current leakage C (this value is the current leaking to ground from the entire spa when the jet pump is on high speed). If the value of current leakage C is 4 - 6 mA and the GFCI does not trip, then the GFCI device is defective.

OUTPUT VOLTAGE TEST

Spa Light

1. Disconnect power to the spa for 30 seconds to restore the default setting – spa light OFF.
2. Reconnect power to the spa and check the main power terminal block for $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC) to see if the spa is receiving the proper source voltage. If the terminal block is not receiving the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Press the **Light** button and measure the voltage in the control box between terminals: For a pre-2002, J25 and J30 (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). For a 2002 – current, measure across, P22 and P23. The spa light flag terminals should supply $12 \pm 10\%$ VAC (10.8 VAC to 13.2 VAC) output voltage.
 - If the flag terminals did not supply $12 \pm 10\%$ VAC when the voltage was measured, perform the Control Panel Test to determine whether or not the control panel is defective. If the control panel is not defective, then the control box is defective.
4. Measure the voltage in the light socket of the light harness, if the flag terminals did supply $12 \pm 10\%$ VAC (10.8 VAC to 13.2 VAC). The light socket should also supply $12 \pm 10\%$ VAC output voltage.
 - If the light socket did not supply $12 \pm 10\%$ VAC when the voltage was measured, the light harness is defective.
 - For 2002-current, the fuse may be blown or defective

Jet Pump

1. Disconnect power to the spa for 30 seconds to restore the default setting – jet pump on STANDBY.
2. Reconnect power to the spa and check the main power terminal block for $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC), depending on the model, to see if the spa is receiving the proper source voltage. If the terminal block is not receive the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Use the **Temp**  button, Pre-2002, or the TEMP (-), 2002-Current, to lower the set point to the minimum setting 80° F (to keep the thermostat from switching the jet pump on unexpectedly).
4. Measure the voltage in the control box between the appropriate terminals, as illustrated in the *Electrical Schematics* section of this manual. The jet pump terminals should not supply any output voltage. If any voltage is supplied the control box is defective (provided that the thermostat did not switch the jet pump to low speed).
5. Press the **Jet** button (to switch the jet pump to low speed) and measure the voltage in the control box between the appropriate terminals as illustrated in the *Electrical Schematics* section of this manual or on the control box cover door. The jet pump terminals should supply $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC), depending on the model, output voltage (continue this test even if the proper output voltage is not supplied).
6. Press the **Jet** button (to switch the jet pump to high speed) and measure the voltage in the control box between the appropriate terminals as illustrated in the *Electrical Schematics* section of this manual. The jet pump terminals should supply $115 \pm 10\%$ VAC (103.5 VAC to 126.5 VAC), depending on the model voltage.
7. For models with dual pumps, test the second pump in the similar manner as above.

Note: *If the terminals did not supply $115 \pm 10\%$ VAC when the jet pump was switched to high and low speed, perform the Control Panel Test to determine whether or not the control panel is defective. If the control panel is not defective, then the control box is defective.*

Heater

1. Use the **Temp**  button, Pre-2002, or TEMP (-) button, 2002-Current, to lower the set point to the minimum setting 80° F, and verify that the set point is at least 2° F below the actual water temperature displayed on the control panel.
2. Check the main power terminal block for 115±10% VAC (103.5 VAC to 126.5 VAC) or 230±10% VAC (207 VAC to 253 VAC), depending on the model voltage, to see if the spa is receiving the proper source voltage. If the terminal block is not receive the proper source voltage, perform the **Source Voltage Test**, correct the failure, and restart this test.
3. Measure the voltage in the control box between the two heater connection (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). The heater connection should not supply any output voltage. If any voltage is supplied the control board is defective (provided that the thermostat did not switch the heater on).
4. Use the **Temp**  button, Pre-2002, or TEMP (+) button, 2002-Current, to raise the set point to the maximum setting (104° F), and verify that the set point is at least 2° F above the actual water temperature displayed in the control panel.
5. Measure the voltage in the control box between the two heater connection bars (refer to the electrical schematic sticker on the control box lid, or the *Electrical Schematics* section of this manual if necessary). The heater connection should supply 115±10% VAC (103.5 VAC to 126.5 VAC) or 230±10% VAC (207 VAC to 253 VAC) output voltage depending on the model of the spa. If the heater connection bars do not supply 115±10% VAC, refer to **Heater Inoperative** in the Diagnosis section of this manual.

CONTROL PANEL TEST

*Refer to **Entire Spa Inoperative** in the Diagnosis section of this manual, if the control panel will not power on, the display will not light up, and the control buttons do not function. If only some of the control panel buttons or control panel LEDs can be switched on and off, the control panel may be defective and should be considered suspect.*

1. Verify that the control panel's serial cable is properly connected to the circuit board.
2. Try to access the spa light, jet pump(s), and temperature set point from the control panel and note which components, buttons, and/or control panel LEDs do not function.
 - a) The **Light** button should switch the spa light and its indicator LED on and off.
 - b) The **Jets** button should access the following jet pump settings:
 - c) The temperature set point has a range from 80° to 104° F. The **Temp**  button, Pre-2002, or TEMP(-), 2002-Current, should lower the temperature set point one degree at a time. The heater and low-speed jet pump should turn off when the set point is ± 2° F below the actual water temperature displayed on the control panel.
 - d) The **Temp**  button, Pre-2002, or TEMP(+), 2002-Current, should raise the temperature set point one degree at a time. The heater and low-speed jet pump should turn on when the set point is ± 2° F above the actual water temperature displayed on the control panel.
3. Disconnect the control panel's cable and connect a test control panel.
4. Repeat step 2 with the test control panel installed, then review the following:
 - If the jet pump, heater, or spa light switch on when the spa is powered on and cannot be turned off with the test control panel, the control box is defective.
 - If the test control panel is fully functional and the suspect control panel is not, then the suspect control panel is defective.
 - If the test control panel responds exactly the same as the suspect control panel, then the control box and/or spa component(s) are defective. Refer to the Diagnosis section of this manual for each component that is inoperative.
 - If the test control panel is not fully functional, but more operational than the suspect control panel, then the suspect control panel is defective in conjunction with the control box and/or

spa component(s). Refer to the Diagnosis section of this manual for each component that is inoperative.

CONTROL SENSOR TEST

Verify that the program jumpers are properly configured.

1. Check the temperature of the spa water with a thermometer.
2. Check the actual water temperature of the spa displayed in the control panel.
 - If the temperature displayed in the control panel coincides with the temperature obtained with the thermometer ($\pm 2^{\circ}\text{F}$) then the sensor is not defective (test complete).
 - If the temperature displayed in the control panel is not within 2°F of the temperature obtained from the thermometer then continue with this test.
3. Disconnect the sensor from the control circuit board.
4. Use the ***Sensor Resistance vs. Temperature Chart or Graph*** to verify that the ohmic value of the sensor coincides with the temperature of the water.
 - If ohmic value and water temperature do not coincide with the values on the graph, the sensor is defective.
 - If the ohmic value corresponds with thermometer but not with the control panel display, the control board is defective.

HIGH LIMIT SENSOR TEST

1. Lower the temperature set point to 40°F , and switch the jet pump to high speed.
2. Check the temperature of the spa water with a thermometer.
3. Check the actual water temperature of the spa displayed in the control panel.
 - If the temperature displayed in the control panel is not within 2°F of the temperature obtained from the thermometer then perform the ***Control Sensor Test***.
4. Use the ***Sensor Resistance vs. Temperature Chart or Graph*** to verify that the ohmic value of the sensor coincides with the temperature of the water.
 - If ohmic value and water temperature do not coincide with the values on the graph, the sensor is defective.
 - If the ohmic value and water temperature do not coincide with the values on the graph, the control board is defective.

PRESSURE SWITCH TEST

*The control panel and jet pump must be operational and there must be good flow to perform this test. (Refer to the **Control Panel Test**, **Flow Test**, and **Jet Pump Inoperative** sections of this manual if necessary).*

1. Disconnect the wires from the pressure switch. If the “FLO” diagnostic message, Pre-2002, or “●●●”, 2002-Current, is not displayed on the control panel, the control box is defective.
2. Make sure the jet pump is turned on and jump the disconnected pressure switch wires. If the “FLO” diagnostic message, Pre-2002, or “●●●”, 2002-Current, is displayed on the control panel, the control board is defective.
3. Test for continuity three times – once with the jet pump off, once with the jet pump on low-speed, and once with the jet pump on high-speed.

Note: The wires from the circuit board to the pressure switch must be disconnected, otherwise continuity through the circuit board will be read.

- The pressure switch is working properly ONLY if there is continuity when there is adequate flow (jet pump on high and/or low speed) and no continuity when there is not pressure in the heater (jet pump on standby). The control box is defective if there are no flow problems and an “FLO” diagnostic messages, Pre-2002, or “●●●”, 2002-Current, appears in the control panel after the pressure switch has been properly installed and the continuity check has determined that the pressure switch is working properly.
- If there is no continuity when there is adequate flow (jet pump on high and/or low speed), or if there is continuity when the jet is pump off, the pressure switch is defective.

FLOW TEST

1. There must be enough water in the spa to cover the filter completely.
2. Turn the jet pump on (perform this test twice, once with the jet pump on low speed and once with the jet pump on high speed). Refer to the **Jet Pump Inoperative** section of this manual if the jet pump cannot be turned on.
3. Feel the water pressure in front of each of the jets. The water pressure should feel the same from each of the jets.
 - If only one of the jets feels weak, it may be in the closed position, or there may be an obstruction behind the jet nozzle. Switch the jet to the open position, or turn the jet pump off, remove the jet nozzle and inspect it.
 - If all of the jets feel weak, probably the filter is dirty or clogged. Turn the jet pump off, remove the filter from the filter compartment, and clean or replace it, depending on how dirty it is. To verify that the filter is responsible for the restricted flow remove the filter from the filter compartment, turn the jets back on, and feel the water pressure from one of the jets (the water pressure should increase significantly when the dirty or clogged filter is removed).
WARNING: DEBRIS IS FREE TO ENTER THE PLUMBING SYSTEM WHEN THE FILTERS ARE REMOVED, BE SURE THAT THE SPA IS CLEAN OR THIS COULD MAKE THE PROBLEM WORSE.
 - If all of jets feel sporadic the air control valve may be closed, or the jet pump may need to be primed if the water level was too low or if the spa was drained recently.
 - If all of the jets feel weak or sporadic and the pump is making an unusual noise the jet pump may be defective.

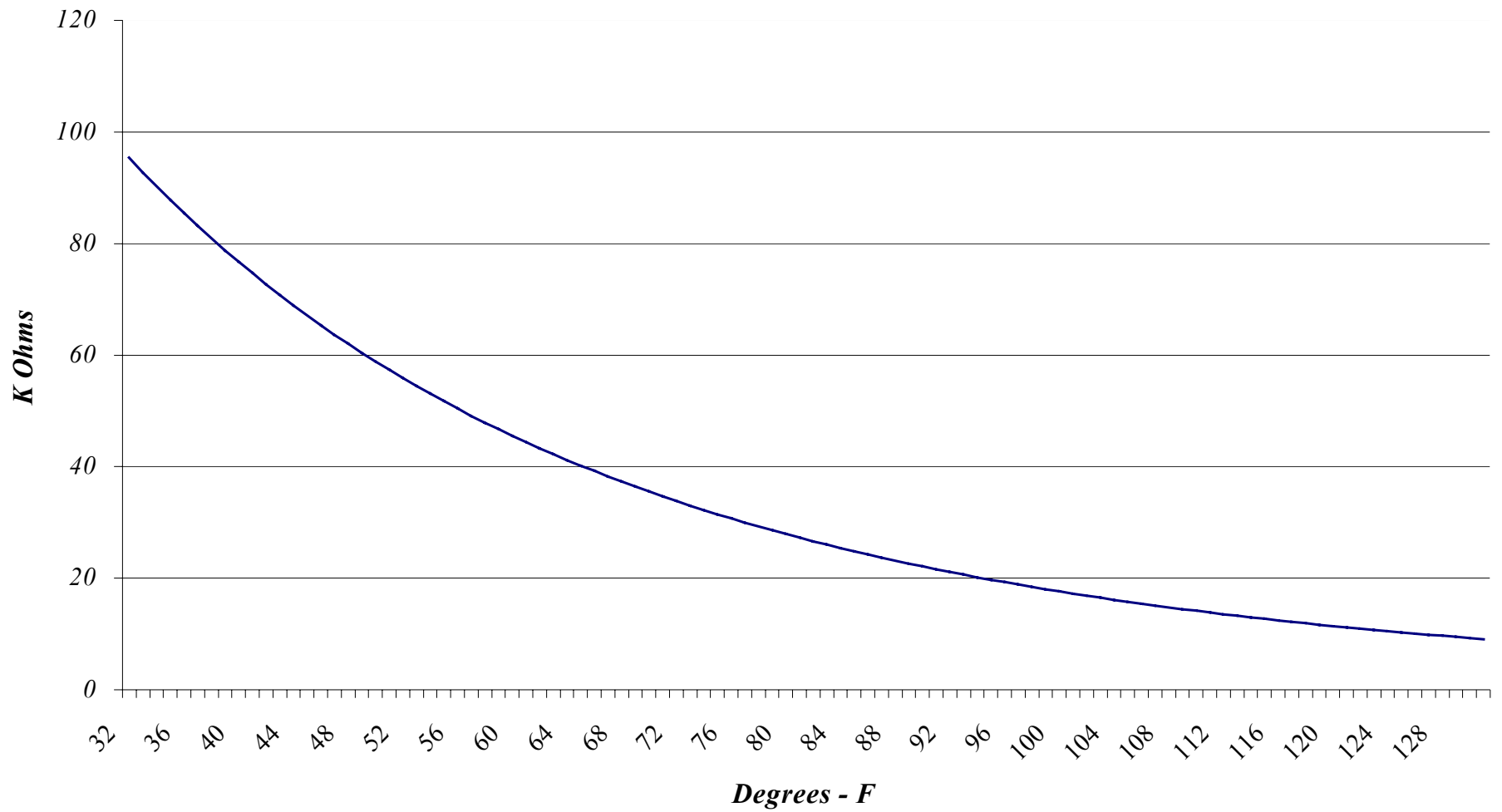
TRANSFORMER TEST (Pre-2002):

1. Power up the spa.
2. At the transformer connector on the board and measure as follows:
For the 120V transformer use the voltmeter set to read volts.
Take a reading between the black (3) and white (4) terminals. The reading will be 120 VAC $\pm$ 10%.
Take a reading between the yellow (2) and yellow (1) terminals. The reading will be 12 VAC $\pm$ 10%.

Sensor Resistance vs. Temperature Chart

<i>Resistance in Degrees – °F</i>	<i>Resistance in Degrees – °F</i>	<i>Resistance in Degrees – °F</i>
<i>K Ohms</i>	<i>K Ohms</i>	<i>K Ohms</i>
95.35618	32	39.20837
92.7654	33	38.25114
90.25305	34	37.32016
87.81653	35	36.41464
85.45334	36	35.5338
83.16106	37	34.67692
80.93736	38	33.84328
78.77999	39	33.03217
76.68679	40	32.24298
74.65567	41	31.4749
72.6846	42	30.72746
70.77165	43	30
68.91493	44	29.29193
67.11263	45	28.60269
65.363	46	27.93171
63.66435	47	27.27848
62.01506	48	26.64244
60.41355	49	26.02613
58.8583	50	25.42005
57.34784	51	24.83273
55.88076	52	24.26072
54.44569	53	23.70358
53.07131	54	23.16087
51.72634	55	22.63219
50.41955	56	22.11714
49.14975	57	21.61532
47.91579	58	21.12636
46.71655	59	20.6499
45.55097	60	20.18558
44.418	61	19.73306
43.31664	62	19.29201
42.24592	63	18.86211
41.2049	64	18.44305
40.19268	65	
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Sensor Resistance vs Temperature





Section 5: Removal & Installation

***IMPORTANT: BEFORE BEGINNING ANY EQUIPMENT REPAIR PROCESS ON THE SPA, ALWAYS
POWER DOWN THE SPA***

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Section 5: Removal and Installation

CONTROL PANEL

Removing the Control Panel

1. Turn the power off to the spa.
2. Remove the equipment compartment door.
3. Access the control box.
4. Disconnect the control panel's modular connector/ribbon cable in the control box.

On the bartop:

4. Make sure the water line is well below, i.e. six inches, the bottom of the control panel.
5. Using a thin metal blade, i.e. a putty knife, gently pry around the edges of the panel.
6. Remove the panel, gasket material, silicone, and cable from the opening in the shell.

Installing the New Control Panel

1. Align the panel up with the opening and thread the ribbon cable through the hole and down to the control box.
2. Rest the control panel in the location where it will be installed.
3. Connect the cable into the appropriate receptacle on the circuit board; refer to the Electrical Schematic.
4. Apply power and verify that the following are operative and make sure that the control panel is operating by running all functions.
 - Indicator Lights
 - Locking Features
 - Jets
 - Light
 - Set Temperature
4. Locate the gasket in the kit.
5. Disconnect the panel and dry fit the gasket. The masking tape can but used to outline proper placement.
6. Remove the remaining backing from the gasket and adhere it to the bartop.
7. Place a 1/8" bead of clear silicone to the lip area of the control panel.
8. Fit the panel onto the gasket and press down evenly to secure in place.
9. Use the clean towel or rag to remove any excess silicone from the area.
10. Inform the spa owner to wait 24 hours before using the spa to allow the silicone to dry completely.

Section 5: Removal and Installation

CONTROL BOX/HEATER

Removing the Control Box and Heater for Pre-2002

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. Disconnect the connections on the right side of the box, i.e. ozone, pump, and spa light.
Optional: Marking and tagging wires for removal - Mark each wire prior to removal by using masking tape and a permanent marker.
4. Disconnect the copper bonding wire(s) from the right side of the control box.
5. On the front of the control box, remove the screws and take off the control box cover.
6. On the bottom of the control box on the left side, disconnect the control panel cable.
7. On the left side of the control board, disconnect the sensors with needle nose pliers from J20.
Note: Mark the direction of the connector.
8. On the left side of the control board, disconnect the pressure switch with needle nose pliers from J9.
Note: Mark the direction of the connector.
9. Split nuts are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the split nut, a pipe wrench can be used.
Note: It is possible that there is approximately 2-5 gallons of water in the plumbing system that will need to be drained when the heater is disconnected.

Installing the Control Box and Heater for Pre-2002

10. Refer to the Electrical Schematic for the correct configuration of the jumpers.
11. Reverse the removal procedure to install the control box / heater in the equipment compartment.

Removing the Control Box and Heater for 2002 - Current

Refer to the following instructions to remove the control box from the equipment compartment.

1. Disconnect power to the spa.
2. Drain the spa.
3. Remove the equipment compartment door.
4. Open the control box lid.
5. Unplug the control panel 8-pin connection wire, the control thermistor 4-pin connection wire, and remove them from the control box.
6. Disconnect the pressure switch connection wires from the pressure switch.
7. With a flathead screw driver, loosen the screws that secure the neutral, line 1, and ground wires to the main power terminal block, loosen the control box main power line input compression fitting, and pull the main power cord out of the control box.
8. With needle-nose pliers, disconnect the spa light hot and neutral wires from the flag terminals, P22 and P23, and remove the spa light wires from the control box.
9. With needle-nose pliers, disconnect the jet pump hot (high and low speed), neutral, and ground wires from the flag terminals, and push the wires out of the control box through the jet pump output hole.
10. With two small wrenches of the same size, loosen the nuts that secure the heater connecting bars. Then disconnect the ground wire attached to the heater.
WARNING: Always use both wrenches to loosen the heater connecting bar. The bottom nut is fused to the heater terminal to prevent the terminal from being twisted when the top nut is turned. If only one wrench is used to loosen the connecting bar, the torque applied to the heating element will be too great, and the heater will be permanently damaged.

11. With a Phillip's head screwdriver, remove the two mounting screws that secure the control box to the equipment compartment wall.
12. Slip the control box off the heater body, and remove it from the equipment compartment.

Installing the Control Box and Heater

Reverse the removal procedure to install the control box in the equipment compartment.

NOTE: *Be sure that the high limit sensor is positioned correctly in its slot before you slip the control box in place.*

Section 5: Removal and Installation

HEATER

Removal for Pre-2002

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. Disconnect the connections on the right side of the box, i.e. ozone, pump, and spa light.
Optional: Marking and tagging wires for removal - Mark each wire prior to removal by using masking tape and a permanent marker.
4. On the right side of the control box, disconnect the copper bonding wire(s).
5. On the front of the control box, remove the screws and take off the control box cover.
6. On the bottom of the control box on the left side, disconnect the control panel cable.
7. On the left side of the control board, disconnect the sensors with needle nose pliers from J20.
Note: Mark the direction of the connector.
8. On the left side of the control board, disconnect the pressure switch with needle nose pliers from J9.
Note: Mark the direction of the connector.
9. Split nuts are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the split nut, a pipe wrench can be used.
Note: It is possible that there is approximately 2-5 gallons of water in the plumbing system that will need to be drained when the heater is disconnected.
10. Inside the control box, disconnect the heater by removing the black heater wires and bonding screws using a nut driver.
11. On the bottom of the equipment compartment, remove the mounting screws.
12. Lift the control box/heater away from the base of the equipment compartment and remove.
13. On the backside of the control box, use the nut driver to remove the screws from the posts that hold the heater in place.
14. Remove the metal straps from around the heater.
15. Grasp the pressure switch and turn to the left, counter clockwise, to loosen. If necessary, channel lock pliers may be used.

Installation for Pre-2002

16. Install the new heater onto the control box in the reverse order of this instruction.
Note: When installing the pressure switch, it may be necessary to apply a non-hardening pipe joint compound or teflon tape to the threads.

HEATER – cont.

Removal for 2002-Current

Refer to the following instructions to remove the heater from the equipment compartment.

1. Disconnect power to the spa.
2. Drain the spa.
3. Remove the equipment compartment door.
4. Drain the spa or use the plug system.
5. Open the drain valve to bleed the water out of the plumbing.
6. Open the control box lid.
7. Disconnect the pressure switch connection wires from the pressure switch and remove the pressure switch (refer to the pressure switch removal procedure).
8. With two small wrenches of the same size, loosen the nuts that secure the heater connecting bars. Then disconnect the ground wire attached to the heater.
WARNING: Always use both wrenches to loosen the heater connecting bar. The bottom nut is fused to the heater terminal to prevent the terminal from being twisted when the top nut is turned. If only one wrench is used to loosen the connecting bar, the torque applied to the heating element will be too great, and the heater will be permanently damaged.
9. Unscrew the heater inlet and outlet compression fittings.
CAUTION: There may still be some residual water in the plumbing.
10. Slip the heater out of the control box, and remove it from the equipment compartment.

Installation for 2002-Current

Reverse the removal procedure to install the heater in the equipment compartment (refer to the pressure switch installation procedure if necessary).

NOTE: Be sure that the high limit sensor is positioned correctly in its slot before you slip the heater in place. After you are finished with the installation, switch the jet pump to high speed and check the compression fittings for leaks.

Section 5: Removal and Installation

PRESSURE SWITCH

Removal of a Pre-2002:

1. Disconnect power to the spa and access the equipment compartment.
2. Drain the spa.
3. The pressure switch is located on the backside of the heater.
4. Using needle nose pliers, disconnect the wires from the flag terminals.
5. Grasp the pressure switch and turn to the left, counter clockwise, to loosen. If necessary, ½" wrench may be used.

Note: Some water, approximately 2-5 gallons, will come out when the pressure switch is removed.

Installation of a Pre-2002:

1. Apply a non-hardening pipe joint compound or teflon tape to the threads of the new pressure switch.
2. Align the threaded end of the switch perpendicular to the opening in the heater.
Important: The threads must be properly aligned so that installation will not break off the switch when tightening it in the heater.
3. Turn the pressure switch right, clock wise, and hand-tighten in place.
4. Replace the black wire into the upper flag terminal and the red wire into the lower flag terminal.
5. Fill the spa.
6. Test the setup as follows:
 - Disconnect the pressure switch from the circuit board.
 - Set your voltmeter to the continuity setting.
 - Place the meter leads onto the two wires of the pressure switch.
 - In applicable spas, open the gate valves before and after the heater.
 - Turn on the low-speed pump.
 - The meter should register continuity.
 - Turn the pump on from low-speed to high-speed if available and run for ten seconds.
 - After ten seconds has passed, turn off all pumps.
 - After four seconds, check the voltmeter and verify that there is no continuity registering.

PRESSURE SWITCH – cont.

Removal of a 2002-Current:

1. Disconnect power to the spa.
2. Remove the equipment compartment door.
3. Drain the spa.
4. Open the control box lid.
5. Disconnect the pressure switch connection wires from the pressure switch.
6. Unscrew the pressure switch from the heater, a ½” wrench may be used, and remove any residual Teflon tape from the threads of the heater body where the pressure switch was connected.

Installation of a 2002-Current:

Reverse the removal procedure to install the pressure switch in the control box.

NOTE: *Wrap one piece of Teflon tape (in the opposite direction of the threads) three times around the pressure switch threads before installation. When screwing the new pressure switch into the heater, DO NOT use pliers, grasp the body of the pressure switch and hand tighten only. After you are finished with the installation, switch the jet pump to high speed and check the pressure switch for leaks.*

Section 5: Removal and Installation

JET PUMP

Removal:

1. Power down the spa.
2. Access the equipment compartment.
3. Drain the water from the spa
4. Drain the water from the pump by removing the freeze/drain plug.
5. Access the control box.
6. Disconnect the connector from the control box.
7. There are approximately three (3) zip ties along the path of the power cord. These will need to be cut in order to free the cord.
8. Remove the bond wire, where applicable, from the jet pump.
9. Remove the four (4) butterfly bolts that hold the pump to the bottom of the equipment compartment.
10. The pump unions are hand-tightened in the factory and should be loosened from the heater union prior to removal from the system. If there is difficulty in loosening the pump union, a pipe wrench can be used.

Note: It is possible that there is a small amount of water in the pump that will need to be drained when the heater is disconnected.

Installation:

1. Reverse the removal procedure to install the jet pump in the equipment compartment.
2. Fill the spa.

Note: After installation, switch the jet pump to high speed and check the compression fittings for leaks.

Section 5: Removal and Installation

SPA LIGHT

Removal:

1. Disconnect power to the spa.
2. Remove the front panel to access the equipment compartment.
3. Locate the connector on the right side of the control box and remove the spa light wires from the control box.
4. Unscrew the spa light one half turn to the left to free it from the back wall of the equipment compartment.
5. Remove the spa light from the equipment compartment.

Installation:

Reverse the removal procedure to install the spa light in the equipment compartment.

Section 5: Removal and Installation

CONTROL/HI LIMIT SENSOR FOR A PRE-2002 SPA

Removal:

1. Disconnect power to the spa.
2. Drain the spa until the water level is below the thermowell in the filter compartment.
3. Remove the front access panel.
4. Open the control box lid.
5. Unplug the control sensor 4-pin connection wire (refer to the Electrical Schematic for the proper jumper location J-20) and remove it from the control box.
6. Trim any zip ties that hold the cords in place.
7. Unscrew the thermowell compression fitting, and remove the sensor. Use a wrench to loosen the compression fitting if necessary.
8. Remove the hi-limit sensor from the clip on the heater.

Installation:

1. Reverse the removal procedure to install the control sensor in the equipment compartment taking special care to apply insulation to the backside of the control sensor.
Note: After installation, switch the jet pump to high speed and check
2. Refill the spa.

CONTROL THERMISTOR FOR A 2002-CURRENT

Removal:

1. Disconnect power to the spa.
2. Drain the spa until the water level is below the thermowell in the filter compartment.
3. Remove the equipment compartment door.
4. Open the control box lid.
5. Unplug the control thermistor 4-pin connection wire and remove it from the control box.
There isn't very much room to access the thermowell from the equipment compartment, if necessary, perform the following procedure to remove the jet pump to heater plumbing.
 - a) Drain the spa or use the plug system.
 - b) Open the drain valve to bleed the water out of the plumbing.
 - c) Unscrew the jet pump outlet compression fitting, unscrew the heater inlet compression fitting, disconnect the bleed line, and then remove the heater to jet pump section of plumbing from the equipment compartment.
6. Unscrew the thermowell compression fitting, and remove the thermistor. Use a wrench to loosen the compression fitting if necessary.
CAUTION: There may still be some residual water in the plumbing.

Installation:

Reverse the removal procedure to install the control thermistor in the equipment compartment.

NOTE: After you are finished with the installation, switch the jet pump to high speed and check the thermowell for leaks.

NOTE: HIGH LIMIT THERMISTOR CANNOT BE REPLACED. IF THE HIGH LIMIT IS NOT FUNCTIONING PROPERLY, REPLACE THE ENTIRE PACK.

Section 5: Removal and Installation

TRANSFORMERS FOR A PRE-2002 SPA

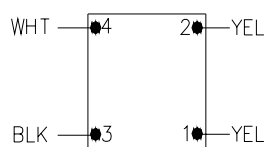
Removing the old transformer:

IMPORTANT: IT IS IMPERATIVE THAT THE POWER TO THE SPA BE DISCONNECTED BEFORE THIS PROCEDURE IS CARRIED OUT.

1. Access the equipment compartment.
2. On the front of the control box, remove the screws and take off the control box cover.
3. Locate the connector, J17, on the board and disconnect it.
4. The transformer is located on the top right part of the box.
5. On the outside of the box, take the screws out of the transformer using a nut driver.

Installing the new transformer:

1. Install the new transformer and using the nut driver, put the screws in place.
2. Install the connector into board using the electrical schematic illustrated below.



Testing the new transformer:

1. Power up the spa.
2. At the transformer connector on the board and measure as follows:
 - For the 120V transformer in the Highland series spa, use the voltmeter set to read volts. Take a reading between the black (3) and white (4) terminals. The reading will be 120 VAC $\pm$ 10%.
 - Take a reading between the yellow (2) and yellow (1) terminals. The reading will be 12 VAC $\pm$ 10%.

Section 5: Removal and Installation

AIR CONTROL VALVE

Removal

1. Disconnect power to the spa and drain the spa.
2. Remove the equipment compartment door.
3. Reach up underneath the bartop (from the equipment compartment) and unscrew the plastic lock nut that holds the air control valve in place against the spa shell.
4. Pull the air valve up through the hole in the spa shell, and cut it off using PVC cutters.
5. Pull the air control valve to free it from the spa.

Installation:

Reverse the removal procedure to install the air control valve.

Section 5: Removal and Installation

PLUMBING PIPE, FITTINGS AND CEMENT

Watkins Manufacturing uses standard sizes schedule 40 pipe and fittings ranging from ½ to 2 inch outside diameters.

Recommended Cements:

- Weld-on 1759 – This cement is used to bond PVC pipe or fitting to other PVC components. This cement has fillers to help seal loose PVC joints.
- Weld-on 794 – This cement is used to bond ABS fittings to rigid PVC pipe and fittings or other ABS components. Do not bond ABS fittings to flexible PVC hose.
- Weld-on 70 – This is a primer that should be used on flexible PVC hose before bonding to PVC fittings with Weld-on 1759 cement.

These cements are available at most hardware or plumbing stores.

Preparing pipes and fittings for bonding

1. Ensure that the pipe ends are free from burrs and cut perpendicular to the length of the pipe.
2. Check the fit of the joint before bonding. The fit of the joint will determine the drying time of the joint. To determine whether the fit of a joint is tight or loose, slip a piece of rigid PVC pipe into a fitting.

The pipe will feel snug half way into the fitting pocket if the fit is tight. The pipe will easily fit to the end of the fitting if the fit is loose.

Note: Flexible PVC tubing will fit into a PVC fitting ¼-inch when dry. The primer and cement will enable the hose to slide to the end of the fitting socket.

Bonding of Pipe and Fittings

1. When cementing PVC pipe, spread primer around the end of the pipe three times. Do not allow the primer to dry.
2. Using the glue applicator, spread the proper cement around the end of the pipe five times for full coverage. Avoid putting cement inside the socket.
Note: For ABS fittings this is imperative because of their thin walls.
3. Insert the pipe in the socket of the fitting. The pipe must seat on the bottom of the socket.
4. Hold the joint for 20 seconds to ensure the pipe does not back out of the socket before the cement is set.

Drying time depends on the fit of the joint. A tight fitting joint should dry within 15 minutes where a loose fitting joint could take 25 minutes or longer. The drying time also depends on weather conditions. Cold weather or high humidity may require longer drying times.

Visually inspect the outside of the glue joint. Be sure the cement is visible all around the joint. If possible, inspect the inside of the joint. Be sure the pipe is seated correctly and the joint is properly cemented.

Section 5: Removal and Installation

VINYL COVERS

The vinyl cover consists of a vinyl encasement and a foam core that insulate the spa when it is not in use. The cover also acts as a safety feature to keep young children out of the spa when the cover is fastened and locked into place. There are four (4) coverlocks hold and secure the cover on the spa.

Puddles on the Vinyl Cover

Puddles on the vinyl cover indicate that the cores are either broken or bowed. A small puddle could gradually grow and cause the cover to bow, which could cause the cover to sag. A sagging cover can create a gap between the cover and the spa, which can lead to heat loss in the water.

- If a core is suspected of being warped on initial delivery, test the cover by applying water on the cover while it is on the spa. If a large puddle forms, take the core out of the encasement, turn it over and reinstall it in the encasement.
- If the core is broken, replace the core.
- If the core is bowed or if there is an isolated depression, take the core out of the encasement, flip it over and reinstall it.

Heavy Foam Cores

A heavy core indicates that the core has absorbed water, usually through a tear in the vinyl. If there is a tear, the vinyl will have to be repaired or replaced.

Depending on the amount of water the core has absorbed it may be able to be dried by the customer. The core should be removed from the encasement and placed in direct sunlight.

Repairing Vinyl Tears

Many times when the encasement has a tear, it can be repaired with one of several vinyl repair kits available. Vinyl repair kits are typically found at major department stores. Refer to the instructions with the kit to make this repair.

VINYL COVERS – cont.

Replacing Foam Core

Refer to the following instructions to replace vinyl cover foam core:

1. Locate the zipper-pull in the pocket found on the side of the cover and open. Do not force the zipper. If the zipper gets stuck, lubricate the zipper with light oil (WD-40) or silicone spray. Gently work a stubborn zipper free.
Note: If the zipper breaks, it will not be covered under warranty.
2. Once the vinyl encasement is unzipped, equally pull on each corner of the foam core and remove it from the encasement.
3. Insert the new foam core into vinyl encasement. Be careful handling the new core so that the plastic covering does not tear. If the plastic rips, use two-inch pieces of quality tape to seal the rip.
 - Insert the core straight into the encasement. This is easier done with two people than with only one. It is easier to accomplish when the vinyl is warm so it will stretch.
 - If the core is slightly too large, use a large rubber mallet to pound the sides of the foam core. This will make the core smaller by compressing the foam along the edge.
 - If the core is too small put a foam strip along the bottom (along the zipper) edge of the foam.***Note: Replacing the foam core is easier on warm days.***
4. Close the zipper. Pull the zipper flap down and pull the zipper halves as close together as possible to make it as easy as possible to pull the zipper around the corners of the foam. Once the zipper is closed, pull up the zipper flap.

Section 5: Removal and Installation

COVERLOCK REPLACEMENT

1. A notch must be made in the portion of the insert where the vinyl cover strap is placed. With a pair of dikes, make a cut at an angle from the edge of the insert to the center.
2. Make a straight cut in the center of the insert to meet the angled cut. Carefully remove the notch at the smallest part of the cut (should be about 3/16 inch).

NOTE: Be careful when making these cuts not to crack or break the insert.

3. Insert the strap from the vinyl cover through the notch.
4. Close the notch with the special tape supplied in the coverlock kit.



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Section 6: Appendix

JUMPER SETTING

The jumpers are located on the main circuit board inside the control box (refer to the Electrical Schematics for the exact locations). The setting of the jumpers determine how the spa's electronic control system will operate. Improper jumper setting will cause erratic spa functions, and may cause the house circuit breaker to trip.

As an example, imagine a control box is installed in a convertible spa wired for 115 VAC. If the jumpers are set for the proper spa model, but wrong voltage of **230 VAC** (converted), the control box will allow the heater to turn on while the jets are on. This will draw too much current, which can trip the house breaker.



Setting the Jumpers

1. Disconnect power from the control box.
2. Refer to the illustrations on the following page to find the jumper setting for your specific spa model.
3. Using a pair of needle-nose pliers, carefully remove or install the jumper in the appropriate location.

IMPORTANT: DO NOT ALTER THE SETTINGS OF JUMPERS NOT CALLED OUT ON THE ELECTRICAL SCHEMATICS.

Section 6: Appendix

JUMPER SETTING: OASIS - 2002 ONLY

Improper jumper positioning may cause permanent damage to the spa. There are three jumpers on the circuit board and two positions for each jumper.

Jumper # 1 (JM 1) – Controls the voltage settings.

Position 1 - (115 VAC 30 Amp spas ONLY): High current mode – enables both the high speed jet pump and the heater to operate at the same time.

WARNING: If the spa is not hardwired for high current mode, the control box and spa components will be permanently damaged.

Position 2 - (115 VAC 15 Amp spas): Low current mode – heater is forced off when the high speed jet pump is turned on.

Jumper # 2 (JM 2) – Controls the temperature designation.

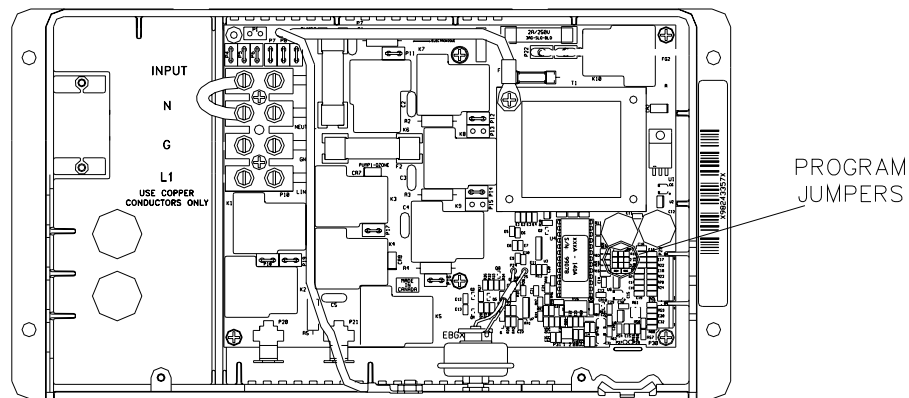
Position 1 - Displays the spa water temperature in Fahrenheit.

Position 2 - Displays the spa water temperature in Celsius.

Jumper # 3 (JM 3) – Developmental

Position 1: Set in position 1 for single pump systems.

Position 2: Set in position 1 for dual pump systems.



GECKO PROGRAM JUMPER

PINS	PINS
1 2 3	1 2 3
JMP 1	JMP 1
JMP 2	JMP 2
JMP 3	JMP 3
230 V, 50 AMP	115 V, 15 AMP
CONNECTED	CORD CONNECTED
CONFIGURATION	CONFIGURATION

JUMPER SETTING: OASIS – PRE-2002 – ONLY

JUMPER SETTINGS

JUMPER	AS LABELED ON BOARD	CALDERA FUNCTION	DEFAULT SETTING
J-23	50A/20A	SYSTEM VOLTAGE	115 POSITION
J-32	----	----	OPEN
J-33	----	----	CLOSED

Section 6: Appendix

SUBPANEL WIRING INSTRUCTIONS – (2002-CURRENT)

SPA MODELS: Oasis

Required Tools:

Standard Screwdriver Needle-nose Pliers
Phillips Screwdriver Strain Relief Compression Tool or Vice-Grips

Required Parts:

38661 (1) Breaker GFCI, Subpanel Box **36021** (1) Program Jumper, Shunt 0.100

1. Disconnect power to the tub and access the control box.
2. Open the control box and locate the program jumpers JP1 – JP3 (on the main circuit board).
3. Using a pair of needle nose pliers, carefully install a jumper at location JP1 (refer to Figure 1).

NOTE: The sub-panel must be placed within 100 feet of the main electrical service panel, and between 5 and 50 feet away from the spa. All electrical connections must be made in accordance with the wiring information contained in this manual and on the back of the field wiring access panel of the control box.

Refer to Figure 1

1. Connect the #8 AWG, WHITE, [NEUTRAL] wire from NEUTRAL/GROUND Bar on the main electrical service panel to the WHITE [NEUTRAL, Panel Neutral, Pigtail] of the sub-panel.
2. Connect the #8 AWG, BLACK, [L1] wire from the main electrical service panel to the sub-panel [terminal L1].
3. Connect the #8 AWG, RED, [L2] wire from the main electrical service panel to the sub-panel [terminal L2].
4. Connect the #10 AWG, GREEN, wire from the NEUTRAL/GROUND Bar on the main electrical service panel to the GROUND terminal of the spa's control box.
5. Connect the #8 AWG, WHITE [NEUTRAL] wire from the sub-panel to the Netral terminal on the spa's control box.
6. Connect the #8 AWG, BLACK [L1] wire from the sub-panel breaker to terminal [L1] on the spa's control box.
7. Connect the #8 AWG, RED [L2] wire from the sub-panel breaker to terminal [L2] on the spa's control box.
8. Bond the spa to all exposed metal equipment or fixtures, handrails, and concrete pad pre N.E.C. and all local codes.

WARNING! Bypassing the G.F.C.I. breaker will result in an unsafe spa and void the warranty.

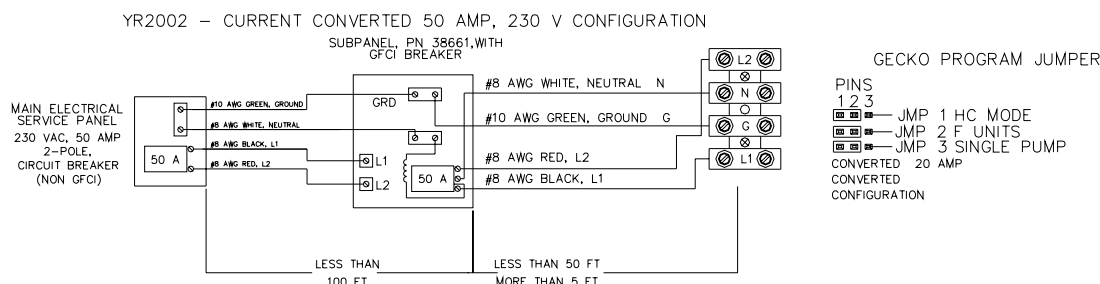


Figure 1

SUBPANEL WIRING INSTRUCTIONS - (PRE-2002)

SPA MODELS: Oasis

Required Tools:

Standard Screwdriver Needle-nose Pliers
Phillips Screwdriver Strain Relief Compression Tool or Vice-Grips

Required Parts:

38661 (1) Breaker GFCI, Subpanel Box **36021** (1) Program Jumper, Shunt 0.100

1. Disconnect the power.
2. Remove the screws and open the equipment compartment door.
3. Remove the screws and open the control box.
4. Locate TB-1, located in the lower right-hand corner of the control box.
If connected, remove the 115 VAC power cord wires from terminals 2, 3 and 5 then remove the strain relief and 115 VAC power cord from the access hole in the control box. - Refer to the picture at below.
5. On the main circuit board (inside the control box), locate the Jumper J-23 (positioned on the upper left of the circuit board, next to the top).

WARNING! Do not allow the pliers to contact any electronic components inside the control box.

IMPORTANT – BE SURE TO CHANGE THE JUMPERS IN LOCATIONS J-23 ONLY. ALTERING OTHER JUMPER LOCATIONS MAY CAUSE DAMAGE TO THE SPA.

6. Use a pair of needle-nose pliers to place J-23 from the 20A position to the 30A.

CAUTION: Damage may result if the jumpers are incorrectly set.

7. The subpanel (38661) containing the GFCI breakers must be installed in series between the spa and the main electrical panel. A LICENSED electrician must install the subpanel.

WARNING! Damage may result if the terminal block or subpanel is incorrectly wired.

